

# Scaffolded Incomplete Fusion Theory: A Cross-Domain Interface-Barrier-Product Framework

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## Abstract

Fusion is usually described as an event in which two entities become one: nuclei merge, membranes merge, datasets merge, materials join, or models combine. This manuscript proposes a more general interface theory: useful fusion is rarely maximized by complete merger alone, but by temporary scaffolded alignment, bounded or incomplete merger, and rapid export or conversion of fusion products. The theory is formulated through an interface-barrier-product functional and initially tested in four dimensionless virtual experiments: adaptive information fusion, bridge-mediated barrier reduction, optimal incomplete fusion, and a product-shadow feedback loop. Version 0.2 adds finite-lifetime scaffold turnover, selective confinement/deconfinement, critical network fusion near percolation, and the exceptional-fusion complementarity ridge. Version 0.3 adds pulsed fusion, recovery-window resonance, nonuniform bridge placement, and diversity-limited information fusion. Version 0.4 adds shadow-to-scaffold recycling, finite interface annealing, impedance-matched scaffolds, and orthogonal product-selective gates. Version 0.5 adds fusion memory, scaffold-shadow autocatalysis, dual-channel percolation, anchor-complementarity windows, and memory-selective reset. Version 0.6 adds shadow-vector balancing, latency-induced fusion oscillation, sparse-contact topology, and hierarchical granularity. Version 0.7 adds non-associative fusion sequencing, scout-fusion value, reciprocal scaffold complementarity, and scarce-scaffold allocation. Version 0.8 adds sacrificial product sinks, defusion preconditioning, finite boundary-layer thickness, and post-selective fusion rectification. Version 0.9 adds compliance-partitioned mismatch allocation, anti-coincidence scheduling for convex product loads, reversibility reserves under hidden shadow, and controlled impurity/dopant windows. Version 1.0 adds witness-scaffold identifiability, action-geodesic scaffold paths, counter-shadow cofusion, and ruin-limited fusion portfolios. Version 1.1 adds mode-locked spectral aperture, selective anti-fusion moats, curvature-enhanced interface windows, and extreme-tail product-shadow throttling. The combined results motivate an expanded design thesis: exceptional fusion is governed not only by drive or contact energy but by scaffold count, scaffold lifetime, fusion degree, product residence, network connectivity, complementarity distance, duty cycle, recovery period, effective diversity, product recycling fraction, annealing amplitude, impedance mismatch, useful-shadow channel geometry, signed memory coefficient, useful/shadow connectivity split, anchor overlap, reset selectivity, residual shadow-vector geometry, feedback latency number, active contact fraction, modular granularity, fusion sequence, scout effort, reciprocal scaffold vectors, scaffold price/budget, sink capacity, defusion number, boundary-layer thickness, post-selection threshold, compliance/action vector, product phase offsets, reversibility reserve, dopant/impurity dose, witness fraction, action-geodesic path, counter-shadow dose, ruin-limited portfolio weights, spectral aperture, anti-fusion moat strength, interface curvature, and tail-shadow index. These results are hypotheses and mathematical reductions, not established physical laws. Their potential value is as a common search frame for blindspots across thermonuclear fusion, lattice-assisted nuclear effects, biological membrane fusion, biomolecular condensates, dissimilar materials joining, and multimodal information fusion.

**Keywords:** fusion theory; interface physics; scaffolds; product shadow; partial fusion; data fusion; membrane fusion; nuclear fusion; dissimilar materials; multimodal alignment; pulsed fusion; recovery windows; diversity-limited fusion; product recycling; interface annealing; impedance matching; orthogonal gates; fusion memory; autocatalysis; dual percolation; anchor complementarity; selective reset; shadow-vector balancing; latency instability; sparse contact; hierarchical granularity; non-associative sequencing; scout fusion; reciprocal scaffold complementarity; scarce scaffold allocation; sacrificial sinks; defusion preconditioning; boundary layers; fusion rectification; compliance partition; anti-coincidence scheduling; reversibility reserve; controlled dopant window; witness scaffolds; action geodesics; counter-shadow cofusion; ruin-limited portfolios; mode-locked aperture; selective moat; curvature window; extreme-tail shadow.

## 1 Epistemic status and scope

This document is a working scientific manuscript. It distinguishes between: (i) established background from the cited literature, (ii) mathematical consequences of stated assumptions, and (iii) proposed principles that require independent testing. The central claim is not that all fusions are governed by one literal microscopic mechanism. The claim is that many fusion systems share a common optimization structure involving an interface, a convex barrier, a controllable degree of merger, a scaffold or alignment map, and a product-removal problem.

The analysis covers representative fusion classes rather than an exhaustive catalogue. The included domains are nuclear fusion, lattice-confined or solid-state nuclear hypotheses, biological membrane and cell fusion, viral fusion, biomolecular condensates, materials joining, and multimodal information fusion. The common blindspot identified here is that many programs optimize the immediate fusion event while under-optimizing a growing set of variables that determine whether the result remains useful:

$$\begin{aligned}
 &\text{scaffold count } k, \quad \text{fusion degree } \lambda, \quad \text{product residence } \tau_P, \\
 &\text{duty cycle } \delta, \quad N_{\text{eff}}, \quad \text{product-recycling fraction } \xi, \\
 &\text{annealing amplitude } \sigma, \quad \text{impedance ratio } z, \quad \text{useful-shadow angle } \vartheta, \\
 &\text{signed memory } g, \quad \text{useful/shadow connectivities } (c_F, c_P), \\
 &\text{anchor overlap } \rho, \quad \text{reset selectivity } q_P/q_M, \quad \text{fusion path } \pi, \\
 &\text{scout effort } e, \quad \text{reciprocal scaffold maps } (u, n), \quad \text{scaffold price and budget } (\nu, S), \\
 &\text{sink capacity } s, \quad \text{defusion number } \mathcal{F}_D, \quad \text{boundary thickness } h, \quad \text{selection threshold } t^*, \\
 &\text{spectral aperture } \Gamma, \quad \text{moat strength } m, \quad \text{curvature } \kappa, \quad \text{tail index } \alpha_T.
 \end{aligned} \tag{1}$$

## 2 Motivating observations across fusion domains

### 2.1 Nuclear fusion

Recent fusion milestones show that ignition and useful power-plant closure are related but not identical goals. The National Ignition Facility reported an April 2025 experiment that delivered 2.08 MJ to the target and produced 8.6 MJ of fusion yield, a target gain greater than four [1]. ITER's public schedule identifies Start of Research Operation in 2034 and deuterium-tritium operation in 2039 [2]. The U.S. Department of Energy's Fusion Science and Technology Roadmap emphasizes

structural materials, plasma-facing components, confinement systems, fuel cycle, blankets, and plant engineering/integration as coordinated technology areas [3]. Wendelstein 7-X's 2025 long-pulse stellarator result is notable because it emphasizes sustained high triple product over tens of seconds rather than only short peak performance [4, 5].

These facts suggest that the barrier-crossing event is not sufficient by itself. A reactor must also handle charged-particle heating, helium ash, neutron damage, heat exhaust, plasma-facing components, tritium breeding and handling, and integrated plant control. In the language of this manuscript, these are product-shadow and nested-loop closure problems.

The Lawson criterion already expresses the need for fusion heating to exceed losses in a plasma through density, temperature, and confinement time [6, 7]. The extension proposed here is to treat fusion products not merely as output energy or passive losses, but as active suppressors of subsequent useful fusion when their residence time is too long.

## 2.2 Scaffolded nuclear concepts

Some nuclear concepts can be interpreted as attempts to alter the interface rather than brute-force the same barrier. Proton-boron-11 fusion is attractive because its primary channel is comparatively aneutronic, but recent analyses emphasize the high Coulomb barrier, approximately 200 keV-scale ion temperatures for appreciable reactivity, and bremsstrahlung limitations under many conditions [8, 9]. NASA describes lattice confinement fusion as deuterium confined within a metal lattice and triggered by energetic radiation, with the metal electrons and lattice environment participating in the reaction setting [10]. ARPA-E's low-energy nuclear reactions exploratory topic has framed that area cautiously, emphasizing hypothesis-driven experiments and evidence convincing to the wider scientific community [11, 12].

The theory here does not assume that speculative solid-state claims are correct. Instead, it asks a safer structural question: when a barrier is convex, can a scaffolded intermediate environment reduce the action enough to justify its cost? The Bridge-Fusion Law in Section 4 gives a simple answer under stated assumptions.

## 2.3 Biological membrane and cell fusion

Cell-cell and viral fusion are not simple coalescence events. Fusogens help overcome energetic barriers to membrane merger [13]. Recent experiments indicate that cell-surface crowding can add an energetic barrier on the order of  $100k_B T$  to membrane apposition and that reducing crowding can increase fusion efficiency [14]. Osteoclast fusion experiments show that decreased plasma-membrane tension is a mechanical prerequisite for fusion in that system; forced elevation of membrane-cortex attachment suppresses fusion [15]. Viral fusion studies continue to resolve how proteins bind lipids, reorganize membranes, and populate intermediate membrane-remodeling states [16, 17, 18].

These findings fit the scaffolded-incomplete view. Biological fusion typically requires regulated approach, local deformation, intermediate hemifusion or pore states, and then product control. Unregulated full merger can be pathological.

## 2.4 Biomolecular condensates

Biomolecular condensates formed by phase separation can fuse, exchange material, age, harden, or transition toward gel-like and aggregate states. Reviews increasingly treat condensates as dynamic material phases with functional and pathological consequences [19, 20, 21, 22]. In the present framework, condensate coalescence is a fusion process in which product shadow appears as aging, gelation, loss of reversibility, or sequestration of components.

## 2.5 Materials joining

Dissimilar materials joining provides a macroscopic analogue. Direct fusion welding or abrupt joining can generate residual stresses, brittle intermetallic compounds, thermal-expansion mismatch, and interfacial failure. Interlayers, diffusion bonding, and functionally graded transitions are used to reduce abrupt mismatch and suppress brittle reaction layers [23, 24, 25, 26]. This is almost exactly the Bridge-Fusion Law in engineering form: introduce intermediate states when direct mismatch is too costly.

## 2.6 Information and sensor fusion

Information fusion often assumes that combining modalities improves accuracy. In practice, multi-modal systems face alignment errors, feature-representation gaps, data quality issues, asynchrony, redundancy, conflicts, and computational bottlenecks [27, 28, 29]. A fusion layer that cannot refuse a bad modality can be worse than a single competent modality. Section 4.5 formalizes this as a no-harm fusion weight.

# 3 General interface-barrier-product formalism

**Definition 1** (Fusion system). *A fusion system is represented as*

$$\mathfrak{F} = (A, B, \mathcal{I}, \phi, \lambda, S, P, \Theta), \quad (2)$$

where  $A$  and  $B$  are the fusing entities,  $\mathcal{I}$  is the interface state,  $\phi$  is an alignment map,  $\lambda \in [0, 1]$  is the degree of merger,  $S$  is an optional scaffold or catalyst,  $P$  denotes products of fusion, and  $\Theta$  is the effective agitation scale: thermal energy, quantum action, mechanical drive, computational uncertainty, or another domain-specific activation scale.

**Definition 2** (Interface-barrier-product objective). *The proposed cross-domain objective is*

$$\boxed{\mathcal{J}(\phi, \lambda, S) = \mathcal{A}_{\lambda, S}(\phi) + \mathcal{M}_{\lambda}(\phi) + \mathcal{P}_{\lambda} - \mathcal{B}_{\lambda} + \mathcal{C}(S)}. \quad (3)$$

Here  $\mathcal{A}$  is barrier action,  $\mathcal{M}$  is mismatch penalty,  $\mathcal{P}$  is product-shadow penalty,  $\mathcal{B}$  is useful binding or function gained, and  $\mathcal{C}$  is scaffold/control cost. The proposed optimal fusion state is

$$(A \oplus B)^{\star} = \arg \min_{\phi, \lambda, S} \mathcal{J}(\phi, \lambda, S). \quad (4)$$

The main difference from ordinary merger language is that  $\lambda = 1$  is not assumed optimal. Similarly,  $S = \emptyset$  is not assumed optimal, and products  $P$  are not assumed passive.

Table 1: Cross-domain interpretation of the interface-barrier-product variables.

| Domain                            | Barrier or mismatch                                         | Scaffold or alignment map                                                             | Product shadow                                                           |
|-----------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Thermonuclear fusion              | Coulomb barrier, transport loss, confinement limits         | Magnetic geometry, compression symmetry, alpha heating profile, fueling control       | Helium ash, neutron damage, heat exhaust, tritium-cycle stress           |
| Lattice-assisted nuclear concepts | Nuclear barrier in condensed matter environment             | Metal lattice, electron screening, external beam trigger                              | Activation products, reproducibility failures, diagnostic ambiguity      |
| Cell and membrane fusion          | Hydration, charge, curvature, membrane tension, crowding    | Fusogens, actin protrusions, lipid reorganization, local tension control              | Genomic instability, pathological syncytia, loss of compartment identity |
| Condensates                       | Interfacial tension, phase compatibility, viscosity         | Multivalent motifs, RNA/protein concentration fields, post-translational modification | Aging, gelation, aggregation, sequestration                              |
| Materials joining                 | Thermal expansion mismatch, brittle phases, residual stress | Interlayers, graded composition, diffusion bonding, process path                      | Intermetallics, cracks, residual stress, heat-affected damage            |
| Information fusion                | Modality gap, time-space misalignment, correlation, bias    | Latent alignment, attention/gating, calibration, uncertainty modeling                 | Negative transfer, overconfidence, lost modality-specific information    |

## 4 Candidate laws and theorems

### 4.1 Bridge-Fusion Law

**Proposed law 1** (Bridge-Fusion Law). *Suppose direct fusion across mismatch distance  $D$  has convex barrier action*

$$\mathcal{A}_{\text{direct}} = D^p, \quad p > 1. \quad (5)$$

*If  $k-1$  intermediate scaffold states split the mismatch into  $k$  equal increments, and each intermediate has cost  $c$ , the staged barrier is*

$$\mathcal{A}_k(D, c) = \frac{D^p}{k^{p-1}} + (k-1)c. \quad (6)$$

*The continuous optimum is*

$$k^* = \left( \frac{(p-1)D^p}{c} \right)^{1/p}. \quad (7)$$

*A two-step bridge is beneficial when*

$$c < D^p \left( 1 - 2^{1-p} \right). \quad (8)$$

*Proof.* For equal splitting, each of  $k$  increments has barrier  $(D/k)^p$ , so the total barrier contribution is  $k(D/k)^p = D^p/k^{p-1}$ . Adding  $k - 1$  scaffold costs gives Eq. (6). Treating  $k$  as continuous,

$$\frac{\partial \mathcal{A}_k}{\partial k} = -(p-1)D^p k^{-p} + c. \quad (9)$$

Setting this to zero gives Eq. (7). The two-step condition follows from  $D^p/2^{p-1} + c < D^p$ .  $\square$

**Interpretation.** When barriers are convex, intermediate states are not decoration; they can be exponentially valuable when rates scale as  $R \propto \exp(-\mathcal{A}/\Theta)$ . This suggests a common design rule: do not only increase the drive  $\Theta$ ; search for scaffolds that reduce  $\mathcal{A}$ .

## 4.2 Optimal Incomplete Fusion Theorem

**Theorem 1** (Optimal Incomplete Fusion Theorem). *Let  $\lambda \in [0, 1]$  measure degree of fusion. Suppose useful benefit is initially linear,*

$$B(\lambda) = b\lambda, \quad (10)$$

*but mismatch, loss of individuality, control cost, or product penalty is convex,*

$$C(\lambda) = a\lambda^p, \quad p > 1. \quad (11)$$

*Then the maximizer of*

$$V(\lambda) = b\lambda - a\lambda^p \quad (12)$$

*on  $[0, 1]$  is*

$$\lambda^* = \min \left[ 1, \left( \frac{b}{ap} \right)^{1/(p-1)} \right]. \quad (13)$$

*Full fusion is optimal only when  $b \geq ap$ .*

*Proof.* For an interior optimum,  $V'(\lambda) = b - ap\lambda^{p-1} = 0$ , giving  $\lambda = (b/ap)^{1/(p-1)}$ . Since  $V''(\lambda) = -ap(p-1)\lambda^{p-2} < 0$ , this interior point is a maximum. Projection to  $[0, 1]$  gives Eq. (13). Full fusion occurs when the unconstrained maximizer is at least one, equivalently  $b \geq ap$ .  $\square$

**Interpretation.** A system can be better fused by remaining partially separated: gated multimodal systems, compartmentalized cells, semipermeable organelles, graded material interfaces, ash exhaust channels, and modular reactor subsystems are all examples of incomplete fusion as a design feature.

## 4.3 Product-Shadow Saturation Law

**Proposed law 2** (Product-Shadow Saturation Law). *Let  $R_0$  be the raw fusion rate in the absence of product inhibition. Suppose product concentration is approximately proportional to useful rate times product residence time,*

$$P \approx \tau_P R, \quad (14)$$

*and suppose products suppress useful fusion by  $\exp(-\alpha P)$ . The self-consistent useful rate satisfies*

$$R = R_0 \exp(-\alpha \tau_P R). \quad (15)$$

Therefore

$$R = \frac{W(\alpha\tau_P R_0)}{\alpha\tau_P}, \quad (16)$$

where  $W$  is the Lambert  $W$  function. For large  $R_0$ ,

$$R \sim \frac{\log(\alpha\tau_P R_0)}{\alpha\tau_P}. \quad (17)$$

*Proof.* Multiplying Eq. (15) by  $\exp(\alpha\tau_P R)$  gives  $R \exp(\alpha\tau_P R) = R_0$ . Multiplication by  $\alpha\tau_P$  gives  $(\alpha\tau_P R) \exp(\alpha\tau_P R) = \alpha\tau_P R_0$ . Applying  $W$  yields Eq. (16). The large-argument expansion of  $W(x)$  gives logarithmic growth.  $\square$

**Interpretation.** If product shadow is strong, brute-force increases in raw fusion rate generate only logarithmic useful gains. In nuclear fusion this points toward ash, heat, neutron, and materials exhaust. In biological fusion it points toward loss of compartment identity or pathological syncytia. In information fusion it points toward overconfidence, bias accumulation, and loss of modality-specific information.

#### 4.4 Product-Shadow Lawson Correction

A dimensionless Lawson-like viability factor can be corrected by product residence:

$$\Lambda_{\text{eff}} = \Lambda_0 \exp[-\alpha R(T) \tau_P]. \quad (18)$$

Here  $\Lambda_0$  is the primary fusion-gain measure,  $R(T)$  is a product-generation rate,  $\tau_P$  is product residence time, and  $\alpha$  is shadow strength. This correction is proposed as a design heuristic rather than a replacement for detailed transport, plasma-wall, or nuclear-reaction modeling.

#### 4.5 No-Harm Fusion Weight

**Theorem 2** (No-Harm Fusion Weight). *Let two estimators have errors  $\epsilon_1$  and  $\epsilon_2$ , and define*

$$E = \begin{pmatrix} e_{11} & e_{12} \\ e_{12} & e_{22} \end{pmatrix}, \quad e_{ij} = \mathbb{E}[\epsilon_i \epsilon_j]. \quad (19)$$

*For fused estimate  $\hat{x} = w\hat{x}_1 + (1-w)\hat{x}_2$ , the mean-square error is*

$$M(w) = w^2 e_{11} + (1-w)^2 e_{22} + 2w(1-w)e_{12}. \quad (20)$$

*The clipped optimum*

$$w^* = \text{clip}\left(\frac{e_{22} - e_{12}}{e_{11} + e_{22} - 2e_{12}}, 0, 1\right) \quad (21)$$

*satisfies*

$$M(w^*) \leq \min(e_{11}, e_{22}). \quad (22)$$

*Proof.* The quadratic  $M(w)$  has unconstrained minimizer Eq. (21). The feasible interval includes the endpoints  $w = 0$  and  $w = 1$ , which correspond to using estimator 2 or estimator 1 alone. Therefore the constrained minimizer cannot be worse than the better endpoint.  $\square$

**Interpretation.** A fusion operator should include the option not to fuse. The same principle appears in physical language as valves, divertors, membranes, graded boundaries, immune checkpoints, and modular interfaces.

## 4.6 Unified useful-fusion rate ansatz

Combining the preceding assumptions yields the following candidate rate equation:

$$R_u = \rho_A \rho_B \nu \exp \left[ -\frac{\mathcal{A}_0/k^{p-1} + (k-1)c - \Gamma_S}{\Theta_{\text{eff}}} \right] \lambda^\eta \exp[-\alpha R_u \tau_P]. \quad (23)$$

The symbols are density factors  $\rho_A, \rho_B$ , attempt frequency  $\nu$ , direct barrier action  $\mathcal{A}_0$ , scaffold count  $k$ , barrier convexity  $p$ , per-scaffold cost  $c$ , scaffold control benefit  $\Gamma_S$ , effective activation scale  $\Theta_{\text{eff}}$ , fusion degree  $\lambda$ , degree-yield exponent  $\eta$ , product-shadow strength  $\alpha$ , and product residence time  $\tau_P$ . Equation (23) is implicit in  $R_u$ . If the prefactor excluding the last exponential is called  $R_0$ , then Eq. (16) gives

$$R_u = \frac{W(\alpha \tau_P R_0)}{\alpha \tau_P}. \quad (24)$$

This equation is not an established law. It is a compact hypothesis: the best design variables in many fusion systems are not only energy, pressure, or accuracy, but the triad  $(k, \lambda, \tau_P)$ .

## 5 Virtual environment and tests

Four dimensionless computational experiments were used as sanity checks. They are not reactor simulations, cell-biophysics simulations, or material-process simulations. Their purpose is to test whether the proposed principles behave plausibly in simple mathematical worlds.

### 5.1 Test 1: information fusion with and without a refusal gate

Two estimators were sampled with random noise scales, biases, and cross-correlation. Equal-weight fusion was compared with the no-harm weight in Eq. (21). The baseline was the better individual estimator in each sampled case.

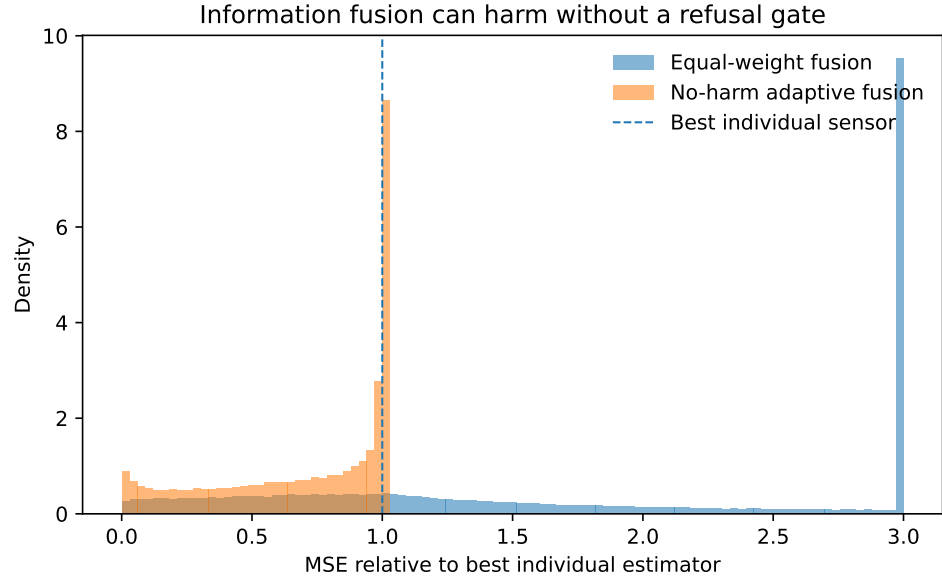


Figure 1: Forced information fusion can be worse than the best input. In this toy test, equal-weight fusion was worse than the better individual estimator in 64.2% of cases. The no-harm adaptive weight was never worse than the better endpoint by construction and had median relative MSE 0.819.

## 5.2 Test 2: bridge-mediated barrier reduction

Random mismatch distances  $D$ , convexity exponents  $p > 1$ , and scaffold costs  $c$  were sampled. The optimum over integer  $k \in \{1, \dots, 60\}$  was computed from Eq. (6).

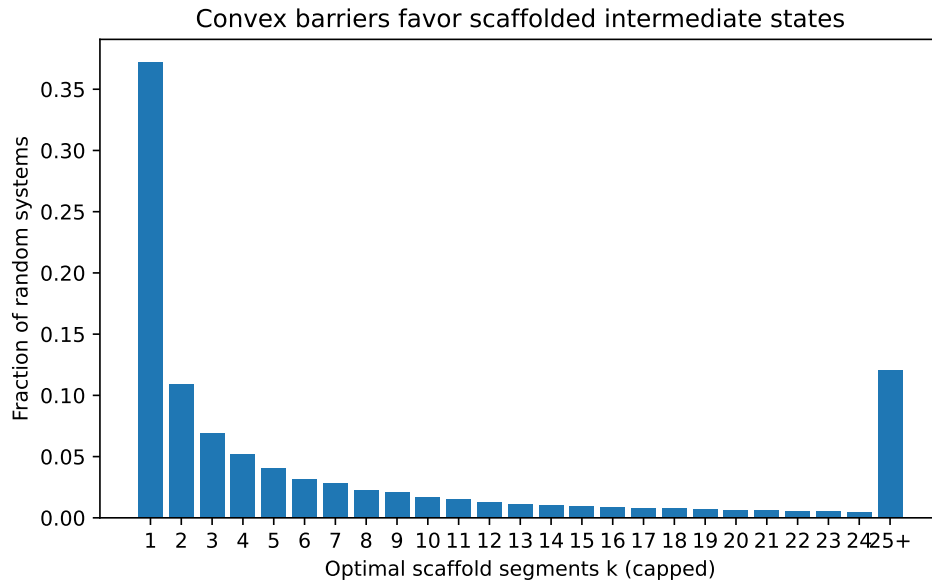


Figure 2: Optimal scaffold count in the bridge-fusion toy search. A bridge was preferred in 62.9% of random cases. The median optimal scaffold count was 3, and the median barrier reduction among bridge-preferred cases was 81.9%.

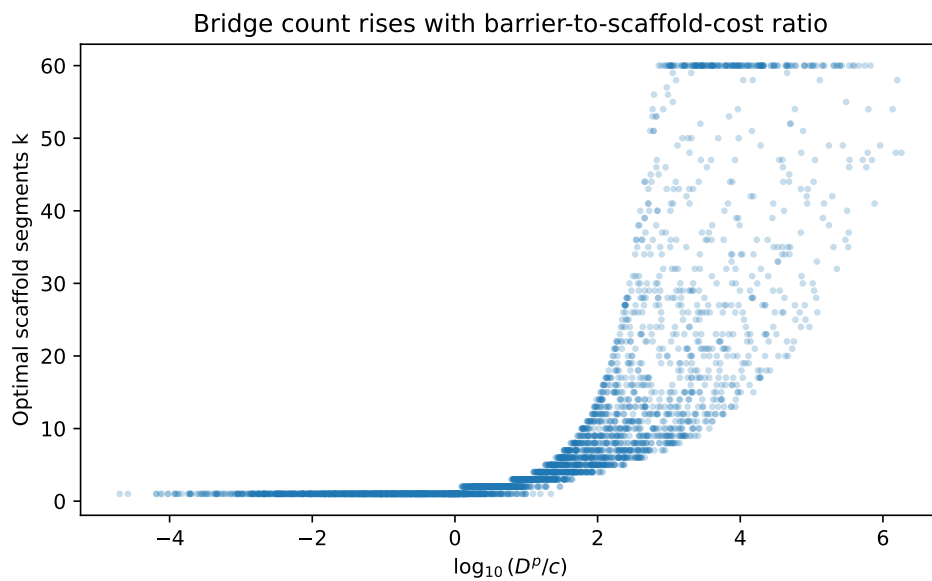


Figure 3: The optimal number of scaffold segments rises with the barrier-to-scaffold-cost ratio  $D^p/c$ , consistent with Eq. (7).

### 5.3 Test 3: incomplete fusion

Random  $a, b, p$  were sampled in  $V(\lambda) = b\lambda - a\lambda^p$ . The optimum  $\lambda^*$  was computed using Eq. (13).

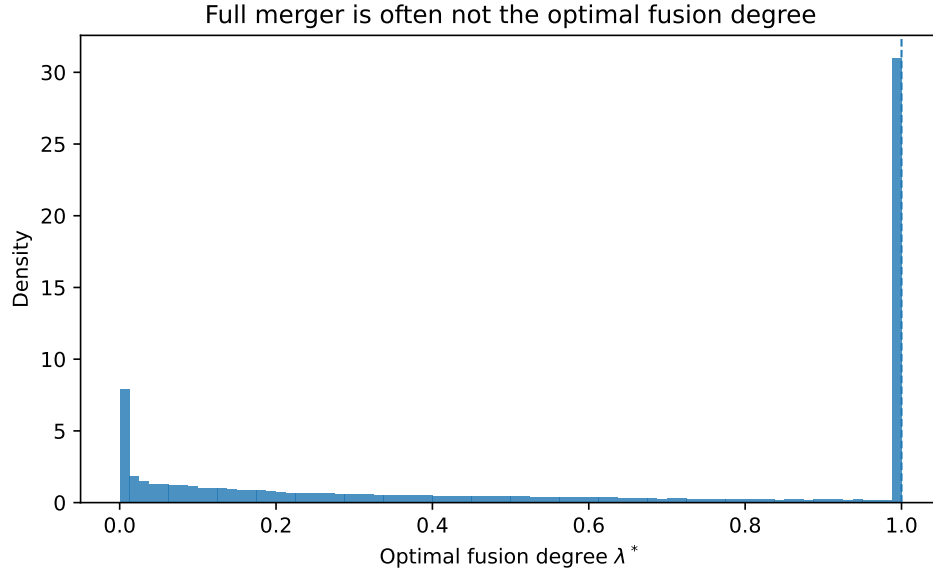


Figure 4: Distribution of optimal fusion degree in the toy incomplete-fusion search. Partial fusion was optimal in 61.8% of sampled cases, and full fusion was optimal in 38.2%. Median  $\lambda^*$  was 0.562.

#### 5.4 Test 4: product-shadow threshold

A dimensionless equilibrium model was used:

$$r(T) = T^2 \exp(-1/T), \quad A = \tau_P r(T), \quad (25)$$

$$G r(T) \exp[-\alpha A] = T + \beta \sqrt{T}. \quad (26)$$

The minimum raw gain required for sustainment is therefore

$$G_{\text{th}}(\tau_P) = \min_T \frac{T + \beta \sqrt{T}}{r(T) \exp[-\alpha \tau_P r(T)]}. \quad (27)$$

This is a toy analogue of a fusion system in which products suppress further useful reactions.

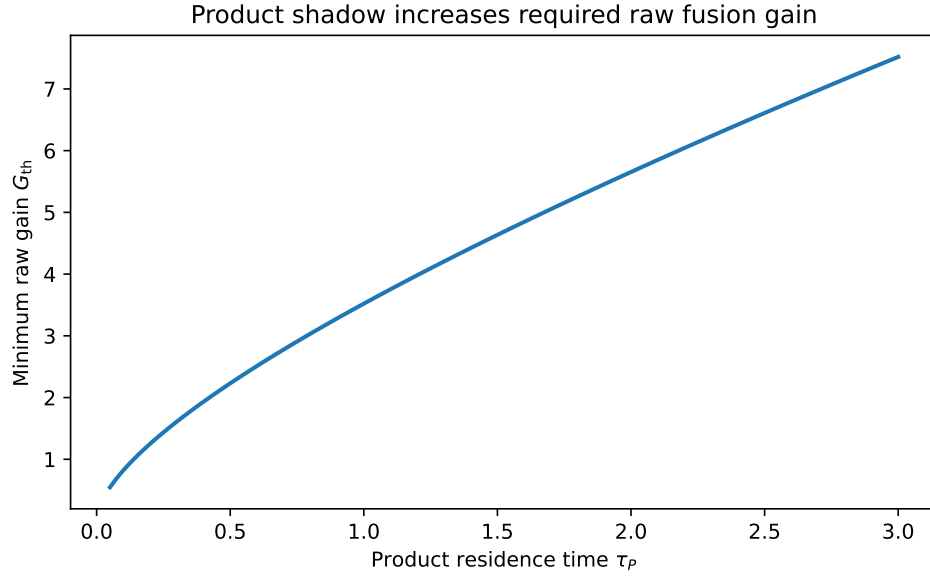


Figure 5: Product residence time increases the raw gain required for sustainment in the toy threshold model. With  $\alpha = 0.65$  and  $\beta = 0.35$ , the threshold gain increased from approximately 0.82 at  $\tau_P = 0.1$  to 5.65 at  $\tau_P = 2.0$ .

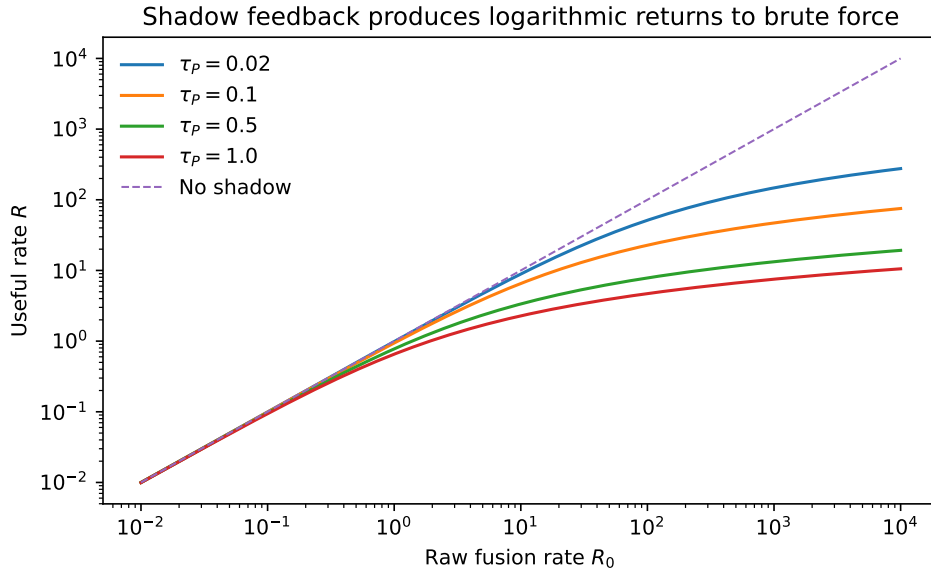


Figure 6: The implicit product-shadow equation  $R = R_0 \exp(-\alpha \tau_P R)$  creates diminishing returns to raw rate. At high product residence time, useful rate grows only logarithmically with raw rate.

Table 2: Summary of virtual tests.

| Test                   | Main numerical result                                                                                                                   | Interpretation                                       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Information fusion     | Equal-weight fusion worse than best input in 64.2% of cases; no-harm fusion worse in 0% by construction; no-harm median MSE ratio 0.819 | Fusion needs a refusal or gating mechanism.          |
| Bridge-mediated fusion | Bridge preferred in 62.9% of cases; median $k^* = 3$ ; median barrier reduction among bridge cases 81.9%                                | Convex barriers reward intermediate scaffold states. |
| Incomplete fusion      | Partial $\lambda^*$ optimal in 61.8% of cases; median $\lambda^* = 0.562$                                                               | Full merger should not be assumed optimal.           |
| Product shadow         | Threshold gain rose from 0.82 at $\tau_P = 0.1$ to 5.65 at $\tau_P = 2.0$                                                               | Product export can dominate primary gain.            |

## 6 Research Sprint 2: Extended candidate laws and virtual experiments

The first virtual environment suggested that useful fusion is controlled by scaffold count, fusion degree, and product residence. A second virtual environment was constructed to search for additional blindspots. Four new structures emerged: finite scaffold lifetime, selective confinement, critical network fusion, and complementarity-distance optimization. These are presented as candidate laws and falsifiable modeling targets.

### 6.1 Candidate law 5: Turnover-Matched Scaffold Law

A scaffold can lower an interface barrier only after it has had time to assemble or align the fusing systems. Yet the same scaffold can become harmful if it persists too long, because it may trap products, immobilize degrees of freedom, induce aggregation, or create overbinding. This is a common pattern in catalysts, fusogens, muon-like mediators, graded interlayers, biomolecular condensate organizers, and attention gates.

Let the log useful rate produced by a scaffold with lifetime  $\tau_S$  be

$$\log \frac{R(\tau_S)}{R(0)} = \eta \left( 1 - e^{-\tau_S/\tau_A} \right) - \gamma \frac{\tau_S}{\tau_P}. \quad (28)$$

Here  $\eta$  is maximum barrier-reduction strength,  $\tau_A$  is scaffold assembly or alignment time,  $\gamma$  is persistence-shadow strength, and  $\tau_P$  is the relevant product-clearance time.

**Proposed law 3** (Turnover-Matched Scaffold Law). *For Eq. (28), the optimal scaffold lifetime is*

$$\tau_S^* = \begin{cases} \tau_A \log \left( \frac{\eta \tau_P}{\gamma \tau_A} \right), & \eta \tau_P > \gamma \tau_A, \\ 0, & \eta \tau_P \leq \gamma \tau_A. \end{cases} \quad (29)$$

*Proof.* Differentiating Eq. (28) gives

$$\frac{d}{d\tau_S} \log R = \frac{\eta}{\tau_A} e^{-\tau_S/\tau_A} - \frac{\gamma}{\tau_P}. \quad (30)$$

An interior optimum exists when the derivative is positive at zero, i.e.  $\eta/\tau_A > \gamma/\tau_P$ . Solving for the stationary point gives Eq. (29); otherwise the optimum on  $\tau_S \geq 0$  is 0.  $\square$

**Discovery from the virtual environment.** In 200,000 random cases, a positive finite scaffold lifetime was optimal in 76.443% of cases. Among those cases, the median optimum was  $\tau_S^* = 3.032\tau_A$ . The median useful-rate gain over no scaffold was 1.794-fold, while the 90th percentile log gain was 6.271, or roughly 530-fold. The exceptional implication is that scaffold design should not seek maximum binding strength alone; it should seek a matched lifetime.

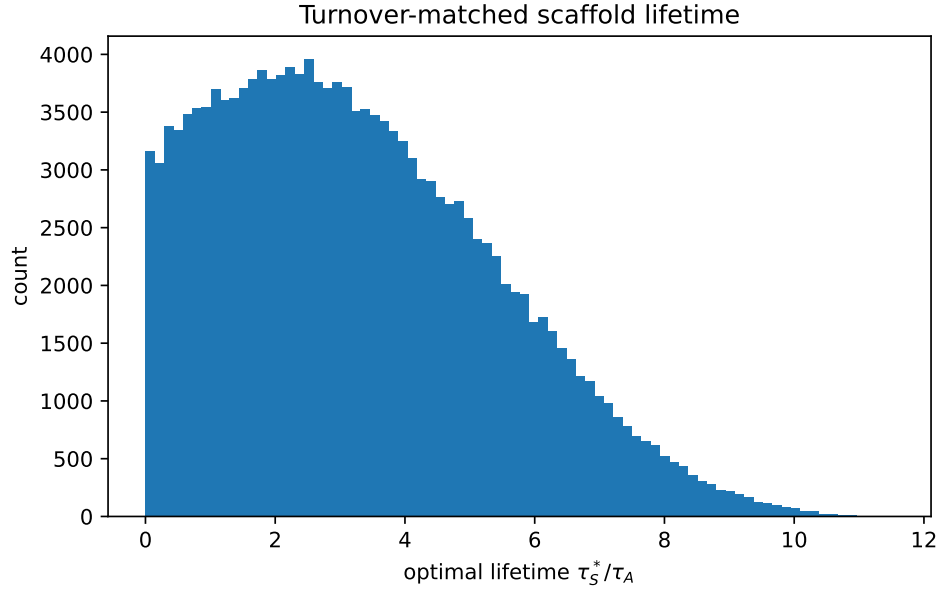


Figure 7: Virtual test of the Turnover-Matched Scaffold Law. When a scaffold has saturating alignment benefit and linear persistence shadow, the optimal lifetime is usually finite rather than permanent.

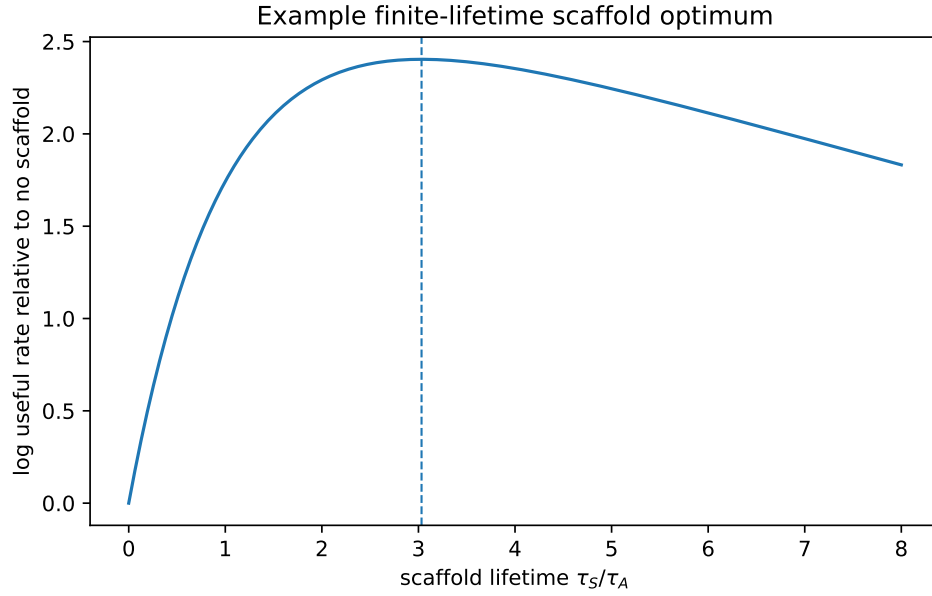


Figure 8: Representative finite-lifetime scaffold optimum. The scaffold must persist long enough to align the interface, but beyond the optimum its residue cost dominates.

## 6.2 Candidate law 6: Selective Confinement Split Law

Fusion systems often need to confine one quantity while deconfining another. A plasma reactor seeks long energy confinement but rapid ash and heat exhaust. A cell seeks local membrane merger without permanent loss of compartment identity. A condensate may need local concentration without aging into a gel. A multimodal model may need shared semantics without retaining correlated noise. Modern fusion pilot-plant studies likewise treat power and particle exhaust as integral to compatibility between the core plasma and the supporting systems, not as an afterthought [30].

Consider a dimensionless activation coordinate  $T$  and product-shadow coordinate  $A$ :

$$\frac{dT}{dt} = Gr(T)e^{-\alpha A} - \frac{T}{\tau_E} - \beta\sqrt{T}, \quad (31)$$

$$\frac{dA}{dt} = \chi r(T) - \frac{A}{\tau_P}, \quad (32)$$

$$r(T) = T^2 e^{-1/T}. \quad (33)$$

At steady state,  $A = \chi\tau_P r(T)$ . Thus the gain required to maintain activation  $T$  is

$$G_{\text{req}}(T; \tau_E, \tau_P) = \frac{T/\tau_E + \beta\sqrt{T}}{r(T) \exp[-\alpha\chi\tau_P r(T)]}. \quad (34)$$

The viability threshold is

$$G_{\text{th}}(\tau_E, \tau_P) = \min_T G_{\text{req}}(T; \tau_E, \tau_P). \quad (35)$$

Equivalently, a design with gain  $G$  is viable in this toy environment when

$$\max_T \left[ \frac{Gr(T)}{T/\tau_E + \beta\sqrt{T}} \exp[-\alpha\chi\tau_P r(T)] \right] \geq 1. \quad (36)$$

**Proposed law 4** (Selective Confinement Split Law). *When useful activation and harmful products have different residence times, the viability criterion is not a single confinement time. It is a split condition: increase the residence time of useful activation variables  $\tau_E$ , decrease the residence time of product-shadow variables  $\tau_P$ , and optimize their ratio under Eq. (36).*

**Discovery from the virtual environment.** On a  $120 \times 120$  grid over  $\tau_E, \tau_P \in [0.05, 20]$ , 57.590% of points were viable at  $G = 8$ . At fixed  $\tau_E = 0.5$ , increasing product residence from  $\tau_P = 0.5$  to  $\tau_P = 5$  raised the threshold from  $G_{\text{req}} \approx 4.97$  to 23.00. At  $\tau_P = 10$ , the minimum  $\tau_E$  required for viability at  $G = 8$  was approximately 4.884. This supports the idea that a lower raw-gain design with fast product exhaust can beat a higher raw-gain design that traps its own products.

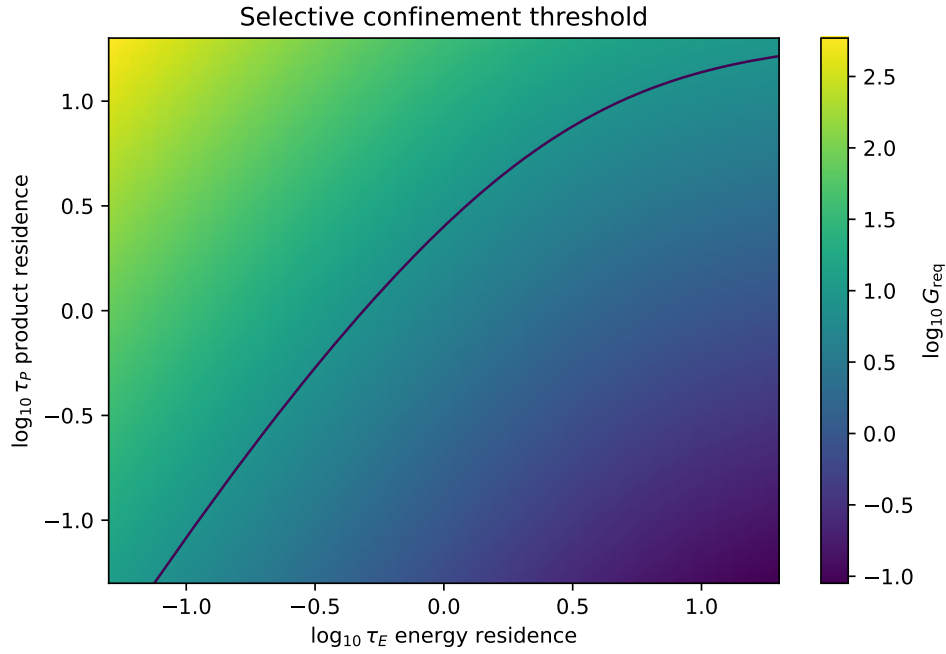


Figure 9: Selective confinement threshold. The contour marks  $G_{\text{th}} = 8$ . Viability improves toward long energy residence  $\tau_E$  and short product residence  $\tau_P$ .

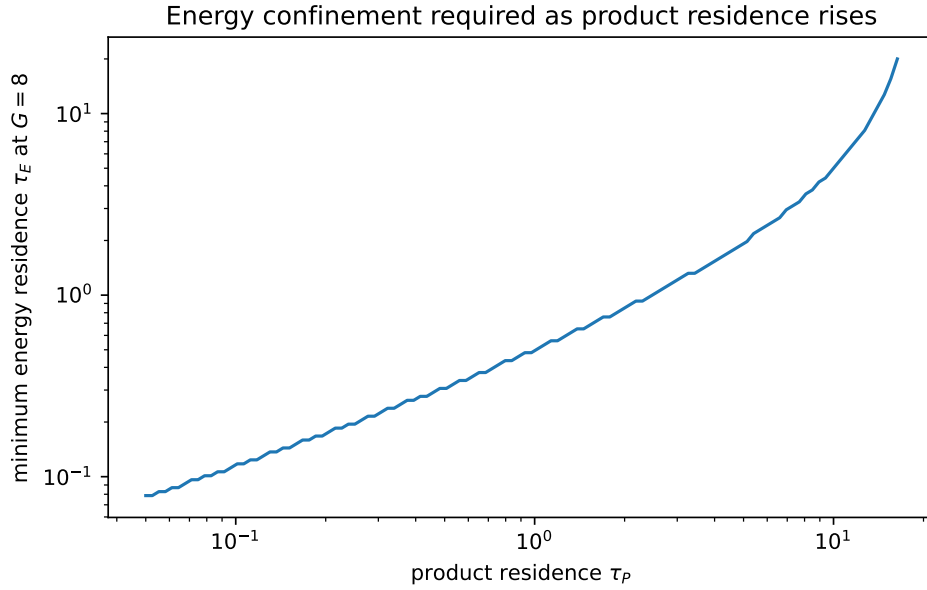


Figure 10: Minimum energy residence needed for viability at  $G = 8$  as product residence increases. The curve gives a reduced-order exhaust requirement.

### 6.3 Candidate law 7: Critical Fusion Percolation Principle

Fusion can occur not only between two objects but across a network: cells form syncytia, droplets coalesce, grains sinter, organizations merge, sensors couple, or models share latent state. In networked systems, local benefits can saturate before global merger becomes beneficial. Global merger then appears as a product shadow: a giant component that destroys diversity, modularity, independent error modes, or controlled compartment boundaries.

Let  $c$  denote the mean fusion degree of an Erdős-Rényi-like network [34]. Let  $S(c)$  be the giant-component fraction:

$$S(c) = 0 \quad (c \leq 1), \quad S(c) = 1 - e^{-cS(c)} \quad (c > 1). \quad (37)$$

A minimal value function is

$$V(c) = b(1 - e^{-c}) - aS(c)^q - dc. \quad (38)$$

The first term captures saturating local sharing. The second captures superlinear giant-component shadow. The third captures link maintenance cost.

**Conjecture 1** (Critical Fusion Percolation Principle). *If local fusion benefit saturates and global merger shadow turns on near a percolation transition, optimal network fusion will concentrate near the critical mean degree  $c \approx 1$ , rather than at maximal connectivity.*

**Discovery from the virtual environment.** In 100,000 random instances of Eq. (38), the median optimum was  $c^* = 1.05$ . The optimum lay in  $[0.8, 1.2]$  in 80.984% of cases, while only 7.759% had  $c^* > 2$ . This suggests that some exceptional fusion systems may live near the edge of a percolation transition: enough contact to share, not enough to collapse into one unstructured component.

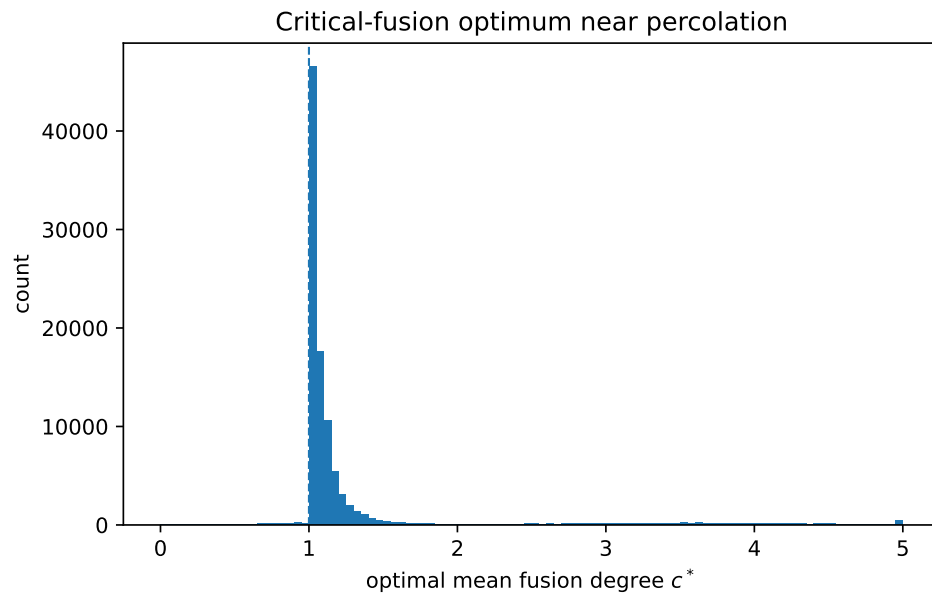


Figure 11: Distribution of optimal mean fusion degree in the network-fusion toy environment. The dashed line is the percolation threshold  $c = 1$ .

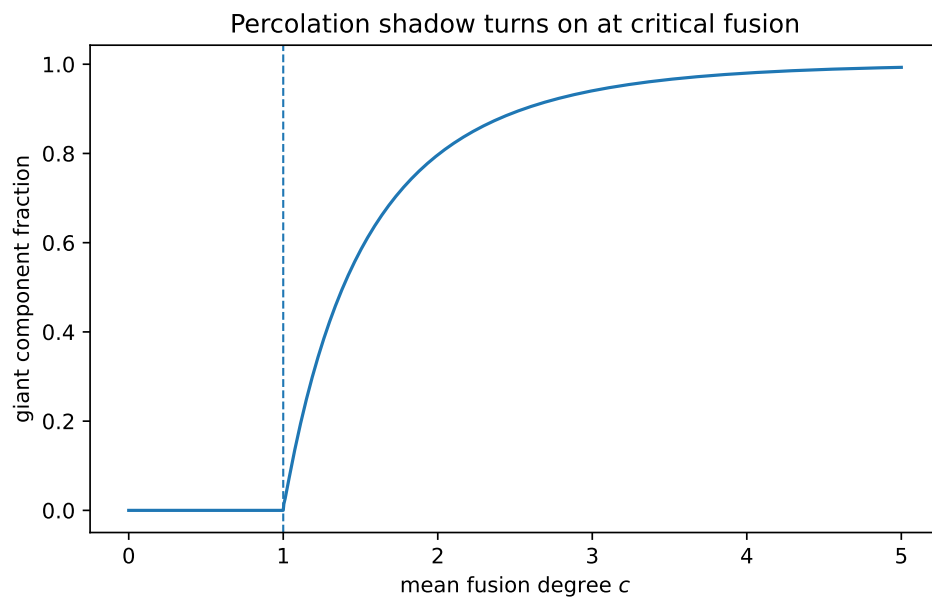


Figure 12: Giant-component fraction  $S(c)$ . In the proposed network-fusion interpretation,  $S(c)$  is a product-shadow variable when global merger suppresses diversity or controllability.

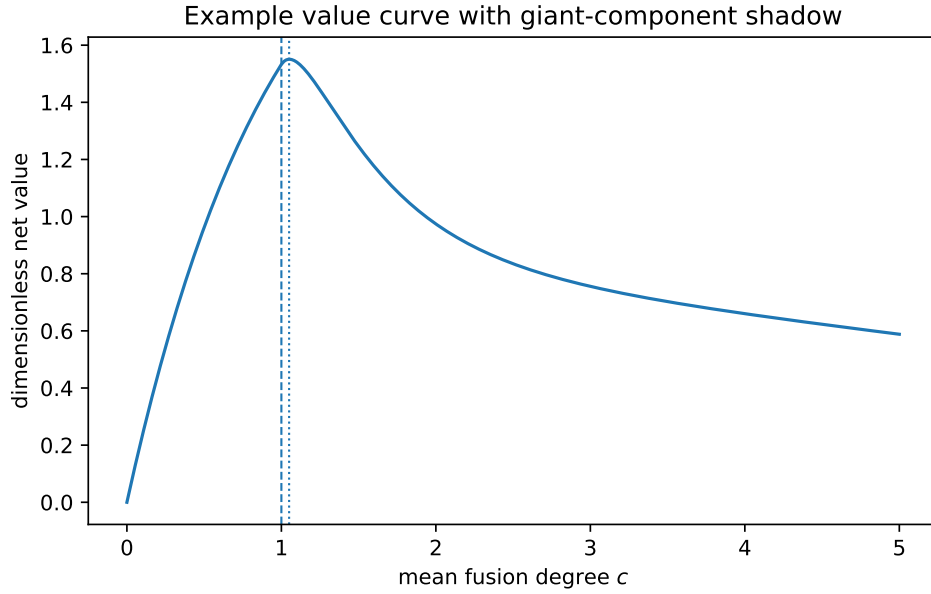


Figure 13: Representative value curve for Eq. (38). Local sharing increases before the percolation shadow dominates.

#### 6.4 Candidate law 8: Exceptional Fusion Ridge

The first bridge law asks how to fuse across mismatch. A different question is which systems should be fused at all. Nearly identical systems have low barrier but little complementarity. Extremely different systems may have high novelty but low alignment probability. This suggests an interior ridge of exceptional fusion.

Let  $d \in [0, 1]$  be normalized difference. Let complementarity scale as  $d^m$ , and let alignment survival decrease as  $\exp(-\mu d^p)$ . Then

$$R_{\text{ex}}(d) = R_0 d^m \exp(-\mu d^p), \quad m > 0, \mu > 0, p > 0. \quad (39)$$

**Proposed law 5** (Exceptional Fusion Ridge). *For Eq. (39), the optimal normalized difference is*

$$d^* = \min \left[ 1, \left( \frac{m}{\mu p} \right)^{1/p} \right]. \quad (40)$$

*Thus the most exceptional fusion is generally neither the most similar nor the most different pairing, but an intermediate complementarity-distance match.*

*Proof.* Taking logs gives  $\log R_{\text{ex}} = \log R_0 + m \log d - \mu d^p$ . Differentiating and setting the derivative to zero gives  $m/d - \mu p d^{p-1} = 0$ , hence  $d^p = m/(\mu p)$ . Clipping to  $[0, 1]$  gives Eq. (40).  $\square$

**Discovery from the virtual environment.** In 200,000 random parameter draws, an interior optimum  $d^* < 1$  occurred in 84.065% of cases. The median optimum was  $d^* = 0.569$ , and 66.930% of optima fell between 0.2 and 0.8. Optimizing over  $d$  gave a median 48.078-fold gain over a

near-identical comparison point  $d = 0.05$ , and a median 3.649-fold gain over the maximum-difference endpoint  $d = 1$ . The target for future study is to measure complementarity distance directly in materials, biology, and multimodal models.

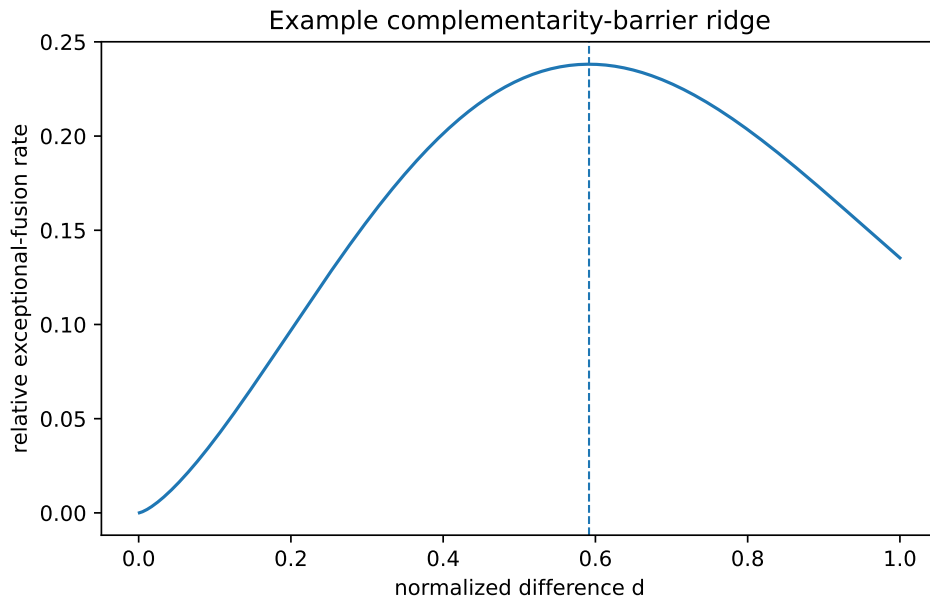


Figure 14: Representative exceptional-fusion ridge. Complementarity rises with difference, but alignment survival falls with difference, producing an interior optimum.

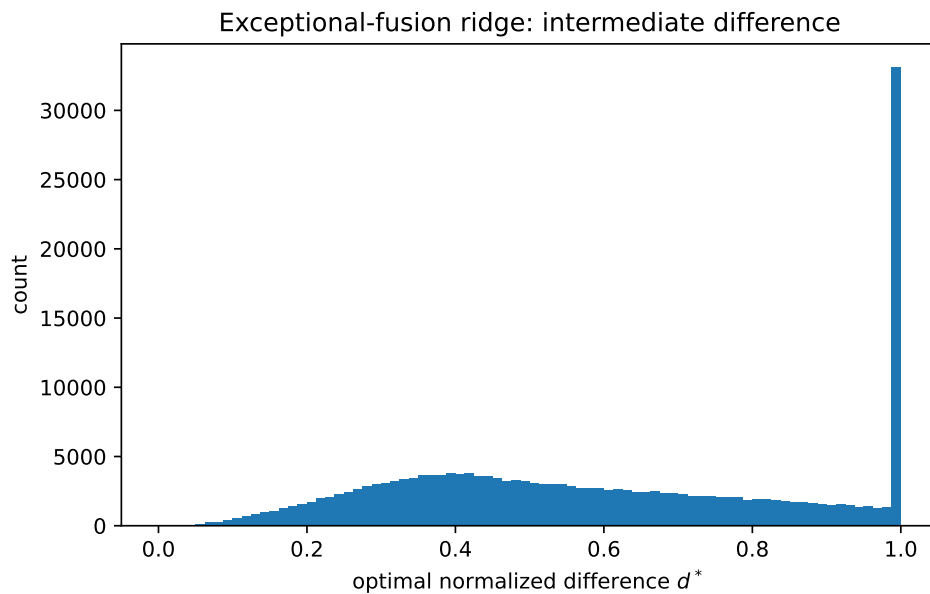


Figure 15: Distribution of optimal normalized difference  $d^*$  in the exceptional-fusion ridge toy environment.

## 6.5 Summary of second virtual tests

Table 3: Second virtual-environment results and proposed discoveries.

| Test                  | Main numerical result                                                                          | Proposed discovery                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Scaffold lifetime     | Positive finite $\tau_S^*$ in 76.443% of cases; median $\tau_S^*/\tau_A = 3.032$ when positive | Scaffold turnover can be more important than scaffold strength.                       |
| Selective confinement | At $\tau_E = 0.5$ , raising $\tau_P$ from 0.5 to 5 raised $G_{\text{req}}$ from 4.97 to 23.00  | Useful variables and product-shadow variables require opposite residence-time design. |
| Network fusion        | Median $c^* = 1.05$ ; 80.984% of optima in $[0.8, 1.2]$                                        | Network fusion may be optimal near percolation, not at maximal connectivity.          |
| Exceptional ridge     | Interior $d^* < 1$ in 84.065%; median $d^* = 0.569$                                            | Most valuable fusion pairings may occur at intermediate complementarity distance.     |

## 6.6 Targets created by the second sprint

The second sprint creates five concrete measurement targets:

1. Measure scaffold alignment time  $\tau_A$ , scaffold lifetime  $\tau_S$ , and product residence  $\tau_P$  separately in at least one biological, material, and computational fusion system.
2. Define a split confinement index, such as  $\tau_E/\tau_P$ , and test whether it predicts viability better than energy confinement alone in reduced fusion-reactor models.
3. Search for near-critical network-fusion regimes in condensate coalescence, cell syncytia, sintering networks, and model ensembles.
4. Build empirical complementarity-distance metrics for dissimilar materials, multimodal data streams, and biological fusogen targets.
5. Test whether forcing  $d \rightarrow 0$ ,  $d \rightarrow 1$ ,  $c \rightarrow \infty$ , or  $\tau_S \rightarrow \infty$  reliably degrades usefulness in systems with measurable product shadow.

## 7 Third virtual sprint: rhythmic, heterogeneous, and diversity-limited fusion

The first two research sprints emphasized scaffold count, scaffold lifetime, incomplete merger, product residence, network connectivity, and complementarity distance. The third sprint tested three additional blindspots. First, continuous drive may be suboptimal when activation is convex and products poison the interface. Second, bridges should not be spaced uniformly when the mismatch landscape is heterogeneous. Third, adding more information sources is not equivalent to adding independent information; correlated sources collapse into a smaller effective diversity.

The pulsed-fusion question is not only abstract. In inertial fusion energy, the scientific burn event is brief, while a power plant would require repeated target delivery, tracking, laser engagement,

chamber recovery, and materials survival; a SLAC discussion frames one practical challenge as moving from NIF-scale daily shots toward order-10 Hz operation [35]. Whole-device modeling efforts similarly emphasize that viable fusion design must couple core plasma, boundary plasma, wall and divertor heat flux, blanket, tritium, thermal, neutronics, and engineering constraints [36]. These are rhythm and recovery problems, not just peak-gain problems.

### 7.1 Candidate law 9: Pulse-Shadow Duty Law

Let a fusion-like system receive fixed average drive  $U$ . If the drive is delivered in pulses with duty fraction  $\delta \in (0, 1]$ , the on-phase drive is  $U/\delta$ . Suppose pre-shadow activation is convex,

$$R_0(\delta) = \delta \left( \frac{U}{\delta} \right)^q = U^q \delta^{1-q}, \quad q > 1, \quad (41)$$

and suppose products suppress useful output with strength  $K$ . A minimal equal-average-drive useful rate is

$$Y(\delta) = U^q \delta^{1-q} \exp \left[ -KU^q \delta^{1-q} \right], \quad \delta \in [\delta_{\min}, 1]. \quad (42)$$

The lower bound  $\delta_{\min}$  represents a peak-drive, damage, safety, synchronization, or actuator limit.

**Proposed law 6** (Pulse-Shadow Duty Law). *For Eq. (42), define  $x = \delta^{1-q}$ . The optimal duty fraction is*

$$\delta^* = \left[ \text{clip} \left( \frac{1}{KU^q}, 1, \delta_{\min}^{1-q} \right) \right]^{1/(1-q)}. \quad (43)$$

*When the optimum is interior, it satisfies*

$$KU^q (\delta^*)^{1-q} = 1. \quad (44)$$

*Thus the optimal pulse does not maximize raw activation. It tunes the product-shadow load to order unity.*

*Proof.* With  $x = \delta^{1-q}$ , Eq. (42) is  $Y = U^q x \exp(-KU^q x)$ . Differentiating with respect to  $x$  gives  $dY/dx = U^q \exp(-KU^q x)(1 - KU^q x)$ . The unconstrained optimum is therefore  $x^* = 1/(KU^q)$ . Projection onto the feasible interval  $[1, \delta_{\min}^{1-q}]$  gives Eq. (43).  $\square$

**Discovery from the virtual environment.** In 150,000 random equal-average-drive cases, pulsed operation was preferred in 74.521% of cases and strong pulsing ( $\delta^* < 0.5$ ) in 56.529%. The median optimal duty fraction was  $\delta^* = 0.396$ . The median useful-rate gain over continuous drive was 2.255-fold; the 90th percentile gain was 91.230-fold. The median optimized shadow load was exactly near one, consistent with Eq. (44). Large gains occur only in the toy setting where peak-drive limits are generous; in physical systems, actuator damage, hydrodynamic instability, target injection, heat cycling, and fatigue would reduce the feasible gain.

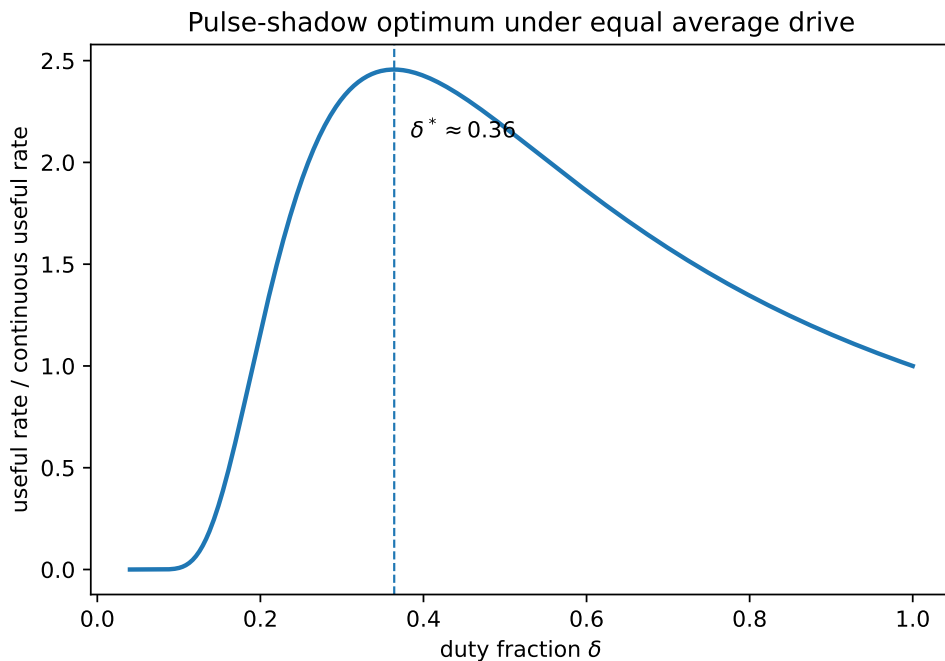


Figure 16: Representative pulse-shadow optimum under equal average drive. Pulsing initially helps because activation is convex; excessive pulsing hurts because product shadow grows faster than useful output.

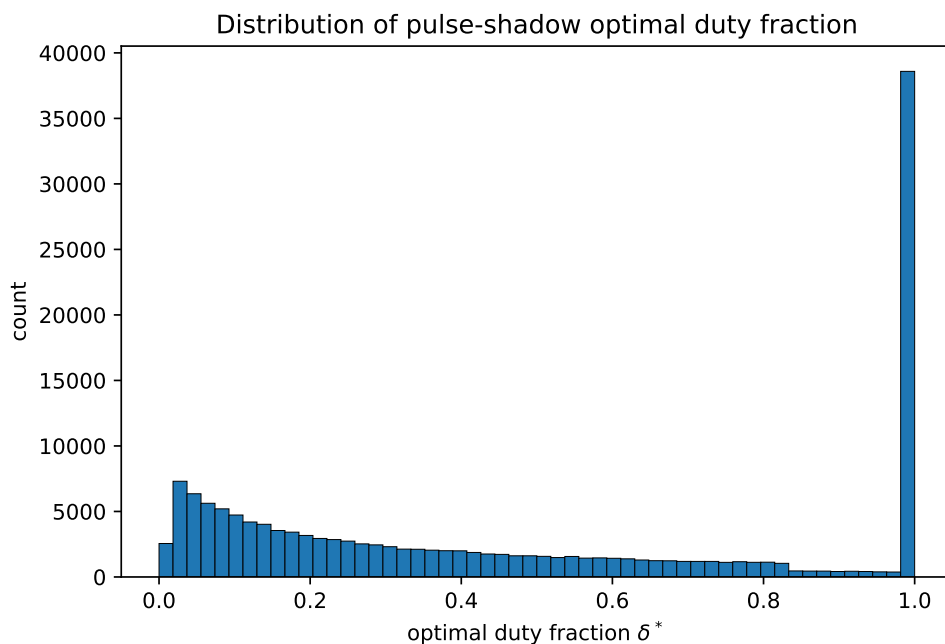


Figure 17: Distribution of optimal duty fraction in the pulse-shadow Monte Carlo environment. Continuous drive is not the default optimum when activation is convex and product shadow is strong.

## 7.2 Candidate law 10: Recovery-Window Resonance

Duty fraction alone is incomplete. A pulsed fusion system also has a period  $T$ , product residence time  $\tau_P$ , alignment build time  $\tau_A$ , and alignment memory time  $\tau_M$ . Let the on-time be  $\delta T$  and the off-time be  $(1 - \delta)T$ . A first-order alignment variable builds during the pulse and decays during recovery. Define

$$a = e^{-\delta T/\tau_A}, \quad b = e^{-(1-\delta)T/\tau_M}, \quad (45)$$

so the periodic start-of-pulse alignment is

$$x_0 = \frac{b(1-a)}{1-ab}, \quad (46)$$

and the mean alignment during the pulse is

$$\bar{x}(\delta, T) = 1 - (1 - x_0) \frac{\tau_A}{\delta T} (1 - a). \quad (47)$$

A reduced product-shadow exposure time is

$$\tau_{\text{sh}}(\delta, T) = \tau_P \left[ 1 - \frac{1 - e^{-\delta T/\tau_P}}{\delta T/\tau_P} \right] + \tau_P e^{-(1-\delta)T/\tau_P}. \quad (48)$$

The first term is within-pulse product exposure; the second is carryover from the previous pulse after off-time clearing. The reduced useful rate is

$$\boxed{Y(\delta, T) = \bar{x}(\delta, T) U^q \delta^{1-q} \exp \left[ -\alpha \tau_{\text{sh}}(\delta, T) \bar{x}(\delta, T) U^q \delta^{1-q} \right]}. \quad (49)$$

**Conjecture 2** (Recovery-Window Resonance). *When activation is convex, products decay faster than alignment memory, and peak drive is bounded, useful fusion can peak in a recovery window whose off-time is long enough to clear products but short enough to preserve or efficiently rebuild alignment:*

$$\boxed{\tau_P \lesssim (1 - \delta)T \lesssim \tau_M} \quad (50)$$

*up to order-one factors and external constraints such as output smoothness, material fatigue, and actuator repetition limits.*

**Discovery from the virtual environment.** In the reduced two-time-scale model with  $\tau_P = 1.6$ ,  $\tau_M = 8.0$ ,  $\tau_A = 0.28$ ,  $q = 2.55$ , and a minimum duty fraction of 0.04, the maximum occurred at  $\delta^* = 0.040$ ,  $T^* = 6.534$ , on-time 0.261, and off-time 6.272. The off-time is roughly 3.9 product residence times and 0.78 alignment-memory times. The optimized useful rate was 15.087 times the continuous-drive reference in this stylized model, and 30.530% of the scanned grid exceeded a twofold gain.



Figure 18: Recovery-window heatmap from Eq. (49). The marked optimum occurs where product carryover is strongly reduced while alignment can still be rebuilt during the on-phase.

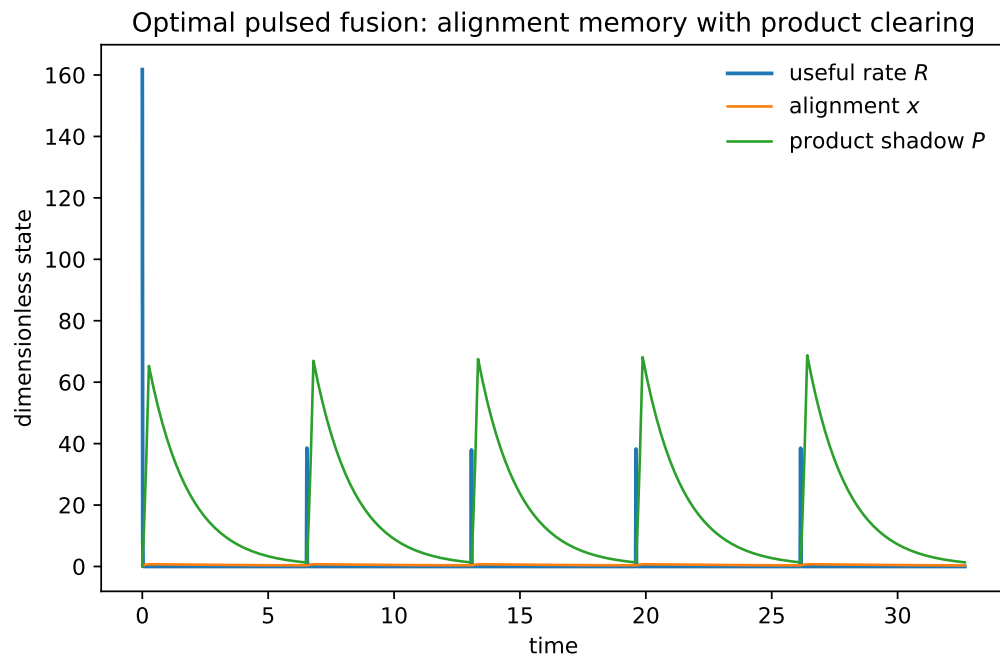


Figure 19: Representative pulsed trajectory at the recovery-window optimum. The pulse raises useful rate while the recovery phase exports product shadow.

### 7.3 Candidate law 11: Marginal-Action Bridge Rule

The original Bridge-Fusion Law assumed equal intermediate steps through a uniform barrier. Real interfaces are heterogeneous: a dissimilar-material joint may have stiff chemical zones; a biological membrane may have patches of different crowding or curvature; a latent alignment path may have regions of high semantic distortion. Let the total mismatch distance be  $D$ , split into  $m$  increments  $\ell_i$ , with local barrier coefficients  $a_i > 0$ :

$$\mathcal{A}(\ell_1, \dots, \ell_m) = \sum_{i=1}^m a_i \ell_i^p, \quad \sum_{i=1}^m \ell_i = D, \quad p > 1. \quad (51)$$

**Theorem 3** (Marginal-Action Bridge Rule). *The minimizer of Eq. (51) is*

$$\ell_i^* = D \frac{a_i^{-1/(p-1)}}{\sum_{j=1}^m a_j^{-1/(p-1)}}. \quad (52)$$

*The minimized action is*

$$\mathcal{A}^* = D^p \left( \sum_{j=1}^m a_j^{-1/(p-1)} \right)^{1-p}. \quad (53)$$

*Thus optimal bridging equalizes marginal action  $pa_i \ell_i^{p-1}$ , not distance.*

*Proof.* Using a Lagrange multiplier  $\eta$  for  $\sum_i \ell_i = D$ , the stationarity condition is  $pa_i \ell_i^{p-1} = \eta$ . Hence  $\ell_i \propto a_i^{-1/(p-1)}$ . Normalizing by the total distance gives Eq. (52). Substitution gives Eq. (53).  $\square$

**Discovery from the virtual environment.** In 100,000 random heterogeneous bridge landscapes, marginal-action allocation reduced cost relative to equal-distance allocation by a median of 53.282%. The reduction exceeded 25% in 70.958% of cases and exceeded 50% in 52.471%. The rank correlation between barrier-coefficient heterogeneity and cost reduction was 0.917. The proposed blindspot is that many fusion designs may use too few measurement points along the interface; the value is not only in adding interlayers or stages, but in placing them where the local marginal barrier is highest.

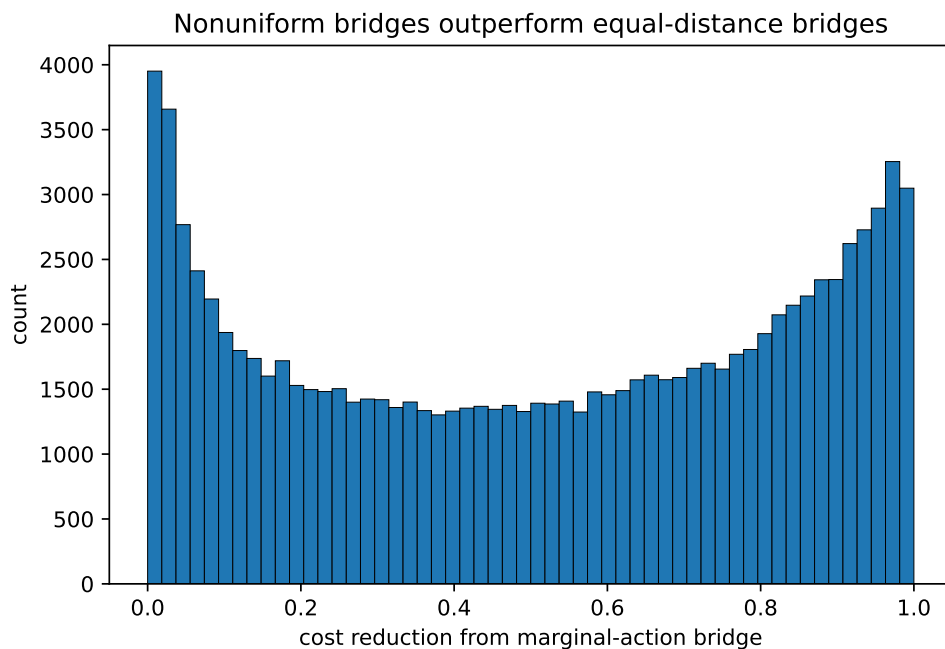


Figure 20: Cost reduction from optimal nonuniform bridge allocation relative to equal-distance bridges in random heterogeneous landscapes.

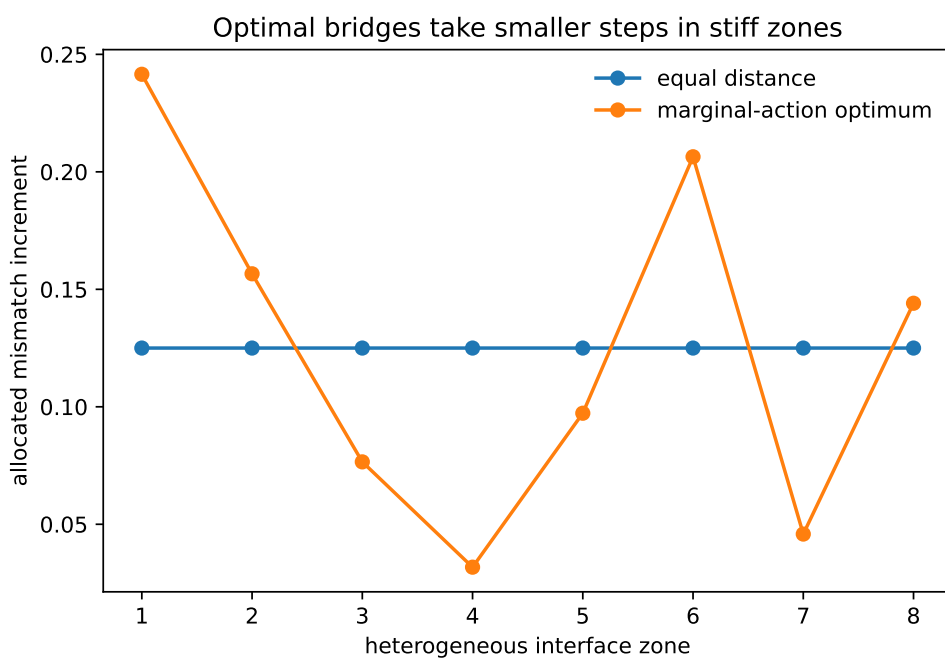


Figure 21: Representative heterogeneous bridge. The marginal-action optimum takes smaller mismatch steps in stiff zones and larger steps in soft zones.

## 7.4 Candidate law 12: Diversity-Limited Fusion Law

Information fusion often counts modalities, sensors, models, or experts. The relevant variable is not the nominal count  $N$ , but the number of independent error modes. Multi-sensor surveys emphasize that fusion can improve robustness while also introducing alignment, synchronization, and architectural challenges [27, 37]. The simplest analytic warning is the equicorrelated estimator model. Let  $N$  unbiased sources have equal error variance  $\sigma^2$  and pairwise error correlation  $\rho$ . The equal-weight fused error variance is

$$M_N = \sigma^2 \left[ \rho + \frac{1 - \rho}{N} \right]. \quad (54)$$

Therefore the gain over one source is

$$G_N = \frac{\sigma^2}{M_N} = \frac{N}{1 + (N - 1)\rho}. \quad (55)$$

This motivates the effective diversity count

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\rho}, \quad \lim_{N \rightarrow \infty} N_{\text{eff}} = \frac{1}{\rho}. \quad (56)$$

**Proposed law 7** (Diversity-Limited Fusion Law). *For equal-quality correlated sources, fusion gain is bounded by effective diversity, not source count. If  $\rho > 0$ , no amount of additional perfectly exchangeable sources can exceed the correlation ceiling  $1/\rho$ .*

**Discovery from the virtual environment.** In 120,000 random equicorrelated information-fusion cases with  $N \in [2, 100]$ , the median nominal number of sources was 51, but the median effective diversity was only 2.697. The median achieved fraction of independent-source gain was 7.035%. In 60.204% of all cases,  $N \geq 20$  but  $N_{\text{eff}} < 5$ . This suggests a design target: do not merely add modalities; engineer decorrelated failure modes, calibrated refusal, and diversity-preserving gates.

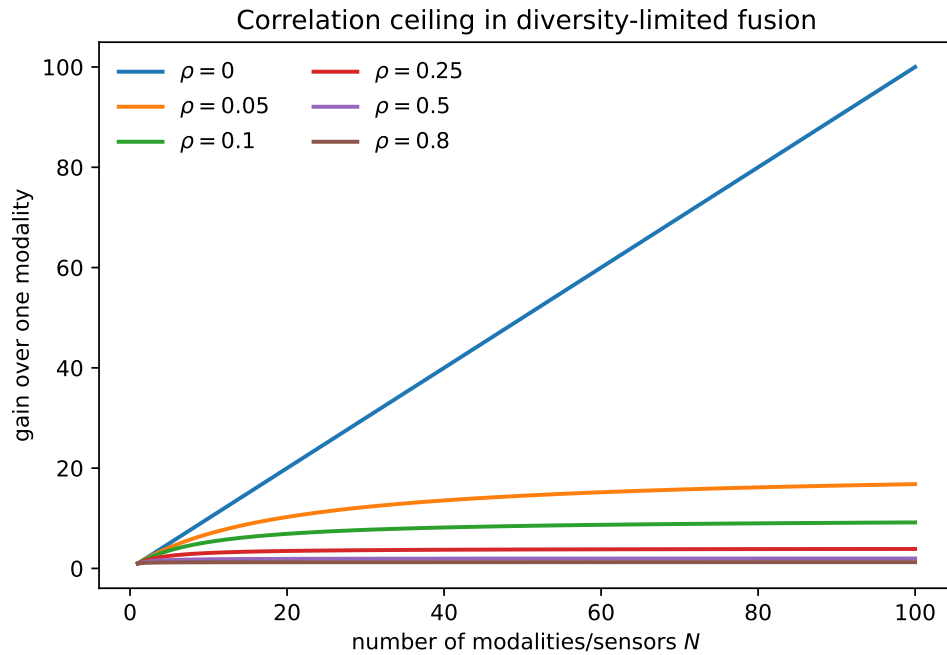


Figure 22: Diversity-limited fusion. Even weak positive correlation imposes a ceiling on the gain from adding more nominal sources.

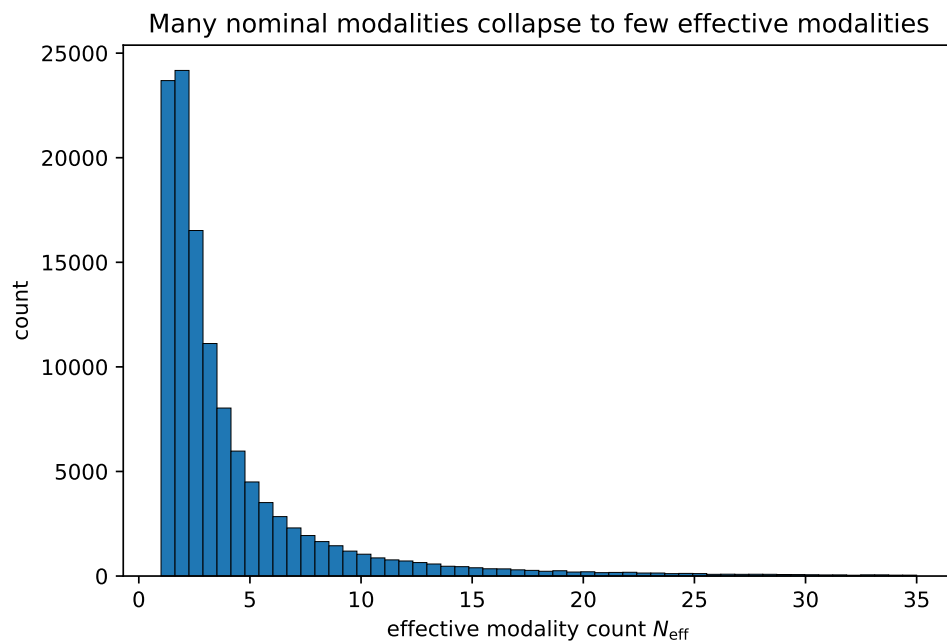


Figure 23: Distribution of effective modality count in the equicorrelated Monte Carlo environment. Many-source systems frequently collapse to a few effective independent sources.

## 7.5 Summary of third virtual tests

Table 4: Third virtual-environment results and proposed discoveries.

| Test                      | Main numerical result                                                                                            | Proposed discovery                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Pulse-shadow duty         | Pulsed optimum in 74.521% of cases; median gain 2.255-fold; median optimized shadow load near one                | Useful pulsing tunes product-shadow load rather than maximizing raw activation.                 |
| Recovery-window resonance | Optimum at off-time 6.272 for $\tau_P = 1.6$ , $\tau_M = 8.0$ ; maximum 15.087 times continuous in reduced model | Fusion can benefit from rhythm: drive, clear products, preserve or rebuild alignment.           |
| Nonuniform bridges        | Median 53.282% cost reduction over equal-distance bridging                                                       | Optimal scaffolds equalize marginal action and concentrate resolution in stiff interface zones. |
| Diversity-limited fusion  | Median $N = 51$ collapsed to median $N_{\text{eff}} = 2.697$                                                     | Fusion gain is limited by independent error modes, not source count.                            |

## 7.6 Targets created by the third sprint

The third sprint creates eight concrete targets for further study:

1. Measure product-shadow load during pulsed operation and test whether useful rate is maximized near the order-one condition in Eq. (44).
2. In pulsed physical, biological, and computational systems, independently vary duty fraction  $\delta$  and period  $T$  while measuring product residence  $\tau_P$ , alignment build time  $\tau_A$ , and memory time  $\tau_M$ .
3. Add fatigue, damage, and output-smoothness penalties to Eq. (49); determine when continuous operation returns as the constrained optimum.
4. For dissimilar materials, map local barrier coefficients  $a_i$  along proposed interlayers and compare equal-thickness gradients with Eq. (52).
5. For biological membrane fusion, test whether local crowding, curvature, tension, or lipid composition acts as a spatially varying  $a_i$  that predicts where fusogens or protrusions should concentrate.
6. For multimodal models, estimate pairwise error correlation  $\rho$  under distribution shift and report  $N_{\text{eff}}$  alongside nominal modality count.
7. Build fusion systems that optimize for decorrelated failure modes, not only for more sensors or larger models.
8. Search for a cross-domain “rhythm map” in which useful fusion is plotted against  $(\delta, T, \tau_P, \tau_A, \tau_M)$ , analogous to a phase diagram.

## 7.7 Additional applications suggested by the third sprint

**Pulsed reactor and chamber design.** The Pulse-Shadow Duty Law suggests screening pulsed designs by the dimensionless product-shadow load  $KU^q\delta^{1-q}$ . In inertial fusion energy this abstractly

corresponds to a coupled target-drive-chamber cycle: high instantaneous burn, debris and heat recovery, target injection, optical reset, and material survival. The law does not assert that lower duty is always better. It predicts that optimal duty emerges from a compromise between convex activation, product shadow, and peak-drive constraints.

**Dissimilar joining and additive manufacturing.** Functionally graded materials, interlayers, gradient porosity, and patterned joining already address warpage, delamination, residual stress, and coefficient-of-thermal-expansion mismatch in large-area and multi-material joining [38]. The Marginal-Action Bridge Rule gives a mathematical target for such gradients: the spacing or composition increments should be densest where the local barrier coefficient is largest.

**Biological and synthetic tissue fusion.** The recovery-window model suggests experiments in which fusogen presentation is pulsed rather than constant. A useful protocol would pulse the alignment scaffold, allow membrane or cytoskeletal recovery, and measure whether fusion specificity improves while pathological over-fusion declines.

**Multimodal AI and sensor systems.** The Diversity-Limited Fusion Law suggests a reporting standard. Alongside accuracy, a system should report estimated error correlation, effective modality count, and no-harm fallback behavior. A four-sensor system with  $N_{\text{eff}} = 3.5$  may be more valuable than a hundred-stream system with  $N_{\text{eff}} = 2$ .

## 8 Fourth virtual sprint: recycling, annealing, impedance, and orthogonal gates

The first three research sprints treated product shadow mostly as a suppressive output to be exported. The fourth sprint searched a different blindspot: in some systems a product is not only waste, but a possible scaffold, heat source, alignment signal, reaction layer, memory trace, or gate-control variable. This creates a design problem: recycle enough product to assist the next fusion event, but not so much that the recycled product becomes a new obstruction. Three additional blindspots were tested in parallel. First, some interfaces may need finite disorder or agitation to find an alignment path. Second, some barriers are multiplicative transmission mismatches rather than additive action costs. Third, product shadow is often a vector channel; a useful boundary should transmit the useful channel while rejecting the shadow channel.

### 8.1 Candidate law 13: Shadow-to-Scaffold Recycling Law

Let  $\xi \in [0, 1]$  be the fraction of a convertible product stream recycled into a scaffold for the next fusion event. Let  $L = R_0\tau_P$  denote the product-shadow load if no product is recycled, and let  $M = R_0\tau_S$  denote the scaffold load if all convertible product is retained. Assume residual products suppress useful rate, recycled scaffold increases alignment, excessive scaffold clogs the interface, and conversion has a direct cost:

$$\log Y(\xi) = \log R_0 - \alpha(1 - \xi)L + \beta\xi M - \gamma(\xi M)^2 - c\xi. \quad (57)$$

The parameters  $\alpha, \beta, \gamma, c > 0$  describe product-shadow strength, scaffold benefit, over-scaffold occlusion, and recycling cost.

**Proposed law 8** (Shadow-to-Scaffold Recycling Law). *For Eq. (57), the optimal recycling fraction*

is

$$\xi^* = \text{clip} \left( \frac{\alpha L + \beta M - c}{2\gamma M^2}, 0, 1 \right). \quad (58)$$

The marginal condition for product recycling to begin is

$$\alpha L + \beta M > c, \quad (59)$$

and full recycling is avoided when

$$\alpha L + \beta M - c < 2\gamma M^2. \quad (60)$$

Thus a product stream can be harmful at  $\xi = 0$ , useful at intermediate  $\xi$ , and harmful again at  $\xi = 1$ .

*Proof.* Differentiate Eq. (57) with respect to  $\xi$ :

$$\frac{d \log Y}{d\xi} = \alpha L + \beta M - c - 2\gamma M^2 \xi. \quad (61)$$

The unconstrained optimum is the zero of this derivative. Projection to  $[0, 1]$  gives Eq. (58).  $\square$

**Discovery from the virtual environment.** In 200,000 random product-recycling cases, some recycling was optimal in 81.274% of cases, an interior recycling fraction occurred in 48.477%, and full recycling was optimal in 32.797%. The median optimal recycling fraction was  $\xi^* = 0.207$ . The median useful-rate gain over no recycling was 1.233-fold overall and 1.556-fold among cases where recycling was active. The 90th percentile gain was 112.604-fold. Full recycling was worse than no recycling in 59.763% of cases, which is the important qualitative result: converting product into scaffold is valuable only inside a window.

This law gives a compact expression for a recurring cross-domain ambiguity. In D-T fusion, alpha particles can help heat the plasma, but helium ash must ultimately be removed. In materials joining, a reaction product can become an adherent interlayer or a brittle intermetallic. In condensates, a product can nucleate a functional droplet or seed pathological aging. In information fusion, errors and residuals can become calibration signals or can reinforce overconfidence.

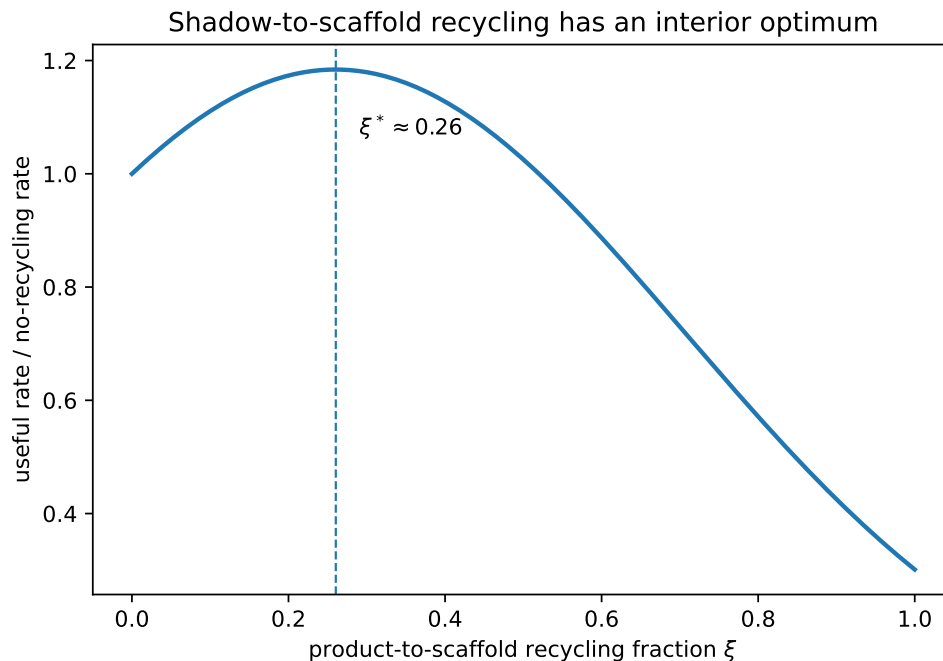


Figure 24: Representative shadow-to-scaffold recycling curve. Partial recycling converts some product shadow into useful scaffold, but full recycling over-clogs the interface.

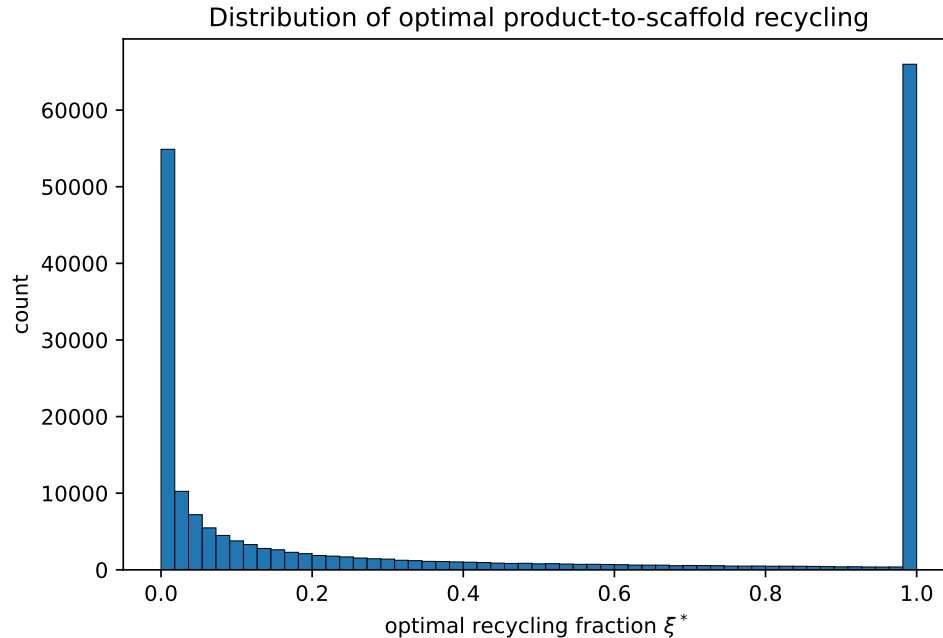


Figure 25: Distribution of optimal product-to-scaffold recycling fraction in the fourth virtual environment. The mass at zero, the interior, and one correspond to discard, partial recycling, and full recycling regimes.

## 8.2 Candidate law 14: Interface Annealing Theorem

The preceding theory often treats alignment as a deterministic map. A contrary possibility is that a small amount of agitation helps the interface discover lower-action contact states. Let  $\sigma \geq 0$  denote controlled annealing amplitude: thermal fluctuation, acoustic vibration, membrane fluctuation, stochastic training perturbation, random modality dropout, or exploratory jitter. A minimal model is

$$\log Y(\sigma) = \log Y_0 - \frac{A}{1 + a\sigma^2} - \mu\sigma^2, \quad 0 \leq \sigma \leq \sigma_{\max}. \quad (62)$$

Here  $A$  is the cold barrier action,  $a$  is annealing efficacy, and  $\mu$  is noise damage.

**Theorem 4** (Interface Annealing Theorem). *For Eq. (62), the optimal annealing amplitude is*

$$(\sigma^*)^2 = \text{clip} \left( \frac{\sqrt{Aa/\mu} - 1}{a}, 0, \sigma_{\max}^2 \right). \quad (63)$$

*A finite nonzero annealing optimum exists when*

$$Aa > \mu. \quad (64)$$

*Proof.* Set  $s = \sigma^2$ . Then  $d \log Y / ds = Aa(1 + as)^{-2} - \mu$ . The unconstrained optimum solves  $1 + as = \sqrt{Aa/\mu}$ . Projection to the feasible interval gives Eq. (63).  $\square$

**Discovery from the virtual environment.** In 150,000 random annealing cases, finite nonzero annealing was optimal in 61.866% of cases, and only 0.829% were limited by the imposed maximum  $\sigma_{\max} = 3$ . The median optimum was  $\sigma^* = 0.560$ . The median useful-rate gain over the cold interface was 1.109-fold overall and 2.530-fold among active finite-annealing cases. Gains exceeded twofold in 34.119% of cases and tenfold in 19.712%.

The proposed discovery is a useful-disorder principle: fusion interfaces can be too cold, too rigid, or too deterministic. A controlled noise source can lower effective barrier action if its alignment-search benefit exceeds its damage term. This does not imply that arbitrary heating or randomization helps. The theorem predicts a finite optimum and a clear failure condition,  $Aa \leq \mu$ .

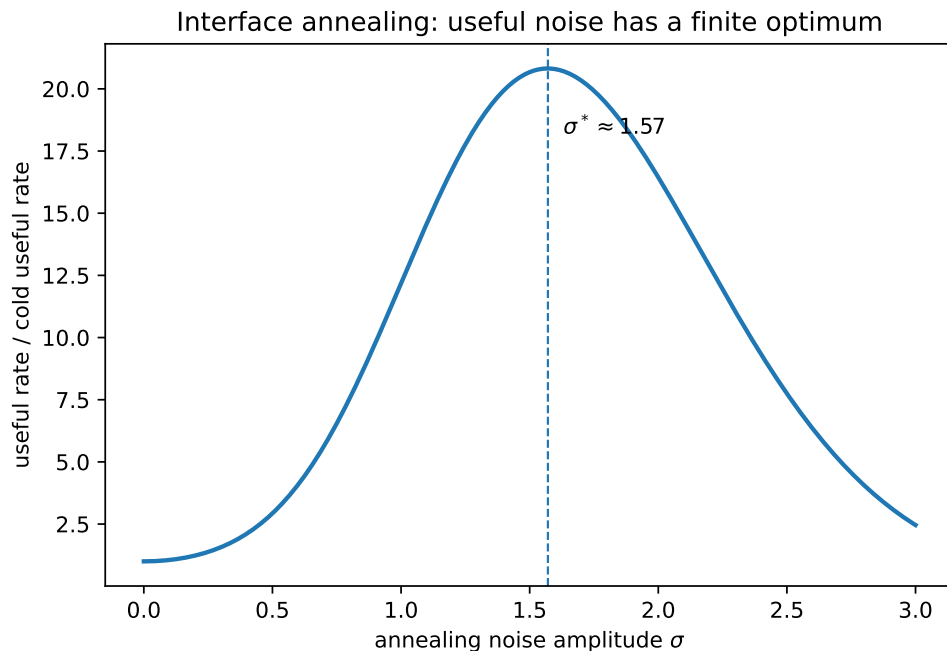


Figure 26: Representative interface-annealing optimum. Low agitation reduces effective barrier action; excessive agitation destroys alignment.

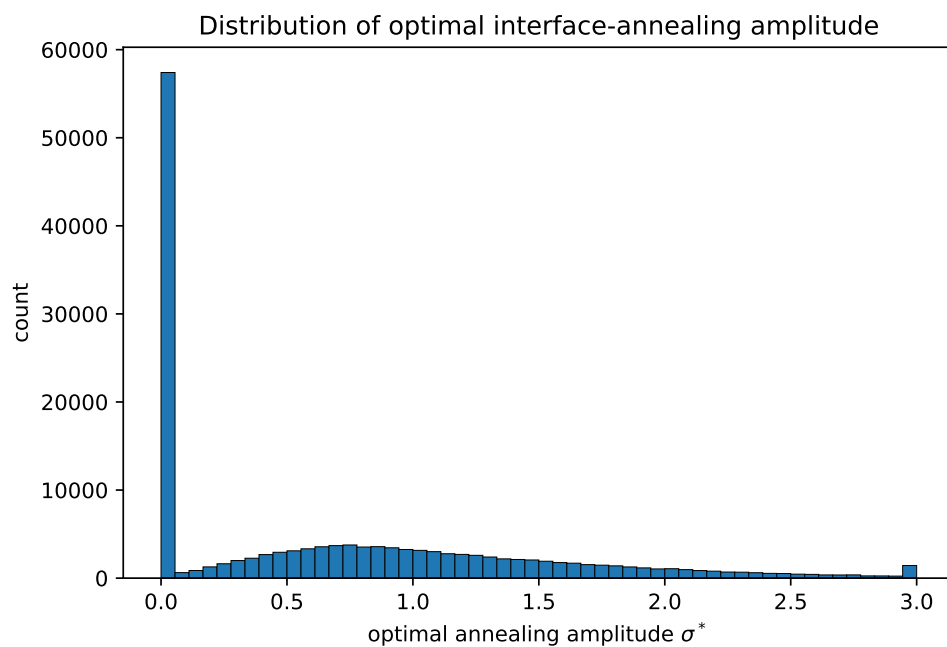


Figure 27: Distribution of optimal annealing amplitude in the fourth virtual environment. The spike at zero identifies cases where the damage term dominates barrier-relief.

### 8.3 Candidate law 15: Impedance-Matched Scaffold Law

The Bridge-Fusion Law treats mismatch cost as additive action. Some interfaces instead behave like multiplicative transmission problems. Let  $z = Z_B/Z_A$  be an impedance-like mismatch ratio between two domains. A standard dimensionless transmission factor is

$$T(z) = \frac{4z}{(1+z)^2}. \quad (65)$$

Now insert  $k$  intermediate layers with geometric ratio  $r = z^{1/(k+1)}$ , and let each layer impose loss  $c$ . The total transmission is

$$\eta_k(z, c) = \left( \frac{4r}{(1+r)^2} \right)^{k+1} e^{-ck}, \quad r = z^{1/(k+1)}. \quad (66)$$

For small per-step log mismatch,  $|\ln z|/(k+1) \ll 1$ ,

$$\log \eta_k \approx -\frac{(\ln z)^2}{4(k+1)} - ck. \quad (67)$$

**Proposed law 9** (Impedance-Matched Scaffold Law). *Under the approximation in Eq. (67), the optimal number of intermediate impedance scaffolds is approximately*

$$k^* \approx \max \left( 0, \frac{|\ln z|}{2\sqrt{c}} - 1 \right), \quad (68)$$

*with integer rounding in realized systems. Large mismatch and low per-scaffold loss favor more graded intermediates; low mismatch or high loss favors direct contact.*

**Discovery from the virtual environment.** In 120,000 random impedance-matching cases, intermediate scaffolds improved transmission in 82.143% of cases. The median optimal scaffold count was  $k^* = 6$  overall and 8 among cases where scaffolding was beneficial. The median gain over direct transmission was 2.506-fold overall and 3.960-fold among beneficial cases; the 90th percentile gain was 40.103-fold. The approximation in Eq. (68) was highly predictive: the rank correlation between the approximate and exact discrete optima was 0.990, and 99.083% of exact optima were within two layers of the approximation.

This is a multiplicative counterpart to the additive Bridge-Fusion Law. It suggests that graded interlayers, refractive-index matching, acoustic/mechanical impedance matching, learned latent bridges, and biological approach states may share a common scaling: optimal scaffold count grows roughly with absolute log mismatch and inversely with the square root of per-layer loss.

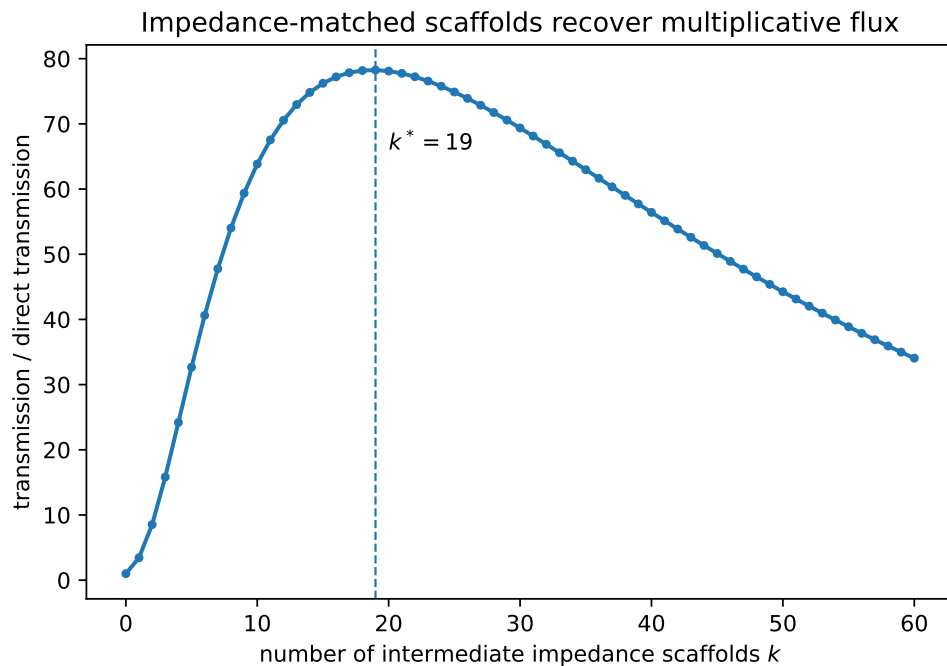


Figure 28: Representative impedance-matching curve. Intermediate scaffolds recover transmission across a large mismatch until per-layer loss dominates.

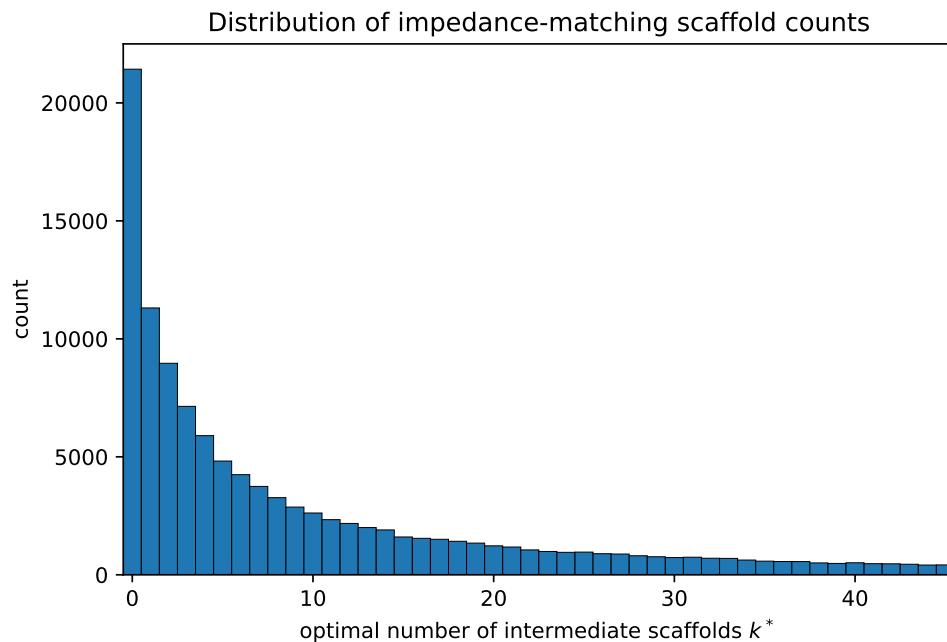


Figure 29: Distribution of optimal impedance-matching scaffold counts. Many systems prefer a finite number of graded intermediates rather than direct contact or infinitely fine grading.

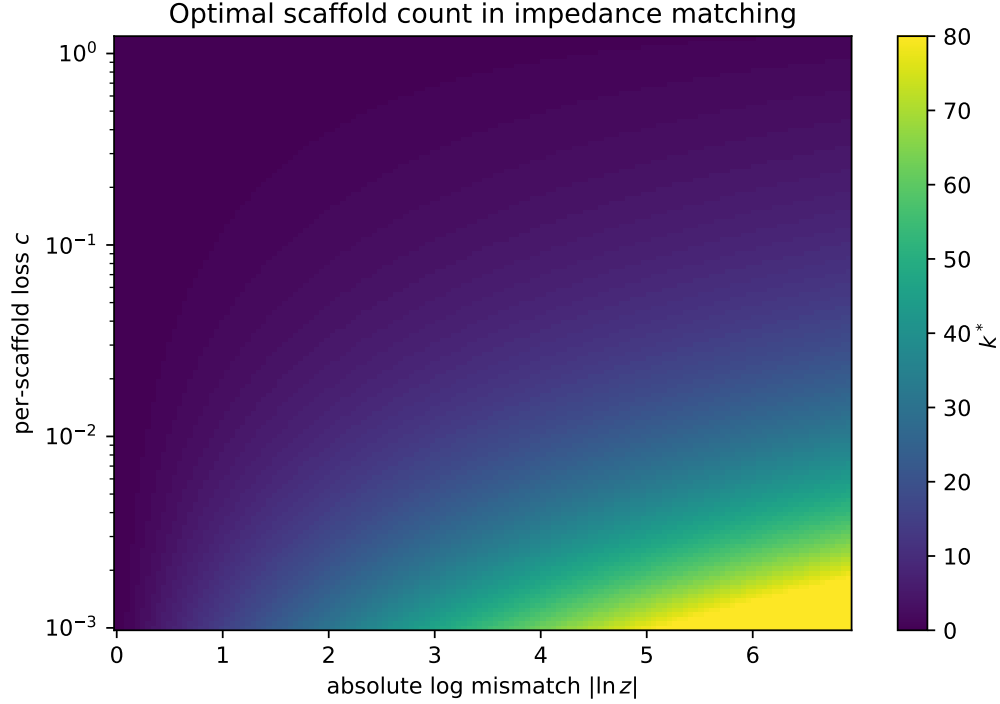


Figure 30: Optimal impedance-matching scaffold count as a function of log mismatch and per-scaffold loss. The boundary follows the square-root scaling in Eq. (68).

#### 8.4 Candidate law 16: Orthogonal Gate Fusion Theorem

The selective confinement law split residence times for useful and shadow products. A more general version treats useful output and product shadow as directions in a channel space. Let  $u$  be a unit vector for useful transmission and  $p$  a unit vector for product-shadow leakage. Let  $G \succeq 0$  be a positive semidefinite boundary conductance or fusion gate with trace budget  $\text{tr } G \leq C$ . Define

$$V(G) = u^T G u - \kappa p^T G p - c_g \text{tr } G, \quad G \succeq 0, \quad \text{tr } G \leq C. \quad (69)$$

Here  $\kappa$  is product-shadow penalty and  $c_g$  is gate-maintenance cost.

**Theorem 5** (Orthogonal Gate Fusion Theorem). *Let  $\lambda_{\max}$  be the largest eigenvalue of  $uu^T - \kappa pp^T$ . The optimizer of Eq. (69) is*

$$G^* = \begin{cases} Cvv^T, & \lambda_{\max} > c_g, \\ 0, & \lambda_{\max} \leq c_g, \end{cases} \quad (70)$$

where  $v$  is a leading eigenvector of  $uu^T - \kappa pp^T$ . If  $\vartheta$  is the angle between  $u$  and  $p$ , then

$$\lambda_+ = \frac{1 - \kappa + \sqrt{(1 + \kappa)^2 - 4\kappa \cos^2 \vartheta}}{2}. \quad (71)$$

Thus a useful fusion gate transmits along an eigen-direction, not necessarily along the raw useful channel, and refuses fusion when product-shadow separation is not worth gate cost.

*Proof.* Eq. (69) can be written as

$$V(G) = \text{tr} \left[ G(wu^T - \kappa pp^T - c_g I) \right]. \quad (72)$$

A linear functional over the positive semidefinite trace ball is maximized at an extreme point, either  $G = Cvv^T$  along a top positive eigen-direction or  $G = 0$  when all eigenvalues are nonpositive. The two-vector eigenvalue formula gives Eq. (71).  $\square$

**Discovery from the virtual environment.** In 200,000 random useful-shadow channel geometries with dimension 8, the directional gate had positive value in 97.745% of cases. A forced isotropic gate was negative in 77.189% of cases, and the directional gate rescued 97.079% of those isotropic-failure cases. Among positive-gate cases, the median useful transmission was 0.960, the median product-shadow transmission was 0.044, and the median useful/product selectivity was 18.531. The 90th percentile selectivity was 355.486; selectivity exceeded 10 in 61.156% of positive-gate cases and 100 in 23.727%.

The proposed discovery is that boundary design should be vectorial. If useful flux and product shadow are nearly aligned, no scalar increase in conductance can help much. If they are angularly separable, a low-rank directional gate can produce high useful transmission while rejecting product shadow. This gives a mathematical reason to measure not only the amount of product shadow, but its channel geometry relative to the desired output.

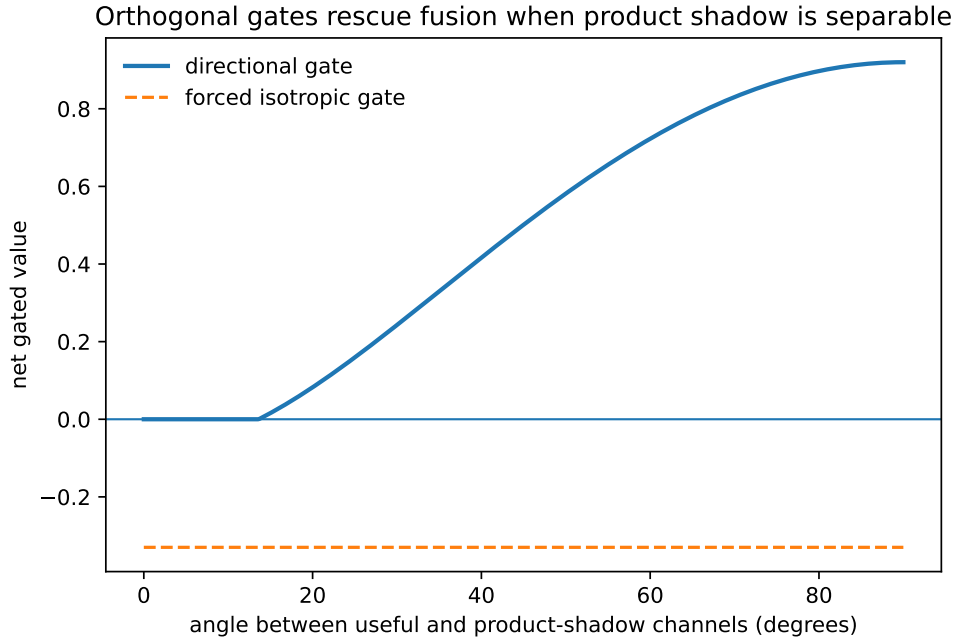


Figure 31: Orthogonal Gate Fusion Theorem. Directional gating can be valuable even when a forced isotropic gate is harmful, provided useful and shadow channels are angularly separable.

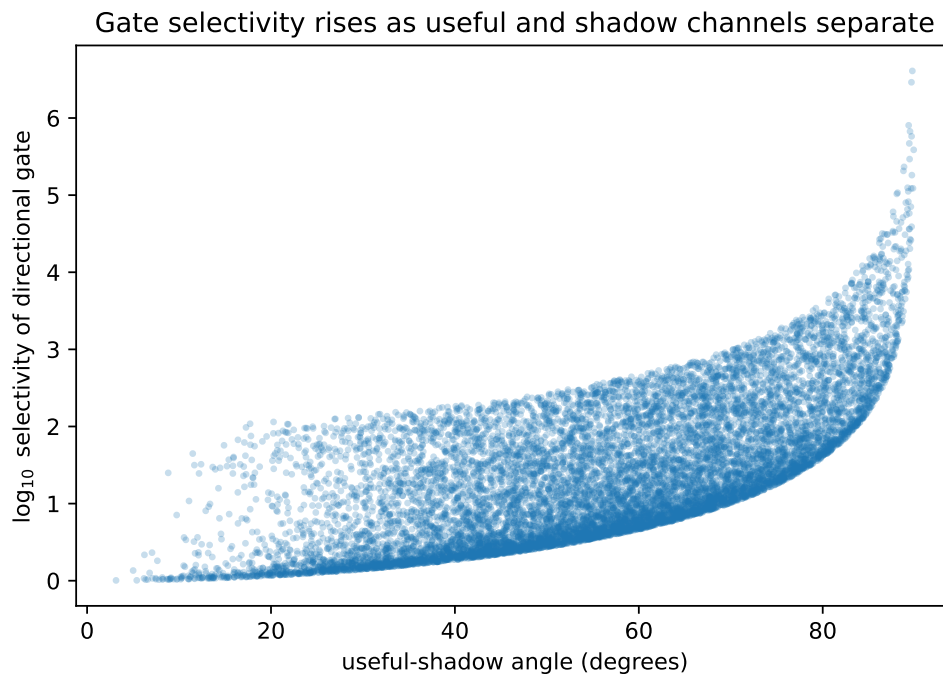


Figure 32: Directional-gate selectivity rises as useful and product-shadow channels separate. The scatter uses random penalty and cost values from the fourth virtual environment.

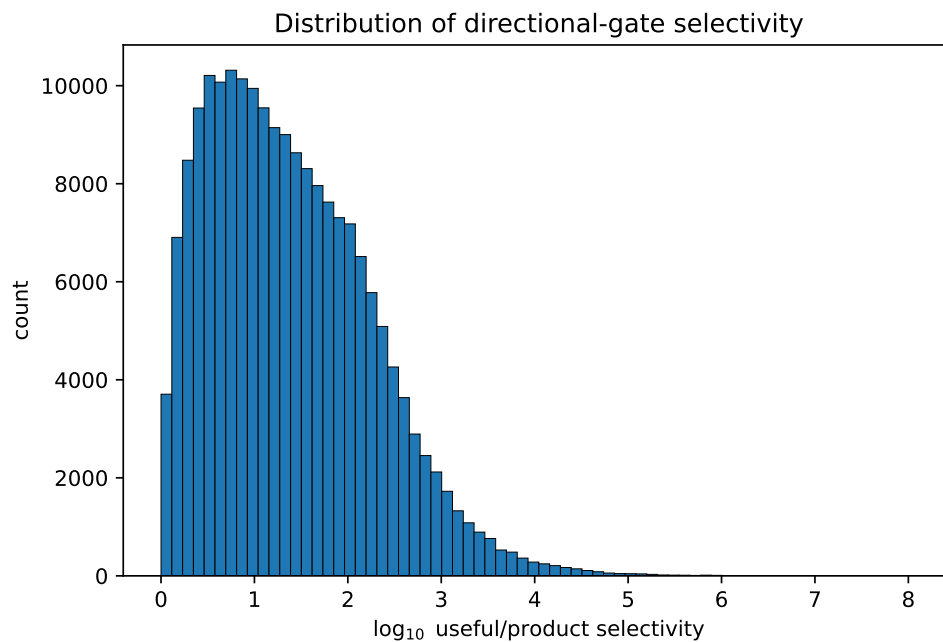


Figure 33: Distribution of useful/product selectivity among positive-value directional gates. The long tail indicates a search target: find domains where product shadow is nearly orthogonal to the desired output.

## 8.5 Summary of fourth virtual tests

Table 5: Fourth virtual-environment results and proposed discoveries.

| Test                | Main numerical result                                                                                                | Proposed discovery                                                                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Product recycling   | Some recycling optimal in 81.274%; interior $\xi^*$ in 48.477%; full recycling worse than none in 59.763%            | Products can be scaffolds, but only within a turnover and occlusion window.                          |
| Interface annealing | Finite $\sigma^* > 0$ in 61.866%; active-case median gain 2.530-fold                                                 | Controlled disorder can lower fusion barriers when alignment-search benefit exceeds noise damage.    |
| Impedance matching  | Intermediate scaffolds beneficial in 82.143%; median gain 2.506-fold; approximate $k^*$ within two layers in 99.083% | Multiplicative mismatch favors finite geometric grading with $k^* \sim  \ln z /(2\sqrt{c}) - 1$ .    |
| Orthogonal gates    | Directional gate rescued 97.079% of forced-isotropic failures; median selectivity 18.531                             | Useful fusion may require vector gates that transmit output while rejecting product-shadow channels. |

## 8.6 Targets created by the fourth sprint

The fourth sprint creates eight concrete targets for further study:

1. Measure when a fusion product is waste, heat, scaffold, memory, reaction layer, or alignment signal; estimate  $(\alpha, \beta, \gamma, c, L, M)$  in Eq. (57).
2. In reactor-relevant models, separate beneficial product heating from product ash poisoning and search for a finite  $\xi^*$ -like retention fraction rather than binary confinement or exhaust.
3. In dissimilar materials, measure whether thin reaction layers improve adhesion before becoming brittle product shadow; compare the optimum to Eq. (58).
4. In membrane and condensate systems, test whether controlled fluctuations, crowding modulation, or mechanical jitter produce the finite-annealing optimum in Eq. (63).
5. In multimodal models, use stochastic modality dropout or representation noise as an annealing variable and test whether accuracy under distribution shift peaks at finite  $\sigma$ .
6. For graded materials, latent spaces, and transport interfaces, measure an impedance-like ratio  $z$  and per-layer loss  $c$ ; compare optimal layer count with Eq. (68).
7. For any selective boundary, estimate the useful-shadow angle  $\vartheta$ , not only the scalar amount of leakage. Search for nearly orthogonal product channels.
8. Implement rank-one or low-rank gates that include an explicit refusal state when  $\lambda_{\max} \leq c_g$ .

## 8.7 Applications suggested by the fourth sprint

**Product recycling and closed-loop fusion.** The Shadow-to-Scaffold Recycling Law suggests that product management should not be only an exhaust problem. Some products should be rapidly

removed, some should be transiently retained, and some should be converted into control signals. Applications include alpha-energy retention versus helium-ash removal, self-limiting interfacial reaction layers in materials joining, condensate client turnover, and machine-learning residuals used for calibration rather than reinforcement of correlated error.

**Interface annealing.** The Interface Annealing Theorem suggests protocols that scan controlled agitation: vibration-assisted joining, thermal cycling below damage thresholds, fluctuating fusogen presentation, stochastic latent alignment, or modality dropout. The target is not maximum noise. It is the finite point where barrier relief and damage balance.

**Impedance-matched fusion paths.** The Impedance-Matched Scaffold Law suggests geometric rather than linear gradients when transmission loss is multiplicative. Candidate applications include graded interlayers, plasma-edge coupling layers, acoustic or electromagnetic coupling, embedding-space curriculum bridges, and staged biological approach states.

**Orthogonal product-selective gates.** The Orthogonal Gate Fusion Theorem suggests designing boundaries as filters rather than doors. A reactor, membrane, condensate, material interface, or multimodal model should ask whether desired output and damaging product occupy separable channels. When they do, a low-rank selective gate can outperform scalar conductance increases.

## 9 Fifth research sprint: fusion memory, dual channels, and anchor windows

The first four sprints treated scaffolds, products, gates, and interfaces as explicit design variables. The fifth sprint asks a sharper question: what happens when fusion changes the future interface on which later fusion events occur? In such systems, the product of one fusion event may be harmful ash, but it may also be alignment memory, a seed, a trained interface, a warmed or conditioned boundary, a catalytic intermediate, or a reusable scaffold. This creates a sign problem: the same event can improve and poison its own future.

This question is motivated by several established but cross-domain observations. Fusion pilot-plant studies treat particle exhaust and core fueling as coupled design requirements because helium ash and edge conditions affect stationary operation [30]. Multimodal systems need enough shared structure for alignment but can suffer from noise, modality gaps, redundancy, and negative transfer when fusion is forced [28]. Biomolecular condensates can create functional microenvironments, but altered material state, immobility, aging, and aggregation can change the future behavior of the condensate itself [22, 39]. The fifth sprint formalizes these as memory-coupled fusion problems.

### 9.1 Candidate law 17: Scaffold-Shadow Autocatalytic Bifurcation

Let a fusion event generate both reusable scaffold and harmful shadow. Suppose useful rate is

$$R = R_0 \exp(\beta S - \alpha P), \quad (73)$$

where  $R_0$  is basal drive,  $S$  is reusable scaffold memory,  $P$  is product shadow,  $\beta$  is scaffold efficacy, and  $\alpha$  is shadow efficacy. Under a quasi-steady approximation,

$$S \approx \rho_S \tau_S R, \quad P \approx \rho_P \tau_P R, \quad (74)$$

so that

$$\boxed{R = R_0 \exp(gR), \quad g = \beta \rho_S \tau_S - \alpha \rho_P \tau_P.} \quad (75)$$

The sign of  $g$  determines the memory regime.

If  $g < 0$ , shadow dominates scaffold and Eq. (75) has one finite saturated branch,

$$\boxed{R = \frac{W((-g)R_0)}{-g},} \quad (76)$$

where  $W$  is the Lambert  $W$  function. In this regime, more basal drive produces sublinear useful-rate growth because each fusion event suppresses future fusion.

If  $g > 0$ , scaffold memory dominates shadow. Finite fixed points exist only while

$$\boxed{gR_0 \leq e^{-1}.} \quad (77)$$

At equality, the two fixed points merge in a fold bifurcation. For  $gR_0 > e^{-1}$ , the uncapped equation has no finite fixed point; a real system must then be limited by some omitted nonlinear loss, depletion, heat load, damage channel, or control saturation.

This gives a compact proposed principle:

$$\boxed{\begin{array}{c} \text{Fusion memory ignites when reusable scaffold feedback exceeds shadow feedback} \\ \text{and crosses } gR_0 = e^{-1}. \end{array}} \quad (78)$$

The novelty is not that positive feedback exists; it is the placement of reusable product and harmful product in the same rate equation. In a reactor analogy, alpha heating and helium ash are not merely separate terms; they are opposite signs in the memory coefficient. In a condensate, seed-like concentration and aging are opposite signs. In an information-fusion system, learned alignment and overfit bias are opposite signs.

## 9.2 Candidate law 18: Dual-Percolation Fusion Window

Networked fusion often has two propagation channels: a useful fusion channel and a harmful product-shadow channel. Let  $c_F$  be the mean degree of the useful-fusion graph and  $c_P$  the mean degree of the shadow graph. In an Erdős-Rényi approximation, their giant-component fractions satisfy

$$S_F = 1 - e^{-c_F S_F}, \quad S_P = 1 - e^{-c_P S_P}. \quad (79)$$

Extensive useful fusion without extensive shadow requires

$$\boxed{c_F > 1, \quad c_P < 1.} \quad (80)$$

If the same edges carry both useful fusion and product shadow, then  $c_F = c_P$ , and the safe extensive-fusion window vanishes. If a rectifying gate makes

$$c_P = \chi c_F, \quad 0 < \chi < 1, \quad (81)$$

then the safe window opens as

$$\boxed{1 < c_F < \chi^{-1}.} \quad (82)$$

This is a network version of selective confinement. The object is not to maximize connectivity. The object is to let useful fusion percolate while keeping shadow subcritical.

A simple value functional used in the virtual test was

$$V(c_F, c_P) = BS_F - HS_P - C_F c_F - C_P c_P, \quad (83)$$

where  $B$  is useful giant-component value,  $H$  is shadow penalty, and  $C_F, C_P$  are connection costs.

### 9.3 Candidate law 19: Anchor-Complementarity Window

Diversity-limited fusion says that redundant sources do not add as much value as their count suggests. The opposite failure also matters: sources that are too unrelated may be difficult to align. Let  $\rho \in [0, 1]$  represent shared anchor overlap. A minimal anchor-complementarity objective is

$$\boxed{V(\rho) = U(1 - \rho^2) - \frac{A}{\rho^2 + \epsilon},} \quad (84)$$

where  $U$  is the scale of useful nonredundant complementarity,  $A$  is alignment difficulty, and  $\epsilon > 0$  prevents a singularity at zero overlap.

Writing  $x = \rho^2$ , the interior optimum satisfies

$$\frac{dV}{dx} = -U + \frac{A}{(x + \epsilon)^2} = 0, \quad (85)$$

so

$$\boxed{\rho^{*2} = \sqrt{A/U} - \epsilon,} \quad (86)$$

projected to  $[0, 1]$ . Thus fusion can have an interior optimum: neither maximum diversity nor maximum overlap.

This law adds a correction to the diversity principle. Decorrelation is valuable only after there is enough common structure to identify what should be fused.

### 9.4 Candidate law 20: Memory-Selective Reset Theorem

Let  $M_t$  be beneficial interface memory and  $P_t$  harmful product shadow. A simple cyclic fusion model is

$$R_t = R_0 \exp(\mu M_t - \alpha P_t), \quad (87)$$

$$M_{t+1} = (1 - \eta_M)M_t + mR_t, \quad P_{t+1} = (1 - \eta_P)P_t + pR_t. \quad (88)$$

Consider a reset, purge, cleaning, dropout, anneal, or recovery operation that removes fractions  $q_M$  and  $q_P$  of memory and shadow. Let  $C_R$  be its log-rate-equivalent cost. The reset changes the next log-rate by

$$\Delta \ln R = -\mu q_M M + \alpha q_P P - C_R. \quad (89)$$

**Theorem 6** (Memory-Selective Reset Theorem). *A reset operation increases the next-cycle useful rate when*

$$\alpha q_P P > \mu q_M M + C_R. \quad (90)$$

*Equivalently, for  $q_M > 0$ , the required selectivity ratio is*

$$\frac{q_P}{q_M} > \frac{\mu M}{\alpha P} + \frac{C_R}{\alpha P q_M}. \quad (91)$$

*Proof.* The reset maps  $M \mapsto (1 - q_M)M$  and  $P \mapsto (1 - q_P)P$ . Substitution into  $\ln R = \ln R_0 + \mu M - \alpha P$ , followed by subtraction of reset cost, gives  $\Delta \ln R = -\mu q_M M + \alpha q_P P - C_R$ . The stated inequalities follow directly.  $\square$

This theorem turns product removal into a selectivity problem. A nonselective reset can erase the very alignment memory that made fusion possible. The ideal operation removes product shadow while preserving scaffold memory.

## 9.5 Fifth-sprint virtual experiments

Four dimensionless experiments were run with random seed 20260517. The goal was not physical prediction but falsification pressure on the candidate laws.

Table 6: Summary of fifth-sprint virtual experiments.

| Experiment                | Setup                                                                                                  | Main result                                                                                                                         |
|---------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Scaffold-shadow catalysis | auto- Random feedback terms in Eq. (75).                                                               | 50.1% shadow-saturated, 42.1% bistable-threshold, and 7.76% runaway-candidate cases. Median positive critical $R_0$ was 2.73.       |
| Dual percolation          | Grid over $(c_F, c_P)$ using Eq. (83).                                                                 | Same-network propagation gave maximum value 0. Rectified propagation with $\chi = 0.25$ reached maximum value 0.866.                |
| Anchor complementarity    | Random $(U, A, \epsilon)$ in Eq. (84).                                                                 | 60.7% of cases had an interior optimal overlap. Median $\rho^*$ was 0.441 overall and 0.461 among interior cases.                   |
| Memory-selective reset    | Vectorized cyclic model with no reset, blind adaptive reset, selective adaptive reset, and over-reset. | Selective reset was best in 38.4% of cases and beat no reset in 38.6%. The 90th percentile selective/no-reset yield ratio was 1.36. |

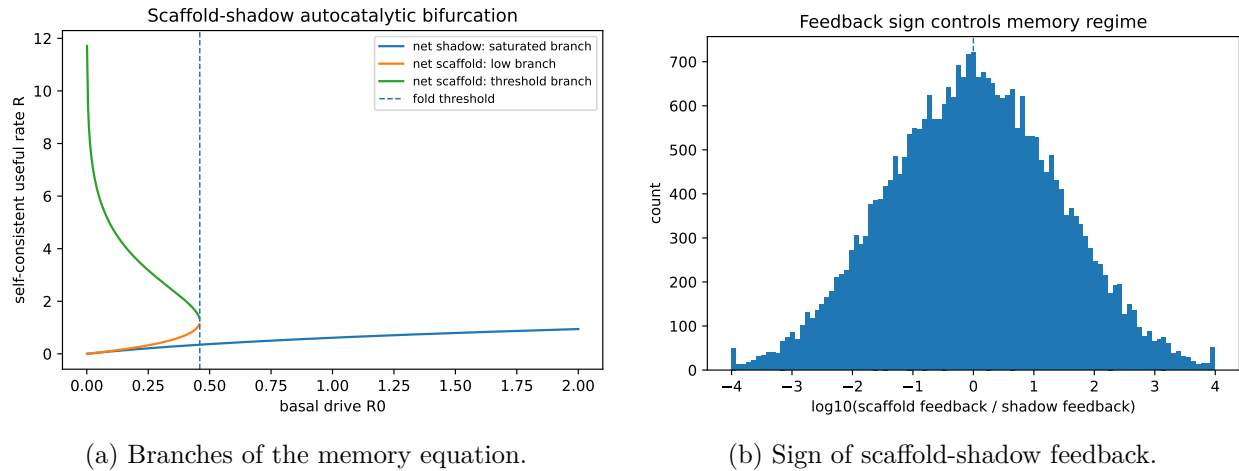


Figure 34: Scaffold-shadow autocatalysis. A product can behave as a reusable scaffold or as shadow. The fifth sprint suggests that the sign of a single effective memory coefficient,  $g$ , controls whether fusion saturates, becomes thresholded, or requires omitted nonlinear losses.

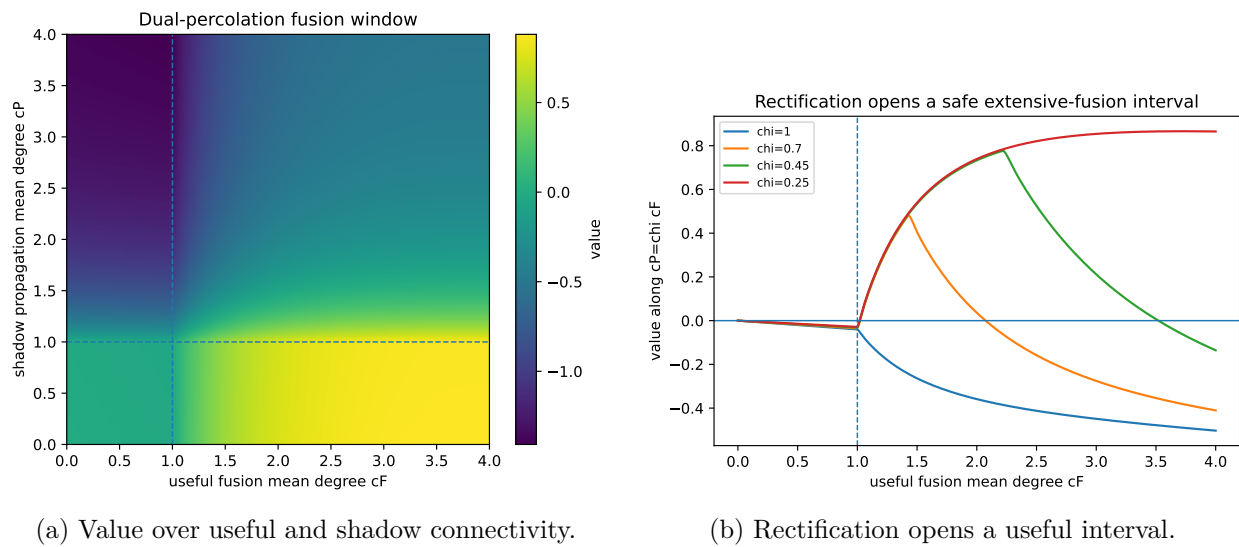


Figure 35: Dual-percolation fusion. The safe region is not high connectivity everywhere; it is high useful connectivity and subcritical shadow connectivity.

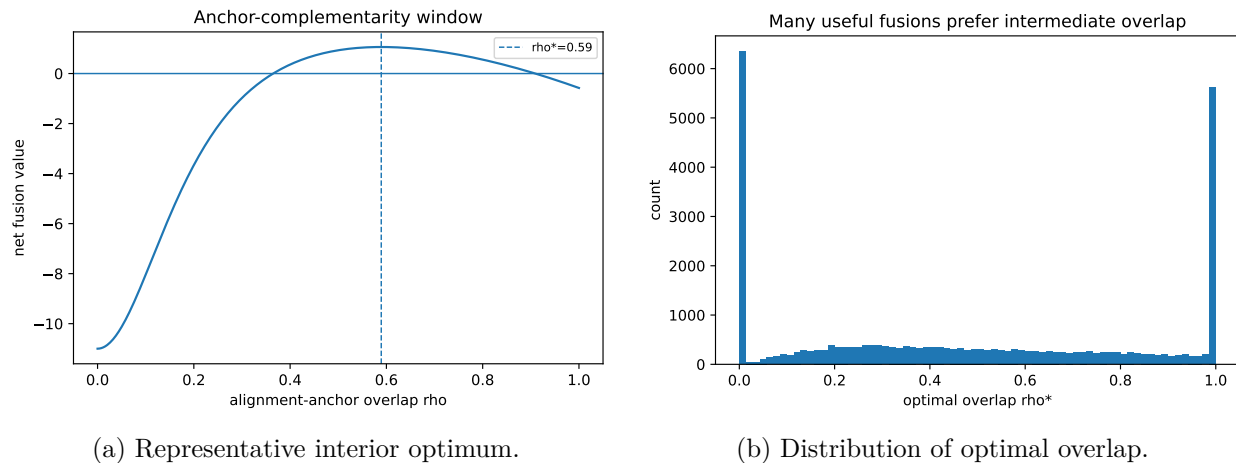


Figure 36: Anchor-complementarity window. Fusion can fail when sources are too redundant, but it can also fail when they lack enough common anchor for alignment.

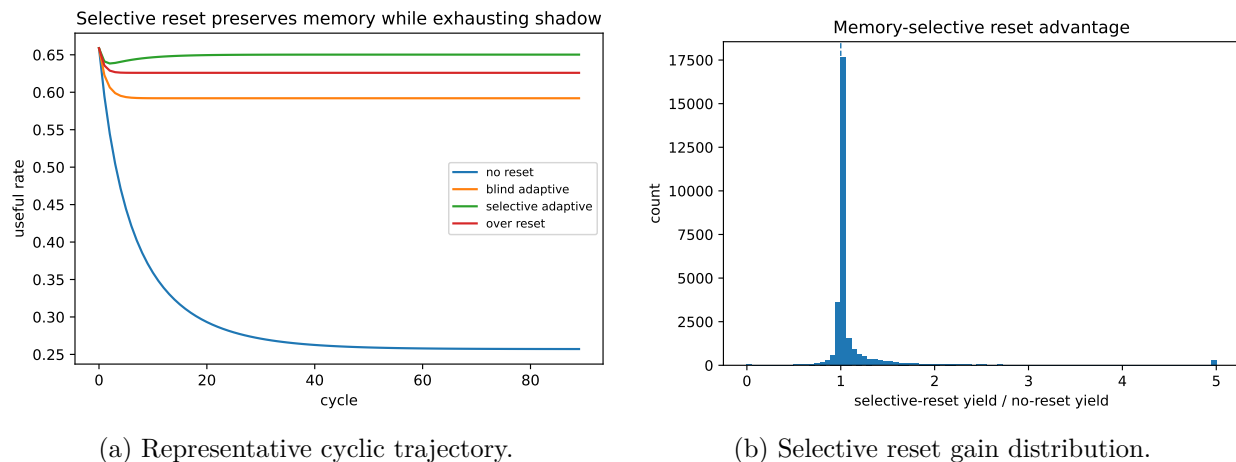


Figure 37: Memory-selective reset. A reset is beneficial only when product-shadow removal outweighs loss of alignment memory and reset cost.

## 9.6 Exceptional findings from the fifth sprint

The most important result is the *product sign split*. Product residence was previously treated mainly as harmful shadow. The autocatalytic model shows that product residence has two signs: one component can become scaffold memory and another can become poison. The decisive variable is

$$g = \beta \rho_S \tau_S - \alpha \rho_P \tau_P. \quad (92)$$

This suggests a sharper experimental target than simply reducing product residence: separate useful product memory from harmful product shadow, then lengthen one and shorten the other.

The second exceptional result is the dual-percolation window. A network can be too connected even when useful fusion needs connectivity. If useful coupling and shadow propagation share the same paths, extensive safe fusion may be impossible. Rectification, quarantining, modular boundaries,

divertor-like paths, immune boundaries, or gated information channels create value by making  $c_F$  and  $c_P$  independently tunable.

The third exceptional result is the anchor-complementarity window. The earlier diversity law could be misread as recommending maximum independence. The new result corrects this: useful fusion often requires a finite overlap anchor. The design target is not independence; it is alignable nonredundancy.

The fourth exceptional result is the reset selectivity inequality. Cleaning, purging, dropout, annealing, or rebooting can be harmful when it erases interface memory. A reset is exceptional only when it removes shadow faster than it removes learned compatibility.

## 9.7 Targets created by the fifth sprint

The fifth sprint creates the following concrete targets for further study.

1. **Measure the memory coefficient  $g$ .** In any repeated fusion system, perturb product residence and scaffold lifetime separately to estimate  $\partial \ln R / \partial S$  and  $\partial \ln R / \partial P$ . Search for the fold condition  $gR_0 = e^{-1}$ .
2. **Build dual-channel diagrams.** Draw separate graphs for useful propagation and shadow propagation. Estimate  $c_F$  and  $c_P$ , not just total connectivity.
3. **Scan anchor overlap.** In data, sensor, biological, and materials fusion, vary shared anchor content while holding source competence approximately fixed. Test for the predicted interior  $\rho^*$ .
4. **Design selective reset experiments.** Compare no reset, blind reset, and product-selective reset while measuring retained alignment memory. Test inequality (90) directly.
5. **Search for product conversion.** Look for mechanisms that convert a shadow product into a scaffold product, or that split one mixed product stream into a high-memory stream and a high-exhaust stream.

## 9.8 Application implications

**Nuclear fusion.** The theory suggests that alpha heating, helium ash, wall conditioning, fueling, and exhaust should be modeled as a signed memory system rather than as independent engineering modules. The target is not simply short product residence; it is selective retention of beneficial heating and rapid removal of ash and damaging products.

**Biological and synthetic membrane fusion.** Fusion protocols should distinguish beneficial local interface priming from harmful global membrane disruption. A useful experiment is to pulse fusogen or calcium-like triggers, then apply selective recovery operations that restore crowding, tension, or composition without erasing productive local alignment.

**Condensates.** The product sign split maps to reversible condensate enrichment versus irreversible aging. A possible intervention is a memory-selective reset that dissolves or mobilizes aged substructure while preserving productive condensate localization.

**Materials joining.** Surface activation, interlayer formation, and thermal history can be beneficial memory, while brittle phases and residual stress are shadow. The proposed target is a process

window that preserves activation and graded compatibility while purging or preventing harmful reaction products.

**Information fusion.** Multimodal learning systems should not maximize independence or overlap alone. They should optimize anchor overlap, preserve alignment memory, and use selective dropout or recalibration to remove modality-specific noise without erasing useful cross-modal maps.

## 10 Sixth virtual sprint: shadow geometry, latency, contact sparsity, and granularity

The sixth sprint searched for blindspots in the control and geometry of fusion rather than in the immediate act of merger. Four questions were emphasized. First, are product shadows scalar penalties, or can they be balanced as vectors across multiple channels? Second, can a control loop that is correct in sign become harmful when delayed? Third, is maximal contact area always beneficial, or can sparse contact outperform full contact? Fourth, when many entities must be fused, is the best architecture atomized, flat, or modular at a finite granularity?

Recent literature makes these questions plausible. Plasma-control work increasingly treats reliable rampdown and instability avoidance as central to high-energy tokamak operation [40, 41]. Dynamic multimodal fusion systems now select fusion stage based on disagreement or reliability rather than applying a fixed fusion depth [42]. Sensor-fusion deployment studies emphasize latency, jitter, memory traffic, and temporal misalignment as practical limits on closed-loop fusion systems [43]. Biomolecular-condensate experiments show that condensates can initially enhance biochemical activity but later lose efficiency through aging toward more solid-like states [44]. These observations suggest that useful fusion depends on vector-valued product effects, feedback timing, active contact topology, and scale granularity.

### 10.1 Candidate law 21: Shadow-Vector Balancing Law

Earlier sections treated product shadow as a scalar suppressor. That is a useful first approximation, but many real systems have multiple shadow channels: heat, ash, wall load, neutron load, bias, latency, residual stress, toxicity, gelation, or lost diversity. Let there be  $m$  possible fusion pathways and  $r$  shadow channels. Pathway  $i$  has useful yield  $y_i$  and shadow vector  $h_i \in \mathbb{R}^r$ . Let

$$H = \begin{bmatrix} h_1 & h_2 & \cdots & h_m \end{bmatrix}, \quad w \in \Delta_m = \left\{ w_i \geq 0, \sum_{i=1}^m w_i = 1 \right\}. \quad (93)$$

The shadow-balanced fusion value is proposed as

$$\boxed{V(w) = y^T w - \alpha \|Hw\|_2^2 - \epsilon \|w\|_2^2, \quad w \in \Delta_m.} \quad (94)$$

The term  $\epsilon \|w\|_2^2$  is a small control-regularization cost. The residual product shadow is not the sum of scalar penalties but the norm of a vector sum.

**Proposed law 21** (Shadow-Vector Balancing Law). *For Eq. (94), an active pathway satisfies*

$$\boxed{w_i^* > 0 \quad \Rightarrow \quad y_i - 2\alpha h_i^T \bar{h} - 2\epsilon w_i^* = \nu, \quad \bar{h} = Hw^*,} \quad (95)$$

while an inactive pathway satisfies

$$\boxed{w_i^* = 0 \quad \Rightarrow \quad y_i - 2\alpha h_i^T \bar{h} \leq \nu.} \quad (96)$$

Thus a pathway should be included not merely when its standalone yield is high, but when its shadow vector is complementary to the current residual shadow. The exceptional region is

$$\boxed{\|Hw^*\|_2 < \min_i \|h_i\|_2,} \quad (97)$$

where mixing pathways creates less residual shadow than every single pathway.

*Proof.* Eq. (94) is a concave quadratic maximization over the simplex. The Lagrangian stationarity condition on the active set is

$$\nabla_i V = y_i - 2\alpha h_i^T Hw - 2\epsilon w_i = \nu. \quad (98)$$

The complementary slackness inequalities give the inactive condition.  $\square$

A simple two-path consequence clarifies the mechanism. If  $y_1 = y_2 = y$ ,  $h_2 = -h_1$ , and  $w = (1/2, 1/2)$ , then  $Hw = 0$ . The mixed value is  $y - \epsilon/2$ , while either single pathway has value  $y - \alpha\|h_1\|^2 - \epsilon$ . The mixture is better when

$$\boxed{\alpha\|h_1\|^2 > \epsilon/2.} \quad (99)$$

Here opposite shadow vectors need not mean negative material products. They can represent opposed residual effects in a controlled channel basis: one pathway heats where another cools, biases one direction where another debiases, stresses one mode where another relieves it, or generates a product that can be consumed by another stage.

## 10.2 Virtual experiment 21: vector-valued product shadows

A Monte Carlo test sampled 30,000 five-pathway systems with four shadow dimensions. For each case, projected-gradient optimization solved Eq. (94) on the simplex. The optimized mixture was compared with the best single pathway.

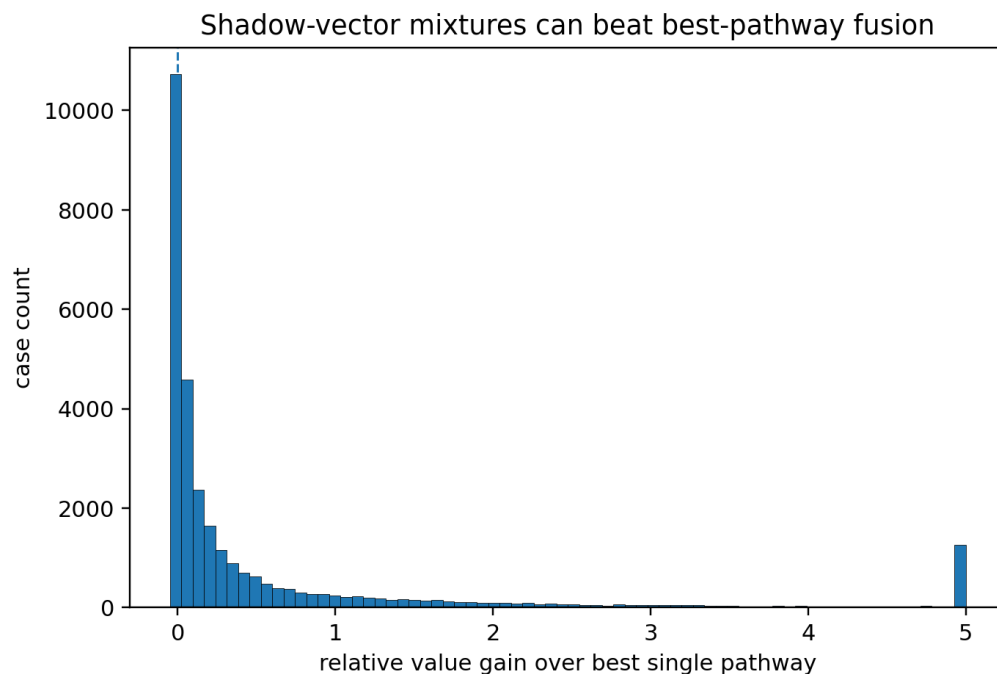


Figure 38: Relative value gain of optimized shadow-vector mixtures over the best single pathway in 30,000 toy cases. The distribution is clipped at a relative gain of 5 for readability.

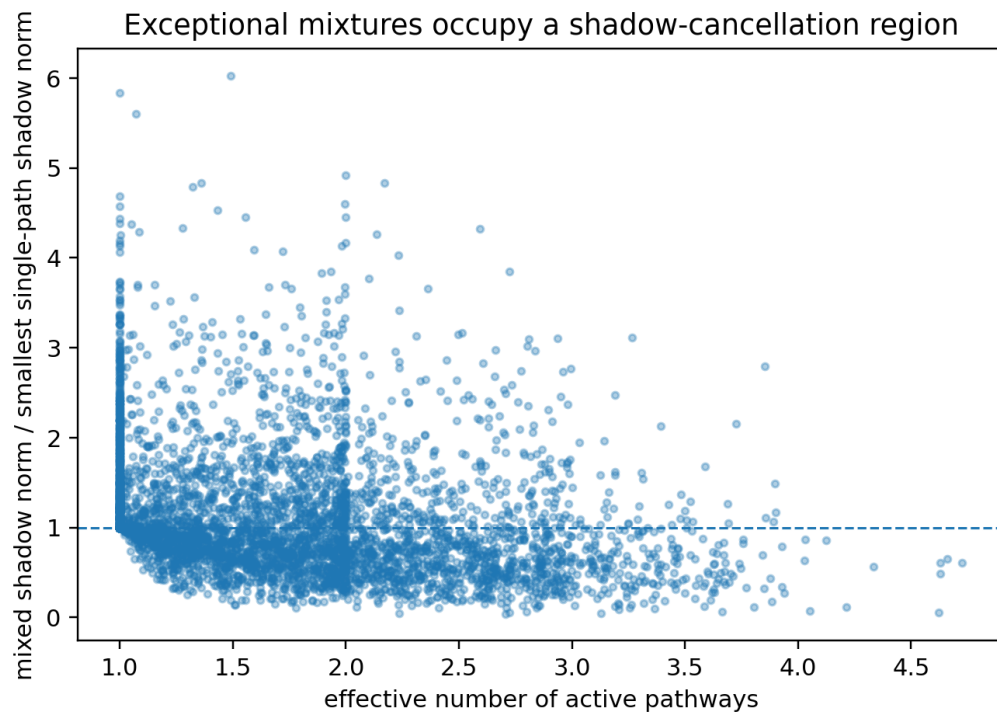


Figure 39: Effective number of active pathways versus residual shadow norm. Points below the dashed line have lower residual shadow than the smallest single-path shadow.

The optimized mixture exceeded the best single pathway in 78.6% of sampled cases, and 74.3% of optima used more than one effective pathway. The median relative value gain was 8.7%; the 90th percentile was 196.5%. Relative gains are inflated when the best single-path value is near zero, so absolute gains are retained in the data files. The residual shadow norm was below that of the best single pathway in 88.9% of cases and below the shadow norm of every single pathway in 59.6% of cases.

**Interpretation.** The notable discovery is that product shadow can be a design geometry, not only a scalar loss. If product effects are represented in the right channel basis, the best fusion protocol may deliberately combine individually imperfect pathways because their shadows partially cancel or spread across nonlimiting channels.

### 10.3 Candidate law 22: Latency-Induced Fusion Oscillation Law

Prior sections emphasized product residence time. The sixth sprint adds control latency. Consider a dimensionless product-shadow state  $P(t)$  and a fusion drive  $\lambda(t)$  suppressed by delayed shadow measurement:

$$\dot{P}(t) = \eta\lambda(t) - \frac{P(t)}{\tau_P}, \quad \lambda(t) = \lambda_0 \exp[-\kappa P(t - \Delta)]. \quad (100)$$

At a steady state  $(P^*, \lambda^*)$ , small perturbations obey

$$\dot{x}(t) = -ax(t) - bx(t - \Delta), \quad a = \frac{1}{\tau_P}, \quad b = \eta\kappa\lambda^*. \quad (101)$$

**Proposed law 22** (Latency-Induced Fusion Oscillation Law). *For the linearized product-shadow controller in Eq. (101), delayed negative feedback is stable for all delays if  $b \leq a$ . If  $b > a$ , the first Hopf boundary occurs at*

$$\Delta_c = \frac{\cos^{-1}(-a/b)}{\sqrt{b^2 - a^2}}. \quad (102)$$

Define the latency number

$$\mathcal{L} = \frac{\Delta}{\Delta_c}. \quad (103)$$

The predicted stability condition is  $\mathcal{L} < 1$ . Above this boundary, a controller that has the correct sign can generate oscillatory product-shadow debt.

*Proof.* Substitute  $x(t) = e^{st}$  into Eq. (101) to obtain

$$s + a + be^{-s\Delta} = 0. \quad (104)$$

At the Hopf boundary,  $s = i\omega$ . Separating real and imaginary parts gives

$$a + b \cos(\omega\Delta) = 0, \quad \omega - b \sin(\omega\Delta) = 0. \quad (105)$$

A real crossing requires  $b > a$ , with  $\omega = \sqrt{b^2 - a^2}$  and Eq. (102).  $\square$

#### 10.4 Virtual experiment 22: delayed product-shadow control

The delayed system in Eq. (100) was simulated for 3,000 randomly sampled parameter settings. The linearized latency boundary was computed for each setting and compared with late-time oscillation amplitude.

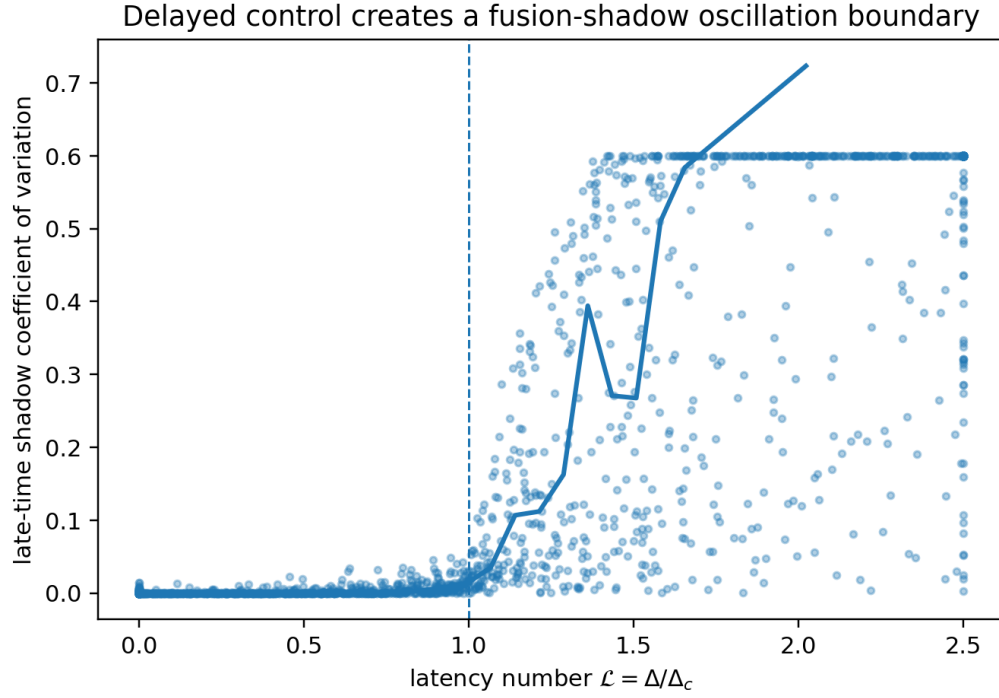


Figure 40: Late-time product-shadow coefficient of variation versus latency number. The dashed line is the predicted linearized boundary  $\mathcal{L} = 1$ .

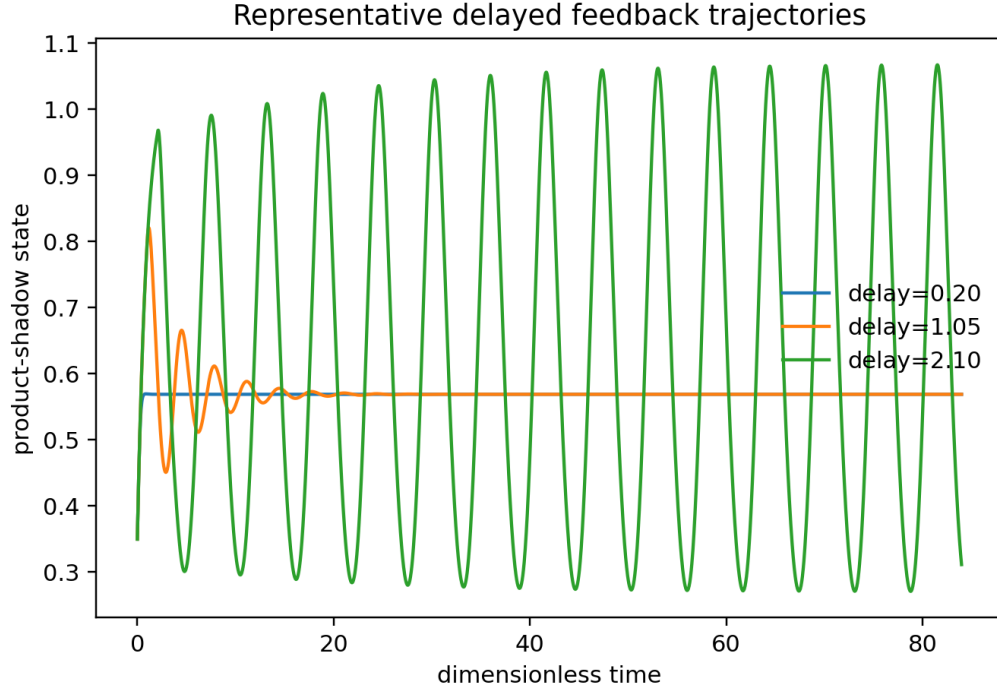


Figure 41: Representative product-shadow trajectories for short, near-critical, and supercritical delays in the same nonlinear feedback system.

The latency boundary was highly predictive in this toy system. Of all sampled cases, 40.8% were above the predicted boundary and 39.9% were oscillatory by the chosen late-time coefficient-of-variation criterion. Below the boundary, only 1.5% were oscillatory. Above the boundary, 95.5% were oscillatory. The median shadow coefficient of variation increased from approximately  $9.5 \times 10^{-6}$  below the boundary to 0.723 above it. Median useful output was also lower above the boundary.

**Interpretation.** Fusion control should not be described only by feedback sign or gain. It needs a latency budget. A system may have a correct suppressive response to product shadow and still generate shadow bursts if measurement, inference, actuation, or transport is too slow.

## 10.5 Candidate law 23: Sparse Contact Topology Law

Fusion is often encouraged by increasing contact area, contact time, or overlap. The following reduction shows why this can fail. Let  $m$  be the number of active microscopic contact sites out of  $M$  possible sites. Suppose useful benefit grows sublinearly or linearly with active contacts, while cross-contact product shadow grows with a larger exponent:

$$V(m) = bm^\beta - am^\gamma, \quad 0 < \beta < \gamma, \quad 1 \leq m \leq M. \quad (106)$$

**Proposed law 23** (Sparse Contact Topology Law). *For Eq. (106), the continuous optimal contact count is*

$$m^* = \left( \frac{b\beta}{a\gamma} \right)^{1/(\gamma-\beta)} \quad (107)$$

clipped to  $[1, M]$ . Full contact is optimal only when  $m^* \geq M$ . When  $1 < m^* < M$ , the best fusion interface is sparse even though additional contact has positive local benefit at small  $m$ .

*Proof.* Treating  $m$  as continuous,

$$\frac{dV}{dm} = b\beta m^{\beta-1} - a\gamma m^{\gamma-1}. \quad (108)$$

Setting this derivative to zero gives Eq. (107). Since  $\gamma > \beta$ , the shadow term eventually dominates.  $\square$

## 10.6 Virtual experiment 23: contact-area paradox

A Monte Carlo test sampled 120,000 contact-topology systems with random  $M, b, a, \beta, \gamma$ . The optimum was rounded to neighboring integer contact counts and compared with single-contact and full-contact baselines.

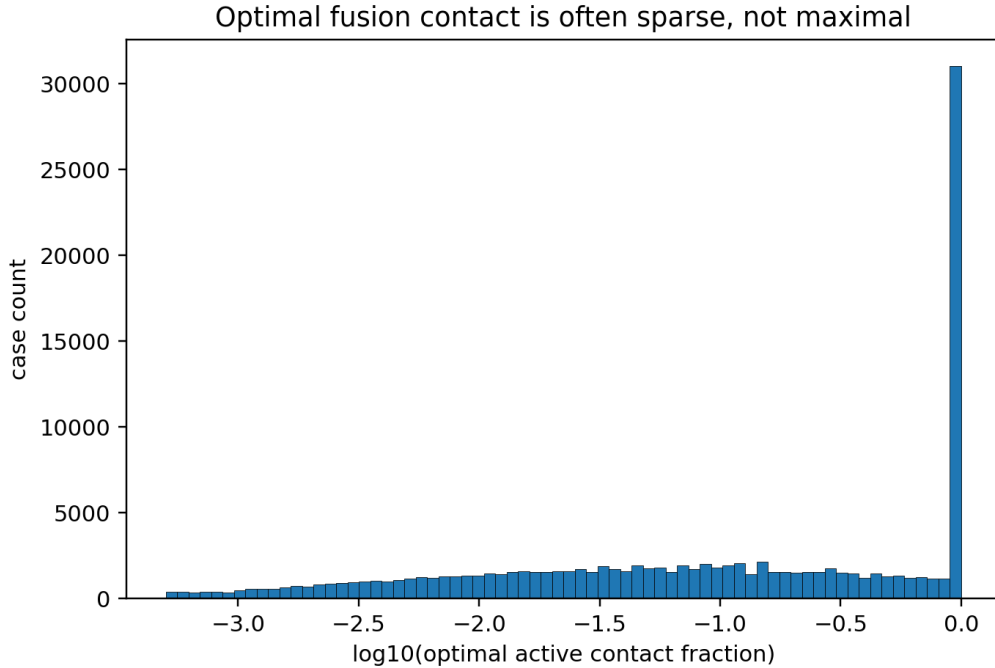


Figure 42: Distribution of optimal active contact fraction  $m^*/M$  in the sparse-contact toy model.

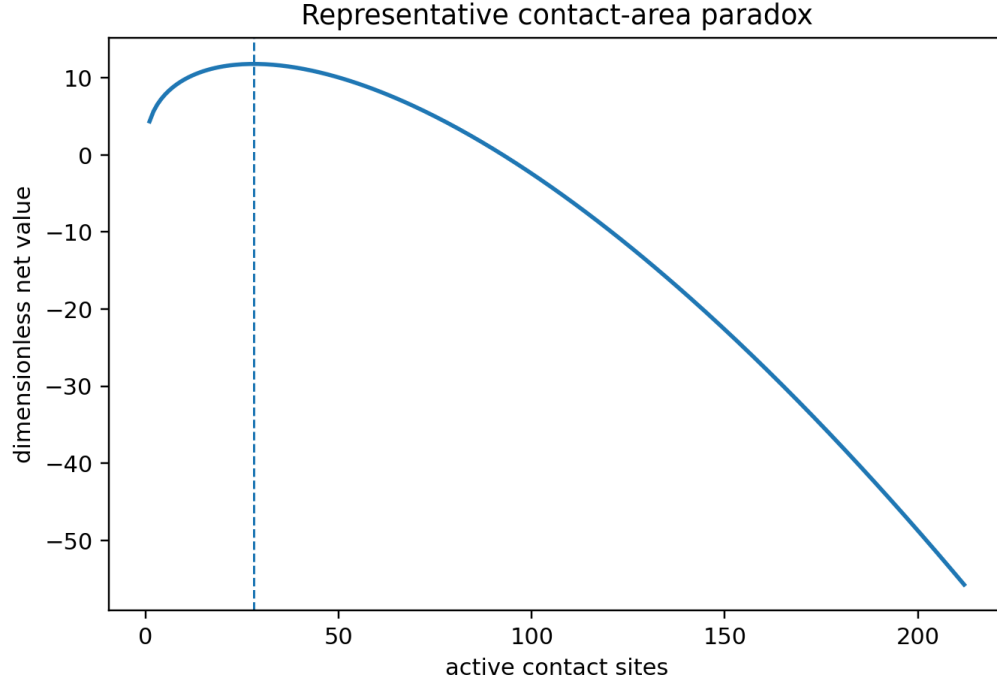


Figure 43: Representative value curve showing a finite sparse-contact optimum. The dashed line marks the selected active-contact count.

An interior sparse optimum occurred in 57.7% of sampled cases. Full contact was suboptimal in 74.8% of cases. The median optimal active-contact fraction was 11.1%. The large value gains in the upper tail reflect samples with severe superlinear shadow under full contact; they should be interpreted qualitatively, not as physical magnitudes.

**Interpretation.** The blindspot is the assumption that interface area is automatically good. When product shadow couples across contacts, fusion should be patterned: punctate, porous, islanded, spotwise, or sparse-attentional rather than maximally overlapped.

## 10.7 Candidate law 24: Hierarchical Fusion Granularity Law

Many fusion tasks require combining not two entities but many. Let  $N$  be the total number of units to be fused. Suppose units are first fused into modules of size  $m$ , after which modules are integrated at a higher level. A simple two-scale cost is

$$C(m) = ANm^{p-1} + BN^q m^{-q}, \quad 1 \leq m \leq N, \quad p > 1, \quad q > 0. \quad (109)$$

The first term is within-module fusion shadow: larger modules have more internal convex cost. The second term is cross-module coordination cost: too many small modules require expensive higher-level integration.

**Proposed law 24** (Hierarchical Fusion Granularity Law). *For Eq. (109), the continuous optimal module size is*

$$m^* = \left( \frac{BqN^{q-1}}{A(p-1)} \right)^{1/(p+q-1)} \quad (110)$$

clipped to  $[1, N]$ . The optimum is atomized when  $m^* \leq 1$ , flat when  $m^* \geq N$ , and genuinely hierarchical when  $1 < m^* < N$ .

*Proof.* Differentiate Eq. (109):

$$\frac{dC}{dm} = AN(p-1)m^{p-2} - BqN^q m^{-q-1}. \quad (111)$$

Setting this derivative to zero yields Eq. (110). The second derivative is positive at the stationary point for the stated exponents.  $\square$

## 10.8 Virtual experiment 24: finite-granularity modular fusion

The two-scale cost in Eq. (109) was tested in 80,000 random systems with  $N$  ranging from approximately 16 to 5,000 and random coefficients and exponents. The optimum was compared with the two extremes: atomized fusion ( $m = 1$ ) and flat all-at-once fusion ( $m = N$ ).

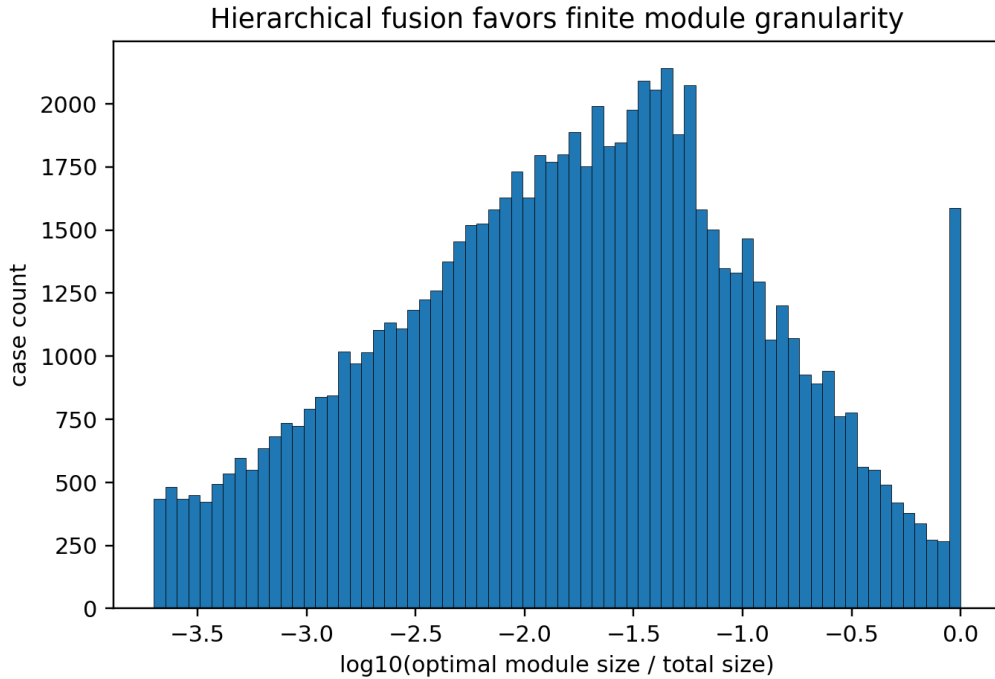


Figure 44: Distribution of optimal module fraction  $m^*/N$  in the hierarchical-granularity toy model.

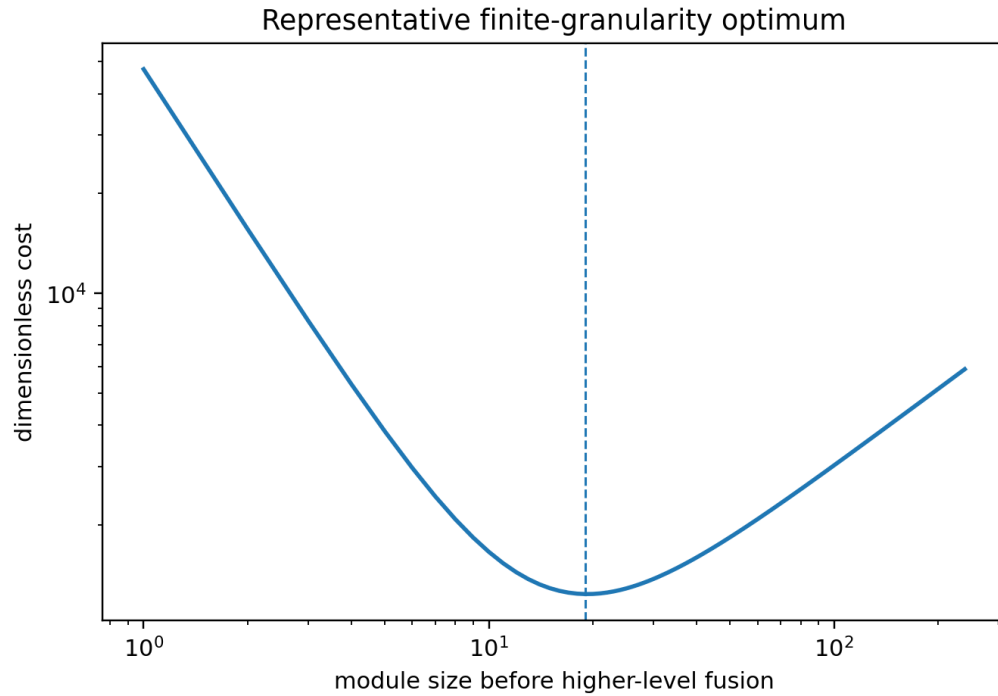


Figure 45: Representative finite-granularity optimum. The optimum beats both atomized and flat fusion in this case.

A genuinely hierarchical interior optimum appeared in 69.1% of sampled cases, and the same fraction beat both atomized and flat extremes. The median optimal module fraction was 1.89%, corresponding to a median optimal module size of four units in the sampled distribution. Median savings relative to the better extreme were 41.1%.

**Interpretation.** The strongest result is that fusion architecture has a scale variable. The useful question may not be whether to fuse or not fuse, but what module size should be fused before higher-level integration. This law is a scale analogue of incomplete fusion: the optimum is neither separated atoms nor complete flat union, but finite modules.

## 10.9 Summary of sixth virtual tests

Table 7: Summary of sixth-sprint virtual experiments.

| Experiment               | Toy setting                                                 | Main result                                                                                                       | Candidate discovery                                          |
|--------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Shadow-vector balancing  | Five pathways, four shadow channels, simplex optimization   | Mixtures beat the best single pathway in 78.6% of cases; residual shadow fell below every single pathway in 59.6% | Product shadow is vectorial and can be balanced or cancelled |
| Latency instability      | Delayed product-shadow feedback                             | Oscillation rate was 1.5% below and 95.5% above the predicted latency boundary                                    | Correct feedback can become a shadow generator when delayed  |
| Sparse contact           | Contact count with sublinear benefit and superlinear shadow | Interior sparse optima occurred in 57.7%; full contact was sub-optimal in 74.8%                                   | Maximal contact area can be worse than patterned contact     |
| Hierarchical granularity | Two-scale fusion of $N$ units into modules of size $m$      | Interior module-size optima occurred in 69.1% and beat both extremes                                              | Fusion has an optimal scale, not only an optimal degree      |

The sixth sprint adds four design variables to the theory:

$$\boxed{\bar{h} = Hw, \quad \mathcal{L} = \Delta/\Delta_c, \quad \rho_c = m/M, \quad \mu = m/N.} \quad (112)$$

Here  $\bar{h}$  is residual shadow geometry,  $\mathcal{L}$  is latency number,  $\rho_c$  is active contact fraction, and  $\mu$  is modular granularity.

## 10.10 Targets for further study from the sixth sprint

**Target 1: shadow-vector metrology.** Build measurement protocols that decompose product shadow into channel vectors rather than scalar loss. In plasma systems this could mean decomposing by ash, heat-exhaust, wall-load, instability, and fuel-cycle channels. In materials it could mean residual stress, brittle phase, porosity, and thermal distortion. In information fusion it could mean bias, variance, calibration error, latency, and missing-modality fragility.

**Target 2: anti-correlated shadow-pair search.** Search for pathway pairs or operating regimes with  $h_i^T h_j < 0$  in the chosen shadow-channel basis. The immediate goal is not to maximize either pathway alone, but to find mixtures satisfying Eq. (97).

**Target 3: latency-number calibration.** Estimate  $a$ ,  $b$ , and  $\Delta$  for real feedback loops and test whether  $\mathcal{L}$  predicts oscillation, disruption, negative transfer, or overshoot. This target is especially relevant to plasma control, closed-loop sensor fusion, robotic perception, and biological fusion inhibition.

**Target 4: patterned-contact experiments.** Replace maximal-contact designs with controlled sparse-contact patterns and test whether product shadow falls faster than useful fusion rate.

Candidate platforms include dissimilar-material spot arrays, patterned fusogen clusters, condensate seeding fields, and sparse attention/gated neural fusion.

**Target 5: module-size sweeps.** In any many-agent fusion system, sweep module size  $m$  explicitly before optimizing other variables. Search for finite-granularity optima and compare against flat all-at-once integration.

### 10.11 Possible applications of the sixth-sprint findings

**Nuclear fusion.** The shadow-vector law suggests searching for mixed operating regimes that distribute heat, ash, wall load, and instability risk across channels rather than maximizing one scalar gain. The latency law suggests that real-time plasma control should report latency numbers alongside gains and prediction accuracy.

**Information fusion and machine learning.** The shadow-vector law maps directly onto mixture-of-experts and modality-fusion systems: modalities should be weighted by complementarity of their error and latency vectors, not only by standalone accuracy. Dynamic stage-selection methods such as disagreement-based fusion can be interpreted as adaptive movement along the sparse-contact and granularity axes [42].

**Materials joining.** Sparse-contact topology suggests islanded interlayers, patterned diffusion paths, and spot arrays that allow stress relaxation or product escape between active sites. Hierarchical granularity suggests joining submodules before final assembly when flat joining would accumulate convex residual stress.

**Biological and condensate fusion.** Patterned fusogen clusters, punctate hemifusion zones, and controlled coalescence schedules may reduce aging or pathological merger. Condensate work showing time-dependent activity loss under aging makes shadow-vector and latency measurements natural targets [44].

**Organizational and knowledge fusion.** The granularity law predicts that many knowledge systems should not merge all components into one flat structure. Small modules should first stabilize internal language, then fuse upward through a controlled interface.

### 10.12 Sixth-sprint limitation statement

The sixth-sprint models are dimensionless reductions. They establish mathematical and computational plausibility, not empirical truth. The shadow-vector model assumes that product effects can be embedded in a linear channel basis. The latency law uses a single delayed feedback loop. The sparse-contact law assumes power-law benefit and shadow. The granularity law uses a two-scale cost. Each assumption should be replaced by domain-specific physics, biology, materials science, or statistical learning before practical decisions are made.

## 11 Seventh virtual sprint: sequencing, scouting, reciprocal scaffolds, and scarce allocation

The seventh sprint searched for blindspots that appear after the main variables of earlier sections have already been recognized. The focus was not on another kind of barrier alone, but on operational

choices that can silently dominate the final outcome: the order in which parts are fused, whether uncertain fusion attempts should be scouted before commitment, whether candidates should be ranked as isolated objects or reciprocal pairs, and how scarce scaffolds should be distributed across many candidate interfaces. These four topics are common across physical, biological, materials, and information-fusion systems, but they are often treated as implementation details rather than as first-order theory variables.

All tests in this section are dimensionless toy models. They are retained because they expose mathematically clean failure modes: non-associativity, value-of-information thresholds, hidden pair complementarity, and water-filling allocation. The results should be read as candidate principles and experiment generators.

### 11.1 Candidate law 25: Non-Associative Fusion Sequencing Law

A fusion operation is often written as if it were associative:

$$(A \oplus B) \oplus C = A \oplus (B \oplus C). \quad (113)$$

This equality can fail whenever intermediate fusion products alter the next barrier or product-shadow state. Let component  $i$  have an interface coordinate  $x_i \in \mathbb{R}^d$  and a product-shadow vector  $s_i \in \mathbb{R}^m$ . For a sequence  $\pi$  of  $n$  fusions, define the path cost

$$L(\pi) = \sum_{t=2}^n \left[ \|x_{\pi_t} - \bar{x}_{\pi_{<t}}\|^p + \alpha \left\| \sum_{r<t} s_{\pi_r} + s_{\pi_t} \right\|^2 \right], \quad p > 1, \quad (114)$$

where  $\bar{x}_{\pi_{<t}}$  is the centroid of the already fused set. The first term is a convex mismatch action; the second is a cumulative product-shadow penalty. Define the order penalty

$$\Omega_{\text{ord}} = \frac{\mathbb{E}_{\pi}[L(\pi)]}{\min_{\pi} L(\pi)}. \quad (115)$$

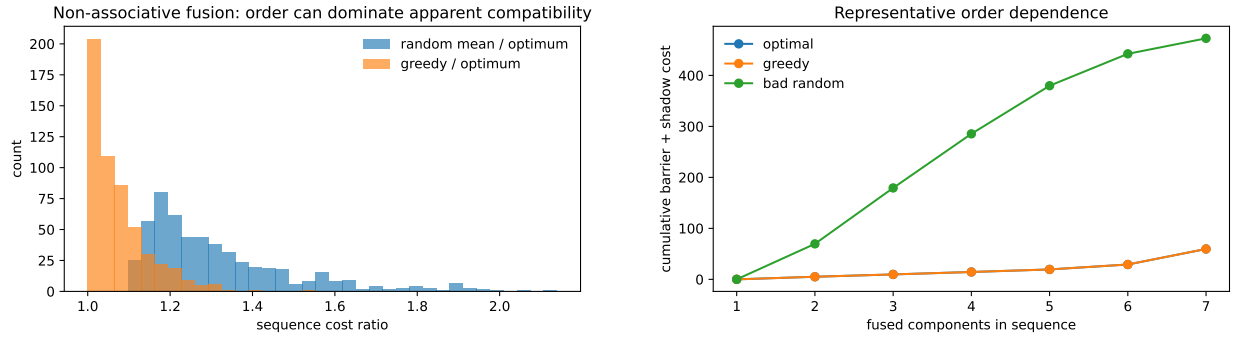
**Proposed law 25** (Non-Associative Fusion Sequencing). *When either mismatch action is convex or product shadow is cumulative and convex, useful fusion is generally path-dependent. The correct design object is not merely a set of entities to fuse, but an ordered fusion path. Low-shadow bridge components and shadow-canceling components should generally be fused before high-shadow or high-curvature components, unless alignment memory imposes the opposite constraint.*

The proposed law is not a statement that every microscopic operation is non-associative. It is a statement about useful fusion under finite product-shadow residence and finite alignment adaptation. In reactor engineering this suggests that ramp order, fueling order, heating order, impurity state, and exhaust conditioning can matter even when the same final nominal plasma state is targeted. In materials joining it suggests that layer order and thermal history can matter even with the same final composition. In information fusion it suggests that early fusion, late fusion, and staged fusion need not be equivalent.

### 11.2 Virtual experiment 25: exact sequencing search

The virtual test generated 550 seven-component systems with random interface coordinates and three-dimensional shadow vectors. For each case, Eq. (114) was minimized exactly by dynamic

programming over subsets and compared with a greedy path and the mean of 50 random paths. The median random sequence cost was 1.265 times the optimum; the 90th percentile was 1.598 times the optimum. A simple greedy sequence was much closer, with median ratio 1.051, and was within 10% of the exact optimum in 74.4% of cases. Random sequencing exceeded twice the optimum in 1.45% of cases.



(a) Distribution of random and greedy sequence costs relative to the exact optimum.

(b) Representative cumulative cost traces for optimal, greedy, and bad random orders.

Figure 46: Seventh-sprint sequencing test. Even with the same components and same final fused set, the path can change barrier-shadow cost substantially.

The exceptional feature of this result is that the same final set of components can have a materially different cost solely because of the order of arrival. This makes “fusion recipe” a separate design axis from “fusion composition.”

### 11.3 Candidate law 26: Scout-Fusion Value Law

Fusion attempts often proceed as if the only options are commit or refuse. A third option is to perform a small probe, diagnostic shot, pilot join, short co-culture, low-dose exposure, or cheap multimodal test before committing to full fusion. Let the unknown net value of a fusion attempt be  $Y \sim \mathcal{N}(\mu, \sigma^2)$ . A scout effort  $e$  observes

$$Z = Y + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2/e), \quad (116)$$

with cost  $ce$ . The posterior mean after observing  $Z$  has variance

$$s_e^2 = \sigma^2 K_e, \quad K_e = \frac{\sigma^2}{\sigma^2 + \sigma_\epsilon^2/e}. \quad (117)$$

If full fusion is attempted only when the posterior mean is positive, the expected scouted value is

$$V_{\text{scout}}(e) = s_e \varphi\left(\frac{\mu}{s_e}\right) + \mu \Phi\left(\frac{\mu}{s_e}\right) - ce, \quad (118)$$

where  $\varphi$  and  $\Phi$  are the standard normal density and distribution. The no-scout value is  $V_0 = \max(\mu, 0)$ , so

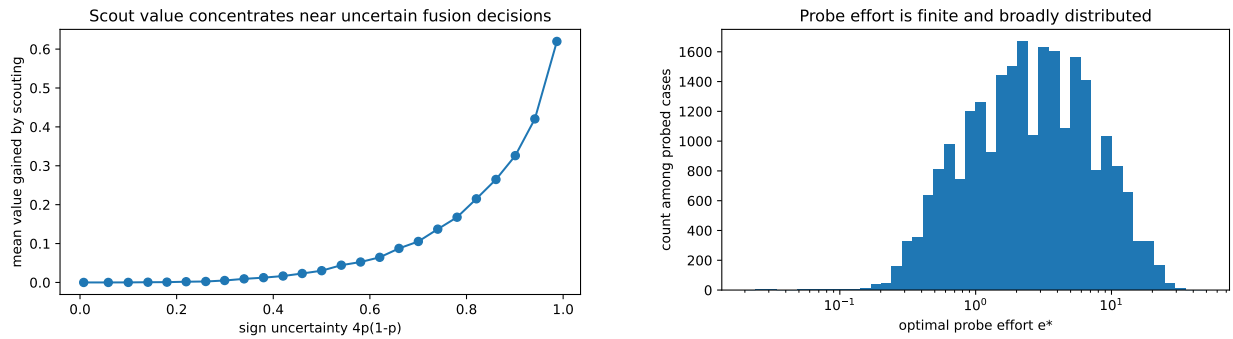
$$e^* = \arg \max_{e \geq 0} \max\{V_0, V_{\text{scout}}(e)\}. \quad (119)$$

**Proposed law 26** (Scout-Fusion Value). *The value of pre-fusion scouting is concentrated near sign uncertainty: cases in which the expected value of fusion may plausibly be positive or negative. Scouting has low value when the fusion attempt is already clearly good, clearly bad, too cheap to matter, or too costly to probe. The missing variable is the pre-commitment value of information.*

This law suggests a measurable target: report not only the cost and yield of full fusion, but also the smallest probe that materially changes the decision to proceed. The probe may be a diagnostic pulse, small batch, micro-weld coupon, low-dose biological perturbation, short multimodal alignment test, or partial deployment.

#### 11.4 Virtual experiment 26: value-of-information scouting

The virtual test sampled 40,000 Gaussian prior fusion values with random uncertainty, probe noise, and probe cost. It optimized Eq. (119) over a finite effort grid. Scouting was beneficial in 61.6% of sampled cases. Among cases where scouting was selected, the median optimal effort was  $e^* = 2.59$  in dimensionless units and the median positive value gained was 0.294. The binned mean gain peaked at sign uncertainty  $4p(1-p) = 0.987$ , where  $p = \Pr(Y > 0)$ , with mean value gain 0.620.



(a) Mean value gained by scouting versus sign uncertainty.

(b) Distribution of optimal probe efforts among cases where scouting is selected.

Figure 47: Seventh-sprint scouting test. The value of a probe is largest near decision boundaries, not necessarily where raw uncertainty alone is largest.

The exceptional finding is that scouting should be treated as a fusion modality. A small noncommittal fusion can be more valuable than either refusing to fuse or committing to full fusion under uncertainty.

#### 11.5 Candidate law 27: Reciprocal Scaffold Complementarity Law

Candidate fusion partners are often ranked by their isolated merit. That can miss pairs in which each partner supplies the scaffold missing from the other. Let candidate  $i$  have benefit  $b_i$ , intrinsic barrier  $a_i$ , scaffold-output vector  $u_i$ , scaffold-need vector  $n_i$ , and shadow vector  $p_i$ . Define pair value

$$V_{ij} = b_i + b_j - a_i - a_j + \gamma (u_i^\top n_j + u_j^\top n_i) - \alpha \|p_i + p_j\|^2 - \eta \|n_i - n_j\|^2. \quad (120)$$

The isolated score is

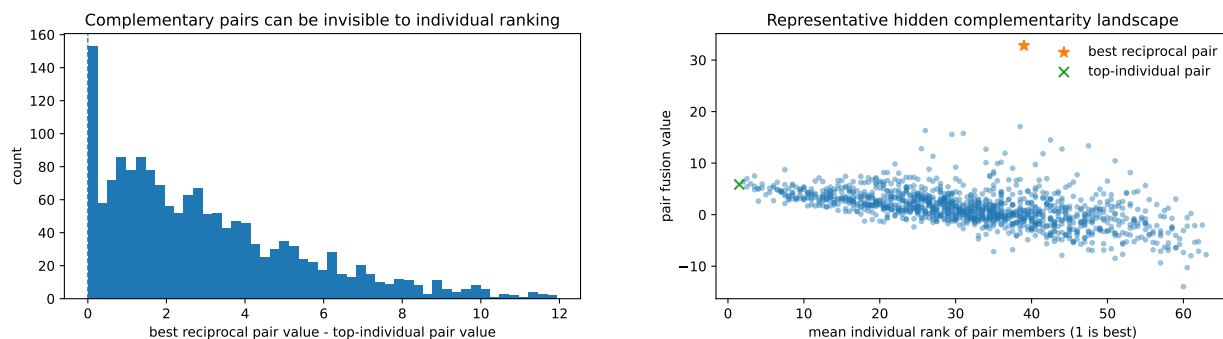
$$v_i = b_i - a_i - \alpha \|p_i\|^2. \quad (121)$$

**Proposed law 27** (Reciprocal Scaffold Complementarity). *The best fusion pair need not contain the best individual candidates. Exceptional pairs occur when reciprocal scaffold terms  $u_i^\top n_j + u_j^\top n_i$  and shadow cancellation overcome mediocre isolated scores. Therefore, discovery searches should rank compositions, pairs, and sequences directly rather than screening only by individual merit.*

The blindspot is especially important where one entity’s product, field, geometry, latent feature, or reaction layer lowers the other’s barrier. This may apply to dissimilar-material interlayers, hybrid catalysts, cell-fusion co-factors, condensate clients, multimodal model components, and mixtures of operating regimes.

## 11.6 Virtual experiment 27: hidden reciprocal pairs

The virtual test generated 1,600 catalogs of 64 candidates with six-dimensional output and need vectors plus three-dimensional product-shadow vectors. It compared the true best pair under Eq. (120) with the pair formed by the top two isolated scores in Eq. (121). The best reciprocal pair beat the top-individual pair in 92.9% of catalogs. The median improvement was 2.46 dimensionless value units, and the 90th percentile improvement was 6.85. In 20.6% of catalogs, the best pair was hidden in the sense that neither member was in the top 20% by isolated score. The median mean individual rank of the best-pair members was 10.5 out of 64.



(a) Value gain of the best reciprocal pair over the top-individual pair.

(b) Representative catalog in which the best pair is not found by individual ranking.

Figure 48: Seventh-sprint reciprocal-complementarity test. Pairwise scaffold exchange can make low-ranking components jointly exceptional.

This result strengthens the complementarity-ridge idea from Sprint 2. Difference alone is insufficient; the difference must be directional. A useful partner should supply what the other needs and should not add the same product shadow.

## 11.7 Candidate law 28: Scarce-Scaffold Allocation Law

Scaffolds, catalysts, fusogens, diagnostic bandwidth, attention, interlayer thickness, active contact sites, and operator time are often scarce. Equal distribution is natural but usually not optimal. Suppose interface  $i$  receives scaffold amount  $s_i \geq 0$  from total budget  $S$  and has response

$$\max_{s_i \geq 0, \sum_i s_i \leq S} \sum_i [w_i \log(1 + \beta_i s_i) - \ell_i s_i]. \quad (122)$$

The Karush-Kuhn-Tucker conditions give

$$s_i^* = \left[ \frac{w_i}{\nu + \ell_i} - \frac{1}{\beta_i} \right]_+, \quad (123)$$

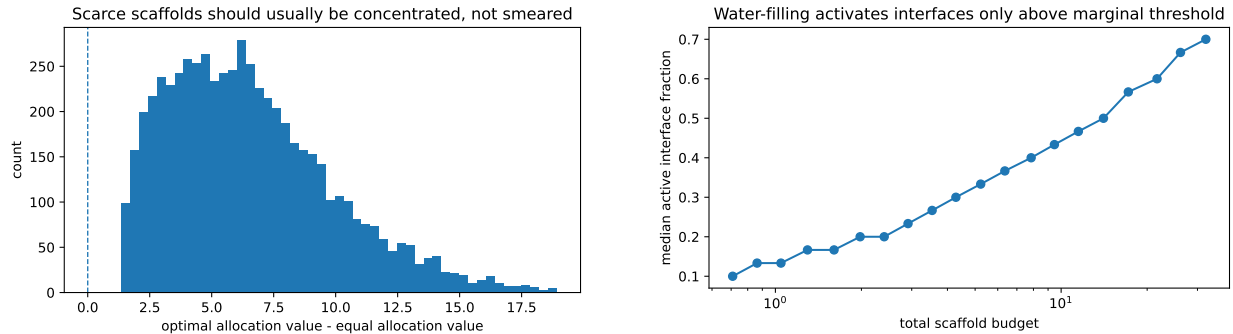
where  $\nu$  is chosen so that the budget constraint is satisfied when binding.

**Proposed law 28** (Scarce-Scaffold Allocation). *When scaffold returns are heterogeneous and saturating, optimal scaffold use is thresholded. Interfaces receive scaffold only if their marginal return exceeds the common shadow-adjusted price  $\nu + \ell_i$ . The generic optimum is a water-filling allocation, not uniform coverage.*

This law connects to the sparse-contact result of Sprint 6, but it is not the same claim. Sparse contact asks how much interface should be active. Scarce allocation asks which of many possible interfaces should receive the limited scaffold that makes activation possible.

## 11.8 Virtual experiment 28: water-filling scaffold allocation

The virtual test sampled 6,000 systems with 30 candidate interfaces each and random  $(w_i, \beta_i, \ell_i, S)$ . The water-filling solution in Eq. (123) was compared with equal allocation and proportional allocation by initial marginal return. The optimal allocation beat equal allocation in every sampled case. The median optimal-minus-equal value was 6.10, and the median gain over proportional allocation was 1.63. The median active interface fraction was 0.300; the 10th percentile was 0.133.



(a) Value improvement of optimal allocation over equal allocation.

(b) Median active interface fraction as total scaffold budget increases.

Figure 49: Seventh-sprint scarce-scaffold allocation test. Optimal scaffolding is selective at low budget and becomes broader only as the marginal threshold falls.

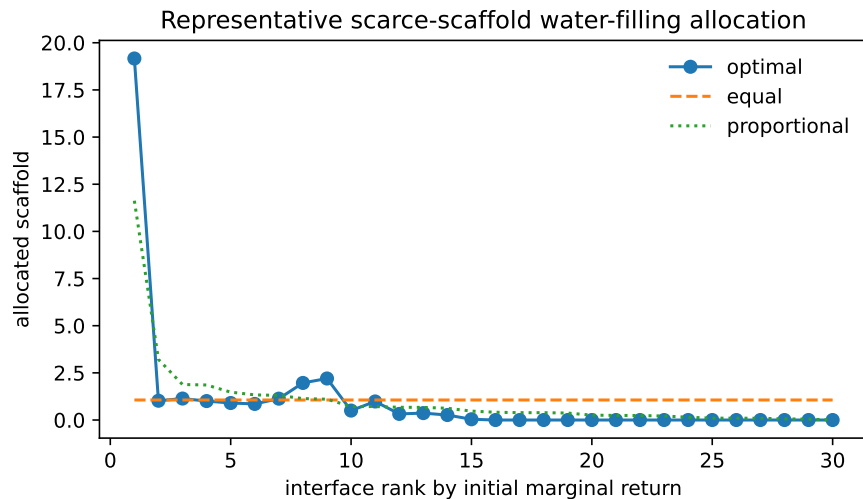


Figure 50: Representative water-filling allocation sorted by initial marginal return. Equal allocation spreads scaffold into low-return interfaces, while the optimum concentrates scaffold on a subset.

## 11.9 Summary of seventh virtual tests

| Experiment                          | Setup                                                                                    | Main result                                                                                           |
|-------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Non-associative sequencing          | 550 seven-component systems; exact subset dynamic program versus greedy and random paths | Random path median cost ratio 1.265; greedy median ratio 1.051; sequencing is a first-order variable. |
| Scout-fusion value                  | 40,000 Gaussian prior decisions with noisy probes and effort cost                        | Scouting beneficial in 61.6% of cases; value peaked at sign uncertainty 0.987.                        |
| Reciprocal scaffold complementarity | 1,600 catalogs of 64 candidates with output, need, and shadow vectors                    | Best pair beat top-individual pair in 92.9% of catalogs; 20.6% hidden-pair rate.                      |
| Scarce scaffold allocation          | 6,000 systems with 30 interfaces and saturating scaffold returns                         | Water-filling beat equal allocation in all sampled cases; median active fraction 0.300.               |

Table 8: Summary of seventh-sprint virtual experiments. Values are dimensionless and model-dependent; the retained results are qualitative design hypotheses.

### 11.10 Exceptional findings from the seventh sprint

Four observations appear especially useful for further theory building. First, fusion composition and fusion sequence are distinct: the same final set can have different cost depending on path. Second, scouting is not merely a diagnostic add-on; it is a mathematically distinct fusion mode that buys option value near decision boundaries. Third, pair complementarity can hide exceptional fusions among mediocre individual candidates. Fourth, scarce scaffolds should usually be priced and allocated by marginal return, not smeared uniformly across all interfaces.

Together these results extend the core thesis. Exceptional fusion is not only scaffolded, incomplete,

and product-managed. It is also ordered, scouted, reciprocally matched, and resource-priced.

### 11.11 Targets created by the seventh sprint

The sprint creates four concrete research targets:

1. **Fusion sequence maps.** Measure whether sequence order changes final performance after controlling for final composition and total drive. Candidate variables are ramp order, layer order, modality order, exposure order, and co-factor order.
2. **Probe-value curves.** For each risky fusion class, estimate  $V_{\text{scout}}(e) - V_0$  as a function of probe effort. The goal is to find the cheapest probe that changes the proceed/refuse decision.
3. **Reciprocal complementarity atlases.** Catalog both what a candidate provides and what it needs. Search pairwise or higher-order compositions for  $u_i^\top n_j + u_j^\top n_i$ , not only for individual scores.
4. **Scaffold price schedules.** Treat scaffold as a scarce resource with a shadow price  $\nu$ . Estimate marginal returns  $w_i \beta_i - \ell_i$  and allocate by threshold rather than by equal coverage.

### 11.12 Possible applications of the seventh-sprint findings

In nuclear and plasma systems, sequencing suggests that ramp trajectories, isotope fueling history, impurity state, wall conditioning, and exhaust preparation may be part of the fusion state rather than merely a route to it. Scouting suggests value in small diagnostic pulses or low-commitment plasma states that update whether a higher-risk scenario should proceed. Scarce allocation suggests that heating, fueling, pellets, magnetic perturbations, diagnostics, and control bandwidth should be treated as limited scaffolds with marginal prices.

In materials joining, sequence maps apply to graded interlayers, thermal cycles, pressure schedules, and diffusion-bonding order. Scout fusion corresponds to coupon-scale tests or micro-joints that update whether a larger join should proceed. Reciprocal scaffold complementarity suggests searching for interlayers whose reaction products reduce, rather than increase, the next barrier.

In biological fusion, sequence effects may appear in the order of fusogen activation, membrane-tension changes, crowding reduction, actin remodeling, and recovery. Scouting corresponds to reversible pre-fusion contacts or low-dose perturbations. Scarce allocation appears as limited fusogen, receptor, membrane area, or cytoskeletal remodeling capacity.

In information fusion, sequencing applies to early, middle, late, and staged multimodal architectures. Scout fusion appears as cheap disagreement scoring, calibration probes, or small routing tests before irreversible feature merging. Reciprocal complementarity suggests model or modality selection by cross-support rather than leaderboards of isolated accuracy. Water-filling gives a formal analogue for attention and compute allocation.

## 12 Eighth virtual sprint: sinks, defusion, boundary layers, and fusion rectification

The eighth sprint searched for a class of blindspots that can be summarized as *anti-fusion operations that improve fusion*. Earlier sections emphasized scaffolds, product management, sequencing, scouting, and allocation. The new question was whether operations that do not look like fusion at all - absorbing products, temporarily separating components, thickening a boundary, or rejecting candidate fusions - can be first-order variables in useful fusion.

All tests in this section are dimensionless toy models. They are retained because they expose four clean mechanisms: sacrificial product sinks, defusion preconditioning, finite boundary-layer thickness, and post-selective rectification. The central discovery target is that a useful fusion system may need deliberately non-fusing subsystems.

### 12.1 Candidate law 29: Sacrificial Sink Law

Consider a fusion rate that is self-shadowed by its own products. A minimal implicit model is

$$R = R_0 \exp[-\gamma(s)R], \quad \gamma(s) = \frac{\gamma_0}{1 + \eta s}, \quad (124)$$

where  $s \geq 0$  is sink capacity,  $\eta$  is sink efficacy, and  $\gamma_0$  is the unsunk product-shadow strength. The sink can represent an exhaust channel, getter, buffer, decoy receptor, chaperone, sacrificial interlayer, quarantine memory, or error-absorbing module. The exact positive solution is

$$R(s) = \frac{W(\gamma(s)R_0)}{\gamma(s)}, \quad (125)$$

where  $W$  is the principal Lambert-W function. If useful value is

$$V(s) = vR(s) - c_1 s - c_2 s^2, \quad (126)$$

then the interior optimum satisfies

$$v \frac{-R(s)^2 \gamma'(s)}{1 + \gamma(s)R(s)} = c_1 + 2c_2 s. \quad (127)$$

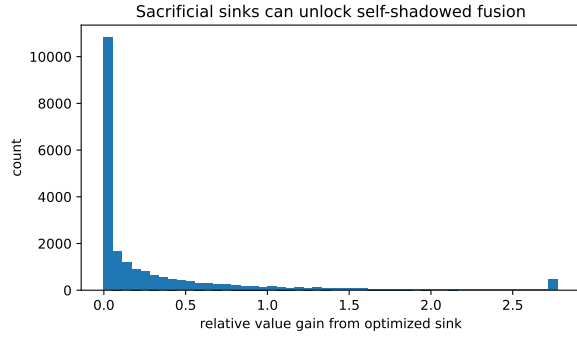
**Proposed law 29** (Sacrificial Sink). *A non-fusing subsystem can increase useful fusion when the marginal value of product-shadow removal exceeds the marginal cost of sink capacity. The sink should not be evaluated by primary yield, but by the counterfactual fusion it unlocks by shortening product residence or lowering product activity.*

This law generalizes the product-shadow correction. It predicts that some apparent dead weight is functional: exhaust surfaces, ash pathways, sacrificial layers, molecular chaperones, scavengers, dampers, error buffers, and rejection memories may be essential fusion components even though they do not participate in the primary merger.

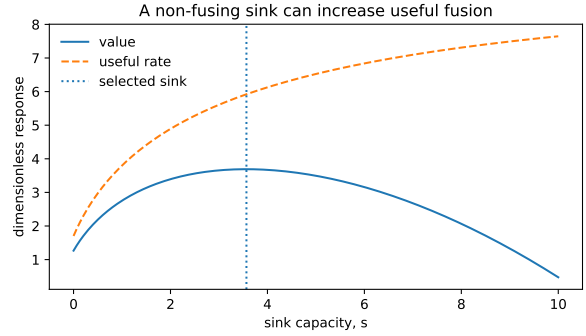
### 12.2 Virtual experiment 29: self-shadowed rate with finite sink capacity

The virtual test sampled 22,000 self-shadowed systems with random raw rate  $R_0$ , shadow strength  $\gamma_0$ , sink efficacy  $\eta$ , value factor  $v$ , and linear/quadratic sink costs. For each case, Eq. (126) was

optimized over sink capacity. A positive sink was selected in 70.0% of sampled cases, and the optimized sink produced more than a 5% relative value gain in 51.8%. The median optimized sink capacity was 0.80 in dimensionless units. The median relative value gain was 6.04%, the median useful-rate gain was 14.3%, and the 90th percentile relative value gain was 104% because some baseline systems were strongly self-poisoned.



(a) Relative value gain from optimized sink capacity.



(b) Representative value and rate curve versus sink capacity.

Figure 51: Eighth-sprint sacrificial sink test. Non-fusing capacity can unlock useful fusion by reducing product self-shadow.

The exceptional observation is that the best fusion component may be a non-fusion component. In a self-shadowed system, increasing drive alone can raise product accumulation, while a sink can raise useful output by making the existing drive less self-poisoning.

### 12.3 Candidate law 30: Defusion Preconditioning Law

Some fusion interfaces begin in a harmful pre-fused state: contaminated, passivated, stressed, correlated, crowded, stuck, saturated, or locked into a bad local minimum. Let  $e \geq 0$  be a temporary defusion or preconditioning effort that lowers the final barrier according to

$$A(e) = \frac{A_0}{(1 + \kappa e)^p}, \quad p > 1. \quad (128)$$

The final value is

$$V(e) = Q \exp[-A(e)] - \mu e - c_2 e^2. \quad (129)$$

The derivative at zero, ignoring the higher-order quadratic penalty, gives a dimensionless defusion number

$$\mathcal{F}_D = \frac{Q e^{-A_0} A_0 p \kappa}{\mu}. \quad (130)$$

Small preconditioning is locally beneficial when

$$\mathcal{F}_D > 1. \quad (131)$$

The full optimum is

$$e^* = \arg \max_{e \geq 0} \left[ Q \exp \left( -\frac{A_0}{(1 + \kappa e)^p} \right) - \mu e - c_2 e^2 \right]. \quad (132)$$

**Proposed law 30** (Defusion Preconditioning). *Before increasing final fusion drive, a system should test whether temporary separation, cleaning, loosening, de-correlation, de-passivation, or controlled disorder reduces the barrier more cheaply. Defusion is useful when its barrier-relief elasticity exceeds its effort cost, approximately when  $\mathcal{F}_D > 1$ .*

In this framing, anti-fusion is not failure. It is an operation in the fusion path. Examples include wall conditioning before plasma operation, removal of surface oxides before joining, relaxation of residual stress, reversible hemifusion before pore formation, condensate dissolution before reassembly, de-correlation before estimator fusion, and low-amplitude annealing before alignment.

## 12.4 Virtual experiment 30: temporary separation before final merger

The virtual test sampled 22,000 systems with random initial barrier  $A_0$ , relief curvature  $p$ , preconditioning efficacy  $\kappa$ , final scale  $Q$ , effort cost  $\mu$ , and quadratic fatigue cost  $c_2$ . Equation (129) was optimized over a finite effort grid. Positive preconditioning was selected in 75.1% of cases and produced more than a 5% relative value gain in 69.2%. The median selected effort was 0.85, and the median relative value gain was 41.9%. The local threshold  $\mathcal{F}_D > 1$  agreed with whether positive preconditioning was selected in 97.2% of sampled systems.

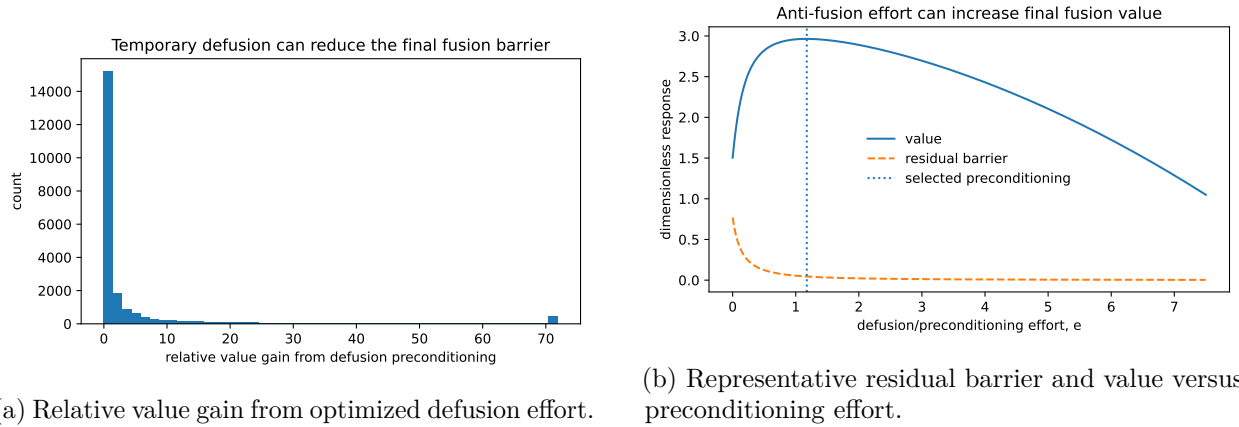


Figure 52: Eighth-sprint defusion test. Temporary anti-fusion can reduce the later fusion barrier enough to increase net value.

The exceptional observation is a reversal of the usual ordering. The optimal path may first move away from fusion in order to make later fusion more useful. This is a formal analogue of cleaning before bonding, deconfounding before inference, and recovery before a high-drive pulse.

## 12.5 Candidate law 31: Finite Boundary-Layer Thickness Law

Many theories idealize the interface as a zero-thickness contact. Suppose a boundary layer of thickness  $h$  spreads mismatch over a finite interphase but also adds volume shadow, coordination cost, or dilution. A minimal cost model is

$$C(h) = \frac{AD^p}{h^{p-1}} + \mu h^q, \quad 1 \leq h \leq H, \quad p > 1, \quad q > 0. \quad (133)$$

Here  $D$  is mismatch distance,  $A$  is mismatch-action scale, and  $\mu$  is layer-shadow scale. The unconstrained optimum is

$$h^* = \left( \frac{(p-1)AD^p}{\mu q} \right)^{1/(p+q-1)}. \quad (134)$$

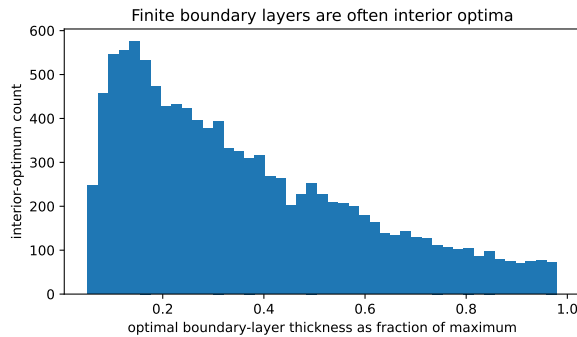
The constrained optimum is  $h^*$  clipped to  $[1, H]$ .

**Proposed law 31** (Finite Boundary-Layer Thickness). *When mismatch relief decreases convexly with boundary-layer thickness but layer volume creates its own shadow, direct contact and maximal buffering are both generally suboptimal. The generic optimum is a finite interphase thickness set by Eq. (134).*

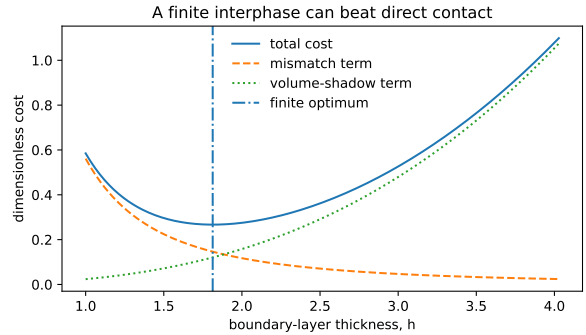
This law unifies graded materials, biological fusion necks, plasma edge regions, catalyst supports, condensate shells, and multimodal adapter layers. A scaffold is often counted as a discrete object; this law treats the scaffold as a finite region. It suggests that fusion design should scan thickness, not merely presence or absence.

## 12.6 Virtual experiment 31: optimal finite interphase

The virtual test sampled 22,000 boundary-layer systems with random mismatch distance  $D$ , action scale  $A$ , layer cost  $\mu$ , exponents  $p, q$ , and maximum feasible thickness  $H$ . Interior optima appeared in 51.1% of sampled cases. A finite optimized layer reduced cost by more than 5% relative to the best endpoint in 38.8% of cases. Among interior-optimum cases, the median cost reduction relative to the best endpoint was 18.0%. The median selected thickness was 26.6% of the maximum feasible thickness.



(a) Interior optima by thickness fraction.



(b) Representative total, mismatch, and volume-shadow costs.

Figure 53: Eighth-sprint boundary-layer test. A finite interphase can outperform both direct contact and maximum buffering.

The exceptional observation is that interface thickness behaves like a fusion variable, not a fabrication detail. Too little layer leaves the mismatch concentrated; too much layer becomes a new product-shadow volume.

## 12.7 Candidate law 32: Post-Selective Fusion Rectification Law

Fusion attempts are often heterogeneous. Some candidate mergers have positive value and others have negative value. Let  $X$  be the true net value of a candidate fusion and  $Y$  a diagnostic score. A threshold  $t$  accepts candidates with  $Y > t$ . The expected thresholded value is

$$V(t) = \int_t^\infty q(y) f_Y(y) dy - c_q, \quad q(y) = \mathbb{E}[X \mid Y = y], \quad (135)$$

where  $c_q$  is the diagnostic cost. If  $f_Y(t) > 0$ , then

$$\frac{dV}{dt} = -q(t) f_Y(t). \quad (136)$$

Therefore any interior optimum satisfies

$$\boxed{q(t^*) = 0}. \quad (137)$$

For a Gaussian latent value  $X \sim \mathcal{N}(m, \sigma_X^2)$  and diagnostic  $Y = X + \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ ,

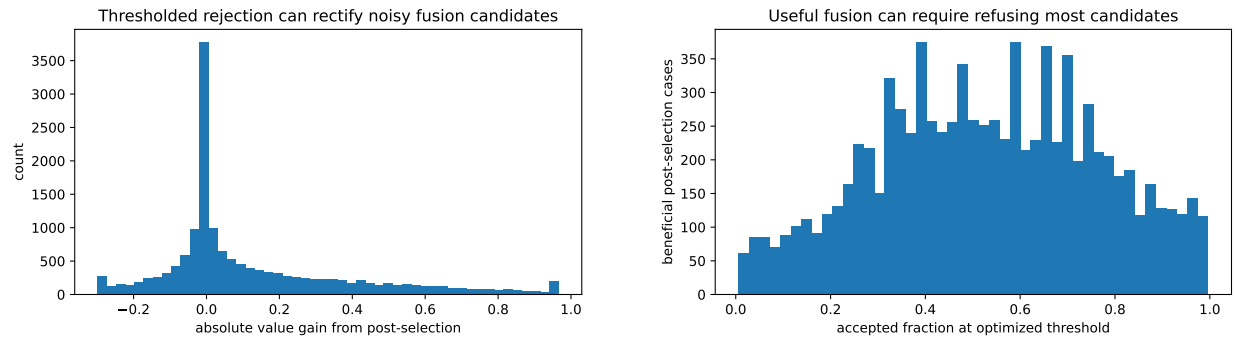
$$q(t) = m + \frac{\sigma_X^2}{\sigma_X^2 + \sigma_\epsilon^2} (t - m). \quad (138)$$

**Proposed law 32** (Post-Selective Fusion Rectification). *When candidate fusion quality is heterogeneous and diagnostics are cheaper than failed mergers, useful fusion can be increased by thresholded refusal. The optimal threshold is not the point that maximizes nominal fusion count, but the point where the marginal accepted candidate has zero conditional value.*

This law converts refusal into an active fusion operation. In physical systems it can represent abort criteria, impurity rejection, defective-coupon screening, or safe ramp-down. In biological and information systems it can represent selective pore completion, quality gates, confidence thresholds, or late-fusion routing.

## 12.8 Virtual experiment 32: thresholded candidate acceptance

The virtual test sampled 16,000 Gaussian candidate populations with random mean value, value variance, diagnostic noise, and screening cost. It optimized the diagnostic threshold and compared against the best no-screening baseline, which either accepted all candidates or refused all candidates. Post-selection beat the best no-screening baseline in 56.0% of cases. Among beneficial cases, the median accepted fraction was 0.530, and the median absolute value gain was 0.204. The median absolute marginal value at the selected threshold was 0.0054, close to the predicted condition  $q(t^*) = 0$ .



(a) Absolute value gain from thresholded post-selection.

(b) Accepted fraction among beneficial post-selection cases.

Figure 54: Eighth-sprint post-selection test. Useful fusion can require refusing a substantial fraction of candidate fusions.

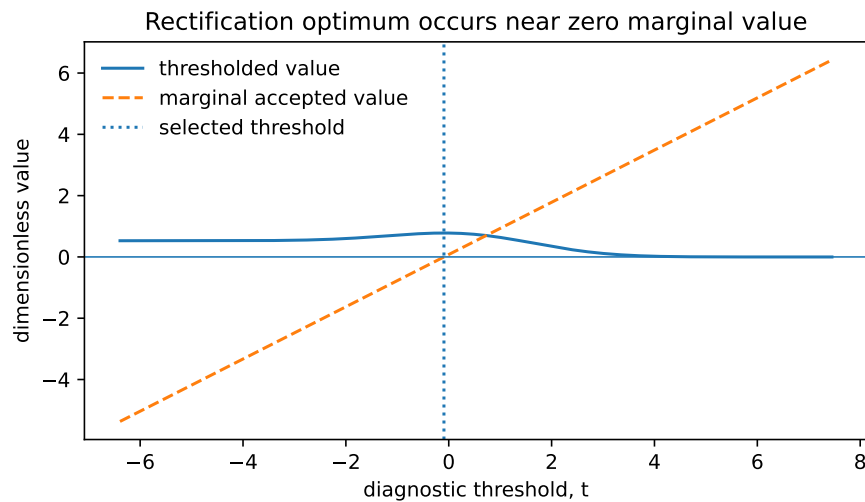


Figure 55: Representative post-selection curve. The optimized threshold occurs near the zero crossing of the marginal accepted value  $q(t)$ .

The exceptional observation is that the optimal fusion system may be a rectifier. It admits positive-value candidate mergers and blocks negative-value candidate mergers. Raw fusion frequency can increase while useful fusion decreases.

## 12.9 Summary of eighth virtual tests

| Experiment                   | Setup                                                          | Main result                                                                               |
|------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Sacrificial sink             | Self-shadowed implicit rate with finite sink capacity          | Positive sink selected in 70.0%; median value gain 6.04%; median useful-rate gain 14.3%   |
| Defusion preconditioning     | Temporary anti-fusion lowers a convex final barrier            | Positive preconditioning selected in 75.1%; threshold $\mathcal{F}_D > 1$ agreed in 97.2% |
| Boundary-layer thickness     | Convex mismatch relief versus volume-shadow cost               | Interior finite layer in 51.1%; median interior cost reduction 18.0%                      |
| Post-selective rectification | Noisy diagnostic threshold for heterogeneous candidate fusions | Post-selection beat no-screening baseline in 56.0%; median accepted fraction 0.530        |

Table 9: Summary of eighth-sprint virtual experiments. Values are dimensionless and distribution-dependent; the retained results are qualitative design hypotheses.

## 12.10 Exceptional findings from the eighth sprint

The sprint produced one unifying observation: *useful fusion may require designed non-fusion*. A sink does not fuse, but can unlock fusion. Defusion moves away from merger, but can lower the later barrier. A boundary layer prevents direct contact, but can distribute mismatch. Post-selection refuses candidate fusions, but can increase total value. These are not exceptions to the theory; they are evidence that the fusion objective is path-and-shadow dependent rather than event-count dependent.

This suggests an expanded principle: the optimal fusion architecture contains both fusion and anti-fusion operators. The aim is not permanent separation, but selective separation at the places where direct merger would amplify barrier, shadow, or negative marginal value.

## 12.11 Targets created by the eighth sprint

The sprint creates four concrete research targets:

1. **Sink dose-response maps.** Add controllable sinks, buffers, exhaust channels, sacrificial phases, or quarantine memories and measure whether useful fusion follows the Lambert-W response of Eq. (125) or another saturating self-shadow model.
2. **Defusion-number assays.** Measure the initial barrier, the local barrier-relief derivative, and the cost of preconditioning. Test whether systems with  $\mathcal{F}_D > 1$  benefit from cleaning, loosening, de-correlation, or stress relaxation before final fusion.
3. **Boundary-layer sweeps.** Fabricate or simulate interphases of multiple thicknesses. Fit Eq. (133) or a domain-specific replacement and test whether finite  $h^*$  predicts the best layer.
4. **Marginal-value post-selection curves.** Estimate  $q(t) = \mathbb{E}[X \mid Y = t]$  for candidate fusion diagnostics. Calibrate the threshold where the marginal accepted event crosses zero rather than maximizing acceptance rate.

## 12.12 Possible applications of the eighth-sprint findings

In nuclear and plasma systems, sacrificial sinks correspond to exhaust, ash removal, impurity control, wall conditioning, and divertor-like functions. Defusion preconditioning corresponds to ramp-down/ramp-up preparation, impurity clearing, surface conditioning, and plasma-state recovery before high-drive operation. Boundary-layer thinking points to the edge and wall interface as a finite design region rather than a zero-thickness loss surface. Post-selection corresponds to abort criteria, pulse gating, and scenario screening.

In materials joining, sink design suggests sacrificial getters or interlayers that absorb brittle phases or contaminants. Defusion preconditioning corresponds to oxide removal, surface activation, stress relief, and transient melting or dissolution before final joining. Boundary-layer thickness directly applies to graded interlayers and diffusion zones. Post-selection corresponds to coupon screening and nondestructive evaluation before full-scale joining.

In biological fusion, sinks include chaperones, decoy receptors, lipid reservoirs, cytoskeletal buffers, and mechanisms that absorb crowding or membrane tension. Defusion preconditioning includes reversible docking, local uncrowding, membrane relaxation, and actin remodeling. Boundary-layer thickness may correspond to hemifusion diaphragms, fusion pores, glycocalyx spacing, and local membrane-cortex geometry. Post-selection appears as regulated pore completion rather than indiscriminate membrane merger.

In biomolecular condensates, sinks correspond to client buffers and aggregate suppressors. Defusion corresponds to transient dissolution before reassembly. Boundary-layer thickness corresponds to shells, interfaces, and multiphase condensate skins. Post-selection corresponds to recruitment thresholds that admit useful clients while excluding aggregation-prone species.

In information and multimodal fusion, sinks correspond to residual channels, uncertainty reservoirs, negative-transfer buffers, and quarantine memories. Defusion corresponds to de-correlation, de-biasing, calibration, and modality-specific preprocessing before fusion. Boundary-layer thickness corresponds to adapter depth and shared-latent dimensionality. Post-selection corresponds to confidence-gated routing, disagreement filtering, and refusal to fuse unreliable modalities.

## 12.13 Eighth-sprint limitation statement

The eighth-sprint models are deliberately minimal. The sink model assumes one scalar self-shadow and one scalar sink. The defusion model assumes monotone barrier relief and simple effort cost. The boundary-layer model assumes power-law mismatch relief and power-law volume shadow. The post-selection model assumes a single diagnostic threshold. The qualitative conclusions should be tested by replacing these forms with domain-specific transport, mechanics, control, and statistical models. The retained claim is narrower: fusion theory should include anti-fusion operations as candidate first-order variables.

## 13 Ninth virtual sprint: compliance partition, anti-coincidence scheduling, reversibility reserves, and dopant windows

The previous sprints emphasized scaffold count, bounded merger, product residence, network structure, sequence, scouting, tail risk, quorum checks, higher-order frustration, and sacrificial sinks.

The ninth sprint asks a more local question: *where should fusion work be paid?* Four new allocation variables were isolated: the compliance distribution of the participants, temporal phase offsets among parallel fusion events, reserve left for reversal after a risky partial merger, and deliberately introduced impurity or dopant dose.

The resulting theme is that exceptional fusion can require asymmetric payment. A soft participant, boundary layer, or adapter should absorb more mismatch than a stiff one. Parallel fusion events should sometimes be de-synchronized so that their product shadows do not coincide. Irreversible commitment should sometimes be delayed even after a favorable partial contact. Purity is not always optimal; a small impurity or dopant can lower barriers or export products, while excessive impurity becomes shadow. These candidates are reductions and virtual tests, not established physical laws.

### 13.1 Compliance-Partition Fusion Theorem

Let a mismatch distance  $D$  be absorbed by  $n$  adaptive elements, such as two participants plus a boundary layer, interlayer, adapter, fusogen field, or scaffold. Element  $i$  has action coefficient  $a_i > 0$  and absorbs deformation or alignment displacement  $x_i \geq 0$ . If the cost is convex,

$$C(\mathbf{x}) = \sum_{i=1}^n a_i x_i^p, \quad p > 1, \quad \sum_i x_i = D, \quad (139)$$

then the optimal allocation is

$$x_i^* = \frac{D a_i^{-1/(p-1)}}{\sum_j a_j^{-1/(p-1)}} \quad (140)$$

and the minimized action is

$$C^* = \frac{D^p}{\left(\sum_j a_j^{-1/(p-1)}\right)^{p-1}}. \quad (141)$$

**Theorem 7** (Compliance-Partition Fusion Theorem). *For convex mismatch action, a direct equal-sharing fusion rule is generally suboptimal. The deformation, adaptation, or alignment burden should be distributed in proportion to inverse compliance power  $a_i^{-1/(p-1)}$ . The softest element should usually absorb the largest share, but all sufficiently compliant elements should participate unless one coefficient dominates.*

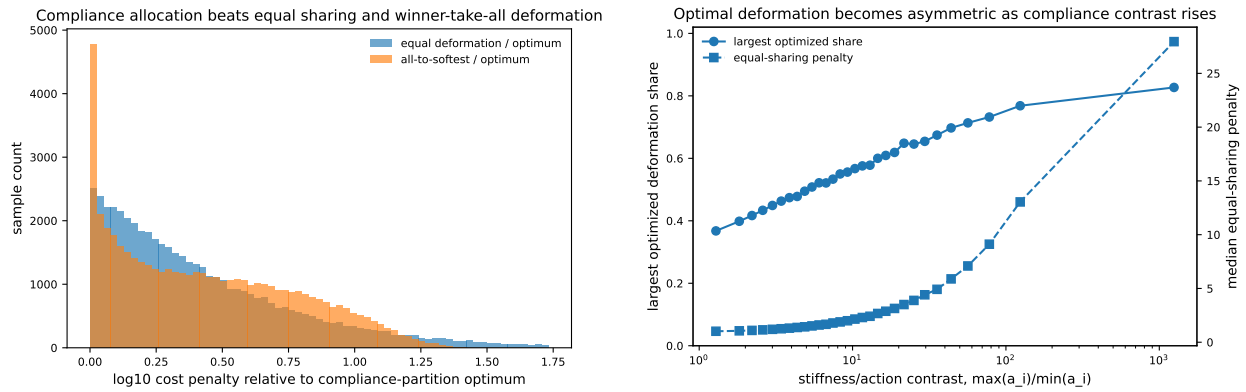
*Proof.* The Lagrangian for Eq. (139) is  $\mathcal{L} = \sum_i a_i x_i^p + \lambda(\sum_i x_i - D)$ . For an interior optimum,  $p a_i x_i^{p-1} + \lambda = 0$ , so all marginal actions are equal and  $x_i \propto a_i^{-1/(p-1)}$ . Normalization gives Eq. (140); substitution gives Eq. (141). Convexity gives global optimality.  $\square$

The theorem is deliberately cross-domain. In materials joining,  $a_i$  may represent stiffness, thermal-expansion resistance, diffusion barrier, or brittleness cost. In biological fusion,  $a_i$  may represent membrane tension, cortex attachment, curvature resistance, or crowding. In information fusion,  $a_i$  may represent the cost of adapting an encoder, changing a latent frame, or distorting a calibrated modality. The law predicts that a fusion architecture with no intentionally compliant element forces expensive deformation into the wrong subsystem.

### 13.2 Virtual experiment 33: compliance partition

A three-element system  $(A, B, S)$  was sampled in 50,000 random cases.  $S$  was interpreted as an optional scaffold or interface layer. Mismatch distance  $D$ , convexity  $p$ , and action coefficients  $a_i$  were randomized. The exact optimum in Eq. (140) was compared with equal deformation and with a winner-take-all rule that assigned all deformation to the softest element.

Equal deformation had median cost  $2.123\times$  the optimum and 90th percentile cost  $8.920\times$  the optimum. The winner-take-all softest-element rule was also suboptimal, with median cost  $2.467\times$  the optimum. Nontrivial partitions, defined by a largest share below 0.95 and all shares above 0.01, appeared in 87.9% of cases. The median scaffold share was 36.7%, and the median largest share was 56.4%. The exceptional result is that both symmetry and winner-take-all adaptation failed: the best fusion was usually an asymmetric but distributed compliance treaty.



(a) Penalty of equal sharing and all-to-softest rules relative to the exact compliance optimum. (b) As stiffness/action contrast rises, the optimal allocation becomes more asymmetric.

Figure 56: Compliance-partition fusion. The virtual test suggests that exceptional interfaces should not merely be soft; they should be soft in the right proportion relative to the participants.

### 13.3 Anti-Coincidence Fusion Scheduling Law

Suppose  $M$  fusion pathways or contact sites generate product loads  $P_i(t)$  and useful output that is approximately independent of their relative phase. If product shadow is convex in instantaneous load,

$$\mathcal{S}[\varphi] = \int_0^T \left( \sum_{i=1}^M P_i(t - \varphi_i) \right)^q dt, \quad q > 1, \quad (142)$$

then product coincidences are disproportionately costly. In the narrow-pulse limit with pulse amplitudes  $a_i$ ,

$$\frac{\mathcal{S}_{\text{sync}}}{\mathcal{S}_{\text{separated}}} \rightarrow \frac{(\sum_i a_i)^q}{\sum_i a_i^q}. \quad (143)$$

For equal amplitudes, this ratio is  $M^{q-1}$ .

**Proposed law 33** (Anti-Coincidence Fusion Scheduling Law). *When useful yield is approximately additive but product shadow is convex in instantaneous product load, parallel fusion events should be*

*phase-staggered to flatten product production. Synchrony is justified only when a coincidence bonus exceeds the convex shadow saved by anti-coincidence.*

A compact decision number is

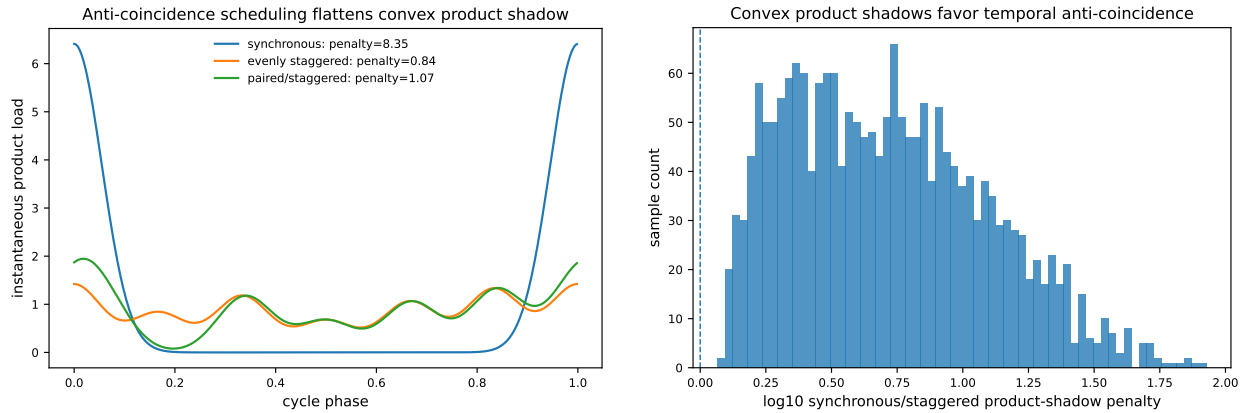
$$\mathcal{C}_{\text{anti}} = \frac{\mathcal{S}_{\text{sync}} - \mathcal{S}_{\text{stagger}}}{\Gamma_{\text{sync}} + \epsilon}, \quad (144)$$

where  $\Gamma_{\text{sync}}$  is the useful complementarity gained by synchronizing and  $\epsilon$  prevents division by zero. Anti-coincidence is predicted when  $\mathcal{C}_{\text{anti}} > 1$ .

### 13.4 Virtual experiment 34: anti-coincidence scheduling

A cycle with three to eight product-producing pathways was simulated. Each pathway produced a periodic Gaussian product pulse with random amplitude. The product-shadow penalty was the cycle average of the summed load raised to exponent  $q$ . Synchronous operation was compared with evenly staggered phases.

Staggering reduced product shadow in all 2,000 sampled cases. The median synchronous/staggered penalty ratio was 4.842, and the 90th percentile ratio was 17.182. The median narrow-pulse prediction in Eq. (143) was 5.624, consistent with the finite-width result after pulse overlap. The result is not that synchrony is never useful; some domains require simultaneity to create the fusion event. The result is that simultaneity should be treated as a costly resource whenever products accumulate superlinearly.



(a) Representative product-load traces for synchronous and staggered schedules.

(b) Distribution of product-shadow penalty reduction from staggering.

Figure 57: Anti-coincidence scheduling. De-synchronization can be a fusion-improving operation when it prevents product-shadow pileup.

### 13.5 Reversibility-Reserve Law

Many fusion systems cross a point after which reversal, disassembly, rollback, or repair becomes expensive. Let  $r \in [0, 1]$  denote a reserve of reversibility preserved during or after partial fusion. A minimal expected-value model is

$$V(r) = B - c_1 r - c_m r^m - \pi L e^{-\eta r}, \quad m > 1, \quad (145)$$

where  $B$  is baseline useful value,  $c_1 r + c_m r^m$  is throughput and control cost,  $\pi L$  is expected hidden shadow loss under full commitment, and  $\eta$  is the efficiency with which reserve suppresses irreversible loss. The optimal reserve solves

$$c_1 + m c_m r^{m-1} = \pi L \eta e^{-\eta r}. \quad (146)$$

The initial condition for selecting a positive reserve is

$$\mathcal{R}_{\text{rev}} = \frac{\pi L \eta}{c_1} > 1. \quad (147)$$

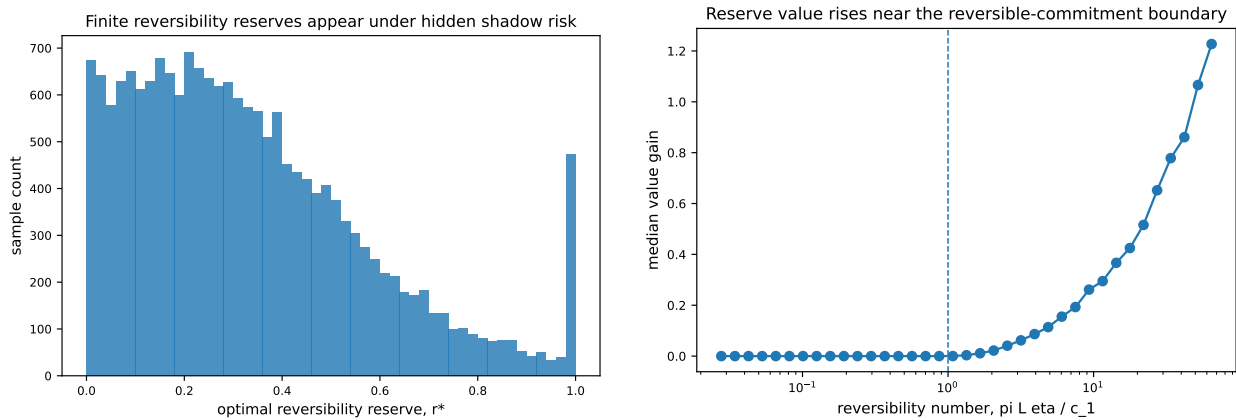
**Proposed law 34** (Reversibility-Reserve Law). *When hidden product-shadow loss is large enough relative to the marginal cost of preserving reversibility, optimal fusion should retain a finite reserve for rollback, defusion, repair, or rerouting. Reversibility is not merely caution; it is a protective scaffold against unknown shadow.*

This law differs from scouting. Scouting acts before commitment. A reversibility reserve remains valuable after a favorable partial test because some shadows only appear after contact, coalescence, heating, training, or operation.

### 13.6 Virtual experiment 35: reversibility reserve

The reversibility model in Eq. (145) was tested in 30,000 random systems. The hidden bad-state probability  $\pi$ , loss scale  $L$ , protection efficiency  $\eta$ , linear reserve cost  $c_1$ , convex reserve cost  $c_m$ , and exponent  $m$  were sampled. The reserve  $r$  was optimized on a grid.

A positive reversibility reserve was selected in 61.7% of sampled cases, and an interior reserve appeared in 57.4%. Conditional on selecting a reserve, the median  $r^*$  was 0.288 and the median value gain was 0.079 in dimensionless value units. The threshold  $\mathcal{R}_{\text{rev}} > 1$  agreed with the grid optimum in 99.2% of cases. The exceptional lesson is that full commitment can be suboptimal even when fusion is expected to be beneficial.



(a) Distribution of selected reversibility reserve. (b) Reserve value rises near the predicted threshold.

Figure 58: Reversibility reserve. The virtual test predicts a broad class of systems where partial reversibility is a productive part of fusion design.

### 13.7 Controlled Impurity/Dopant Window Law

Fusion design often treats impurity, dopant, heterogeneity, noise, or additive concentration as monotone: remove all impurities, or add as much beneficial additive as possible. A minimal counter-model is

$$V(y) = R_0 \exp\left(\frac{\beta y}{K+y} - \alpha y^m\right) - cy, \quad 0 \leq y \leq 1, \quad m > 1. \quad (148)$$

Here  $y$  is dopant or impurity dose,  $\beta y/(K+y)$  is saturating barrier relief or exhaust benefit,  $\alpha y^m$  is superlinear poisoning, dilution, instability, brittleness, or over-regularization, and  $cy$  is direct cost. An interior optimum satisfies

$$\boxed{\frac{\beta K}{(K+y)^2} - \alpha m y^{m-1} - \frac{c}{R_0 \exp\left(\frac{\beta y}{K+y} - \alpha y^m\right)} = 0.} \quad (149)$$

A positive dose is favored near the origin when

$$\boxed{\mathcal{D}_0 = \frac{R_0 \beta}{Kc} > 1.} \quad (150)$$

**Proposed law 35** (Controlled Impurity/Dopant Window Law). *If a small impurity or dopant lowers barrier action or exports products while high dose creates convex shadow, the optimum is a finite impurity window. Both purity and maximal doping can be inferior.*

The law is relevant to impurity-seeded exhaust and detachment concepts in fusion plasmas, where radiating impurities can help manage power exhaust but must be controlled; to interlayers and alloying additions in dissimilar joining; to fusogenic lipid or crowding modifiers in biological membranes; and to noise, dropout, calibration additives, or retrieval modules in information fusion [30, 45, 25, 15, 28].

### 13.8 Virtual experiment 36: controlled dopant window

Equation (148) was tested in 30,000 random systems. Doping was selected in 68.7% of cases, and an interior optimum appeared in 64.6%. Conditional on selecting a dopant, the median optimal dose was  $y^* = 0.210$  and the median value gain was 0.294. The optimized dose beat the high-dose endpoint in 99.5% of cases. The threshold  $\mathcal{D}_0 > 1$  agreed with the grid optimum in 99.2% of cases.

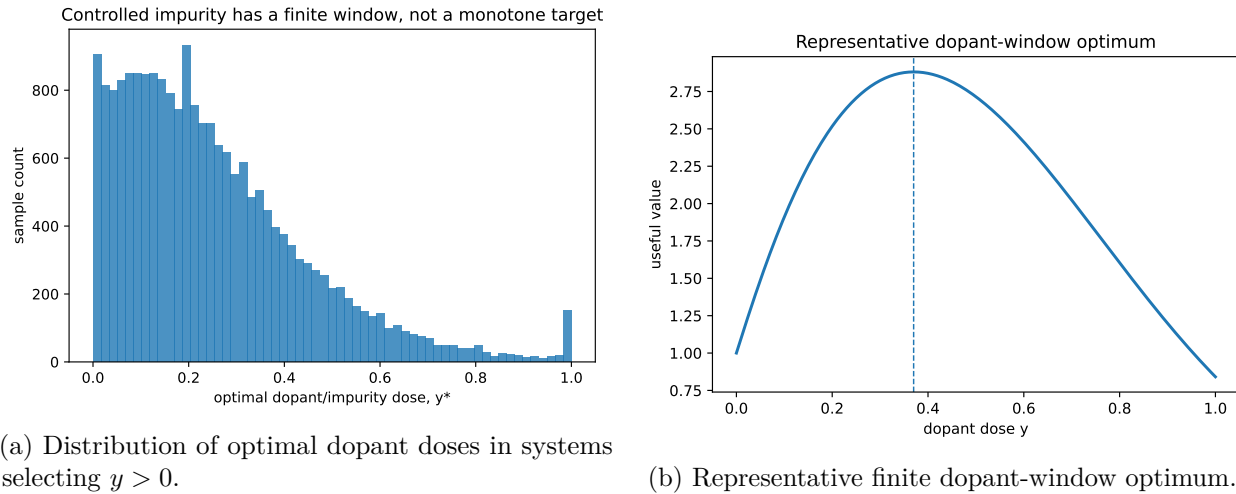


Figure 59: Controlled dopant window. The virtual experiment indicates that impurity is sometimes best treated as a finite scaffold variable rather than as pure contamination.

### 13.9 Ninth-sprint synthesis

| Candidate law               |  | Virtual test |                                  |           | Main result                                                                                                  |
|-----------------------------|--|--------------|----------------------------------|-----------|--------------------------------------------------------------------------------------------------------------|
| Compliance partition        |  | 50,000       | convex allocation systems        | mismatch- | Equal deformation cost was median $2.123\times$ optimum; nontrivial asymmetric partitions appeared in 87.9%. |
| Anti-coincidence scheduling |  | 2,000        | multi-pathway pulse systems      | product-  | Staggering beat synchrony in all sampled cases; median shadow penalty reduction was $4.842\times$ .          |
| Reversibility reserve       |  | 30,000       | hidden-shadow systems            |           | Positive reserve selected in 61.7%; threshold $\mathcal{R}_{\text{rev}} > 1$ agreed with optimum in 99.2%.   |
| Controlled dopant window    |  | 30,000       | barrier-relief/poisoning systems |           | Finite dopant selected in 68.7%; interior optima in 64.6%; high-dose endpoint was inferior in 99.5%.         |

Table 10: Summary of ninth-sprint virtual experiments.

The exceptional finding is a fourfold allocation principle:

Useful fusion improves when mismatch, product time, irreversibility, and impurity are allocated rather than maximized.

(151)

This extends the central thesis. Fusion is not only scaffolded, incomplete, product-managed, and path-aware; it is also *compliance-partitioned*, *anti-coincident* when products are convex, *reversibility-reserved* under hidden shadow, and *dopant-windowed* when small impurity is beneficial but high impurity is poison.

### 13.10 Targets for further study from the ninth sprint

1. **Compliance atlases.** For each fusion interface, estimate action coefficients  $a_i$  for all participants and boundary layers. Test whether observed deformation shares follow Eq. (140).
2. **Anti-coincidence scans.** In repeated, multi-pathway, or distributed fusion systems, vary phase offsets  $\varphi$  while holding average drive fixed. Measure whether product shadow follows Eq. (142).
3. **Reversibility dose-response curves.** Engineer partial rollback, repair, defusion, detachment, or routing reserves. Test the boundary  $\mathcal{R}_{\text{rev}} = 1$ .
4. **Dopant-window maps.** Sweep impurity, additive, interlayer, lipid, noise, dropout, or retrieval dose  $y$ . Fit the small-dose relief parameter  $\beta/K$  and high-dose poisoning term  $\alpha y^m$ .
5. **Interaction tests.** Examine whether compliance partition and dopant dose couple: a dopant may change the action coefficients  $a_i$ , while compliance allocation may change the safe dopant window.

### 13.11 Ninth-sprint applications

**Nuclear fusion and plasma-facing systems.** Compliance partition maps to target, ablator, wall, liquid-metal, blanket, or divertor structures that deliberately absorb stress and mismatch. Anti-coincidence maps to plant-level pulse cadence, exhaust cycling, multi-chamber scheduling, and impurity-seeding timing where product or heat loads are convex in instantaneous rate. Reversibility reserve maps to safe rampdown, detachment control margin, and abortable operating scenarios. The dopant-window law maps to impurity seeding and radiative exhaust: the question is not whether impurity is good or bad, but which finite dose controls exhaust without collapsing useful performance.

**Materials joining and additive manufacturing.** The compliance theorem predicts that interlayer stiffness and geometry should be optimized as deformation-sharing variables, not chosen only by chemical compatibility. Anti-coincidence suggests staggered heat, pressure, current, or deposition pulses when thermal or phase-product shadow is convex. Reversibility reserve suggests tack, reversible bonding, or low-energy pre-joins before irreversible high-energy joining. Dopant windows map to alloying additions, diffusion barriers, getters, and transient liquid phases.

**Biological fusion and condensates.** Compliance partition predicts that successful cell fusion should allocate curvature, membrane tension, cytoskeletal rearrangement, and crowding relief unevenly across the two cells and the fusion site. Anti-coincidence suggests that repeated fusion events or condensate coalescence may be healthier when product loads are temporally dispersed. Reversibility reserve maps to hemifusion arrest, reversible docking, and repairable membrane contacts. Dopant windows map to lipid composition, fusogen density, crowding molecules, ions, and chaperones.

**Information fusion and machine learning.** Compliance partition predicts that adaptation should be concentrated in flexible adapters, calibration layers, or low-risk modalities rather than applied uniformly to all encoders. Anti-coincidence maps to staggered model updates, asynchronous sensor refresh, and load-flattened retrieval or reasoning pipelines. Reversibility reserve maps to reversible routing, rollback, abstention, and late fusion before irreversible latent merger. Dopant windows map to dropout, noise injection, retrieval augmentation, calibration heads, and regularization strength.

### 13.12 Ninth-sprint limitation statement

The ninth-sprint models assume separability: compliance coefficients are fixed, product pulses do not change useful yield, reversibility suppresses shadow exponentially, and dopant benefit and poisoning enter through a scalar dose. Real systems may violate each assumption. Compliance can be history-dependent, anti-coincidence can destroy necessary synchrony, reversibility can itself create shadow, and impurities can open entirely new reaction or failure channels. These violations should be treated as next targets, not as reasons to discard the allocation frame.

## 14 Tenth virtual sprint: witness scaffolds, action geodesics, counter-shadow cofusion, and ruin-limited portfolios

The ninth sprint treated fusion work as an allocation problem across compliance, time phase, reversibility, and dopant dose. The tenth sprint shifts to a different class of blindspots: infrastructure that may not look like fusion at all. Four variables were isolated: a witness scaffold that makes hidden shadow observable, a curved action-minimizing path through scaffold state space, a secondary fusion channel that generates a compensating product, and a portfolio of fusion pathways optimized for repeated survival rather than one-shot expected yield.

This sprint was motivated by several cross-domain observations. Whole-device fusion modeling increasingly requires coupled plasma, boundary, engineering, neutronics, tritium, uncertainty, and provenance workflows rather than isolated peak-performance calculations [36]. Multi-modal perception systems increasingly treat uncertainty as a feature-level object to be quantified and routed, not merely an output confidence [46, 27, 37]. Condensate practice guidelines emphasize phase diagrams, transport, viscosity, FRAP, and microenvironment probes, suggesting that fusion state must be measured dynamically rather than inferred from coalescence alone [47, 39]. Materials joining reviews similarly indicate that interface history, gradients, porosity, residual stress, and patterned contact can dominate final joint value [38, 25, 26]. The resulting hypothesis is that exceptional fusion may require deliberate investment in observability, path geometry, compensating byproducts, and risk portfolios.

### 14.1 Witness-Scaffold Identifiability Law

A scaffold is usually defined by its ability to lower a barrier or align participants. A different scaffold can be defined by its ability to make hidden product shadow observable. Let  $\omega \in [0, 1]$  denote a witness-scaffold fraction. The witness consumes resources and may reduce immediate throughput, but it decreases the posterior variance of a hidden shadow variable:

$$v(\omega) = \frac{v_0}{1 + \kappa\omega}, \quad (152)$$

where  $v_0$  is prior uncertainty and  $\kappa$  is witness informativeness. If local control loss is quadratic in the unobserved shadow error with curvature  $H$ , a minimal value function is

$$V(\omega) = V_0 - \frac{1}{2}H \frac{v_0}{1 + \kappa\omega} - c\omega - d\omega^2. \quad (153)$$

The interior optimum satisfies

$$\frac{1}{2}Hv_0\frac{\kappa}{(1+\kappa\omega^*)^2} = c + 2d\omega^*. \quad (154)$$

The initial witness number is

$$\mathcal{W}_0 = \frac{Hv_0\kappa}{2c}. \quad (155)$$

**Proposed law 36** (Witness-Scaffold Identifiability Law). *When the initial witness number satisfies  $\mathcal{W}_0 > 1$ , useful fusion should allocate a finite fraction of resource to a non-fusing or weakly fusing witness scaffold, even if that witness lowers immediate output. The witness is valuable because it reduces shadow uncertainty before the uncertainty is converted into wrong drive, wrong alignment, wrong merger degree, or wrong product-removal action.*

The exceptional implication is that a diagnostic, reporter, sentinel layer, calibration head, test coupon, or synthetic digital twin can be a true fusion scaffold. It need not contribute to the fusion product directly. It can increase useful fusion by shrinking the hidden state space over which the fusion controller would otherwise act blindly.

## 14.2 Virtual experiment 37: witness scaffolds

Equation (153) was tested in 30,000 random systems. The curvature  $H$ , prior variance  $v_0$ , informativeness  $\kappa$ , linear cost  $c$ , and convex cost  $d$  were sampled across several orders of magnitude. The witness fraction  $\omega$  was optimized on a grid.

Positive witness investment appeared in 70.9% of sampled systems, and an interior witness fraction appeared in 61.1%. Conditional on positive witness use, the median selected witness fraction was 0.236 and the median value gain was 0.240 dimensionless units. The threshold  $\mathcal{W}_0 > 1$  agreed with the grid optimum in 99.8% of cases. The result suggests a blindspot: systems that appear to be underperforming because they lack drive or contact may instead lack identifiability.

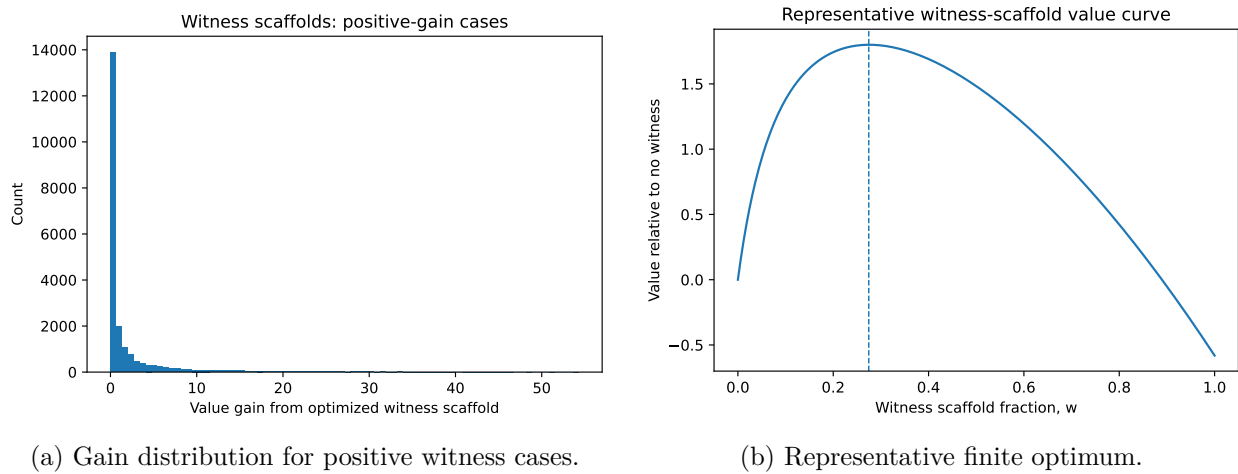


Figure 60: Witness-scaffold identifiability. A witness scaffold lowers expected control loss by reducing hidden-shadow variance, but excessive witness allocation consumes useful resources.

### 14.3 Action-Geodesic Scaffold Law

Previous bridge laws treated scaffold stages as points along a scalar mismatch distance. In a heterogeneous interface, the scaffold is a path through a state space, not a line segment. Let  $\gamma$  be a path through scaffold states,  $a(x) > 0$  a local action density, and  $\kappa_\gamma$  a curvature or abrupt-transition penalty. Define

$$\mathcal{A}[\gamma] = \int_\gamma a(x) ds + \zeta \int_\gamma \kappa_\gamma^2 ds. \quad (156)$$

The optimal scaffold path is

$$\boxed{\gamma^\star = \arg \min_\gamma \mathcal{A}[\gamma].} \quad (157)$$

A detour path  $\gamma_d$  beats a direct path  $\gamma_0$  whenever

$$\boxed{\bar{a}_0 L_0 - \bar{a}_d L_d > \zeta(K_d - K_0),} \quad (158)$$

where  $\bar{a}_i$  is mean action density,  $L_i$  is path length, and  $K_i = \int_{\gamma_i} \kappa_{\gamma_i}^2 ds$ .

**Proposed law 37** (Action-Geodesic Scaffold Law). *In a heterogeneous action landscape, useful fusion follows the minimum-action scaffold geodesic rather than the shortest apparent path. A longer path with lower mean action density should outperform direct contact when Eq. (158) is satisfied.*

This law reframes graded interlayers, pulse trajectories, compression histories, membrane intermediates, and latent alignment curricula as path-finding problems. The path can matter as much as the endpoint.

### 14.4 Virtual experiment 38: action-geodesic scaffold paths

Random two-dimensional action landscapes were generated from high-action obstacles, low-action corridors, and small roughness. Fusion was defined as passage from one side of the interface to the other. A straight path through the midline was compared with a Dijkstra action geodesic that included a small turn penalty. This is not a physical geometry; it is a stress test of the path principle.

In 160 landscapes, the action geodesic beat the straight path in all cases because the sampling deliberately included heterogeneity. The median action saving was 46.1%, the 90th percentile saving was 68.1%, and the median path length ratio was 1.154. Thus the better fusion path was usually longer in ordinary distance but shorter in action. This is an exceptional design inversion: increasing geometric distance can lower fusion cost when it avoids high-shadow or high-barrier regions.

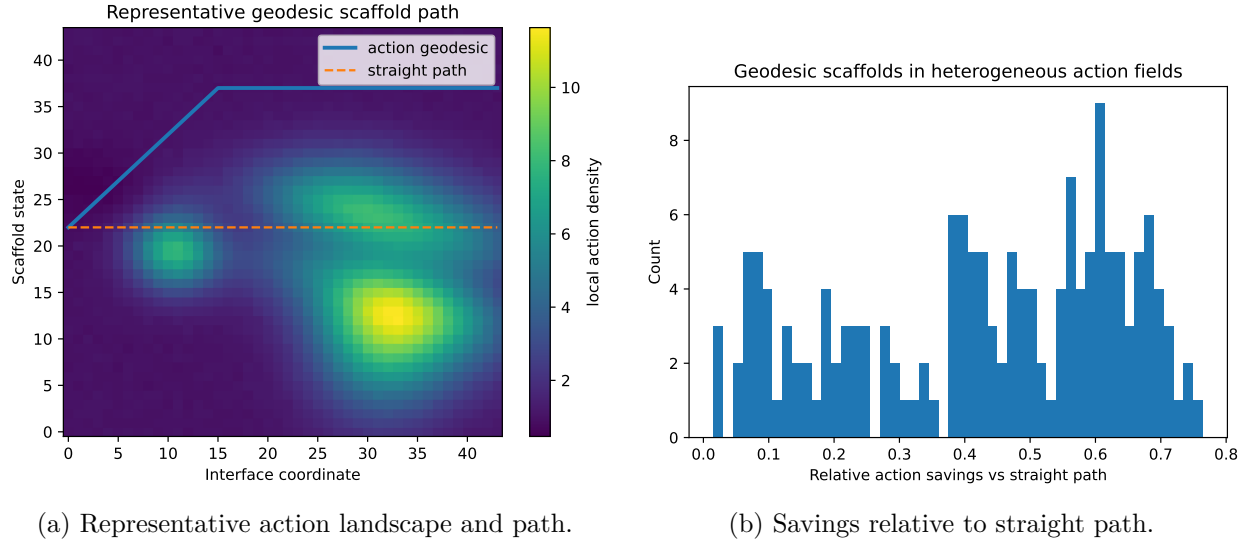


Figure 61: Action-geodesic scaffold paths. The geodesic can detour through low-action states and avoid high-action obstacles.

## 14.5 Counter-Shadow Cofusion Law

A sacrificial sink removes product shadow. A counter-shadow cofusion channel does something different: it produces a second product that offsets the first product. Let primary fusion drive be  $x$  and secondary cofusion drive be  $y$ . In the simplest quadratic model,

$$V(x, y) = bx - ax^2 + \epsilon y - cy - dy^2 - \ell(\alpha x - \beta y)^2, \quad (159)$$

where  $\alpha x$  is primary shadow,  $\beta y$  is counter-shadow,  $\ell$  penalizes residual imbalance,  $c$  and  $d$  are cofusion costs, and  $\epsilon$  allows the cofusion channel to have small direct value or harm. Ignoring box constraints, an interior optimum satisfies

$$\begin{pmatrix} a + \ell\alpha^2 & -\ell\alpha\beta \\ -\ell\alpha\beta & d + \ell\beta^2 \end{pmatrix} \begin{pmatrix} x^* \\ y^* \end{pmatrix} = \frac{1}{2} \begin{pmatrix} b \\ \epsilon - c \end{pmatrix}. \quad (160)$$

If  $x_0$  is the no-cofusion optimum, the initial cofusion number is

$$C_0 = \frac{2\ell\alpha\beta x_0 + \epsilon}{c}. \quad (161)$$

**Proposed law 38** (Counter-Shadow Cofusion Law). *A secondary fusion channel with low or even negative direct yield can increase useful fusion when its product cancels, neutralizes, decorrelates, or exports the primary product shadow. Positive cofusion should appear when  $C_0 > 1$ , provided the secondary product does not introduce a larger independent shadow.*

The cofusion channel is neither a fuel in the usual sense nor a passive sink. It is an active counter-product generator. Candidate analogues include getter-like reaction products in materials, radiative or ash-balancing species in plasma-facing contexts, lipid or chaperone products that reverse crowding or aging, and de-biasing modalities in information fusion.

## 14.6 Virtual experiment 39: counter-shadow cofusion

Equation (159) was tested in 30,000 random systems. The exact constrained optimum on  $0 \leq x, y \leq 1.5$  was found by solving the concave quadratic and checking boundaries. The optimized system was compared with the best no-cofusion setting  $y = 0$ .

Positive cofusion appeared in 65.4% of sampled systems, with an interior cofusion dose in 61.2%. Conditional on positive cofusion, the median cofusion drive was 0.243, the median value gain was 0.098, and the median absolute residual-shadow reduction was 45.1%. The threshold  $C_0 > 1$  matched the constrained optimum in this quadratic experiment to numerical precision. The exceptional lesson is that an apparently inefficient secondary fusion path can be valuable if its product has the right sign in shadow space.

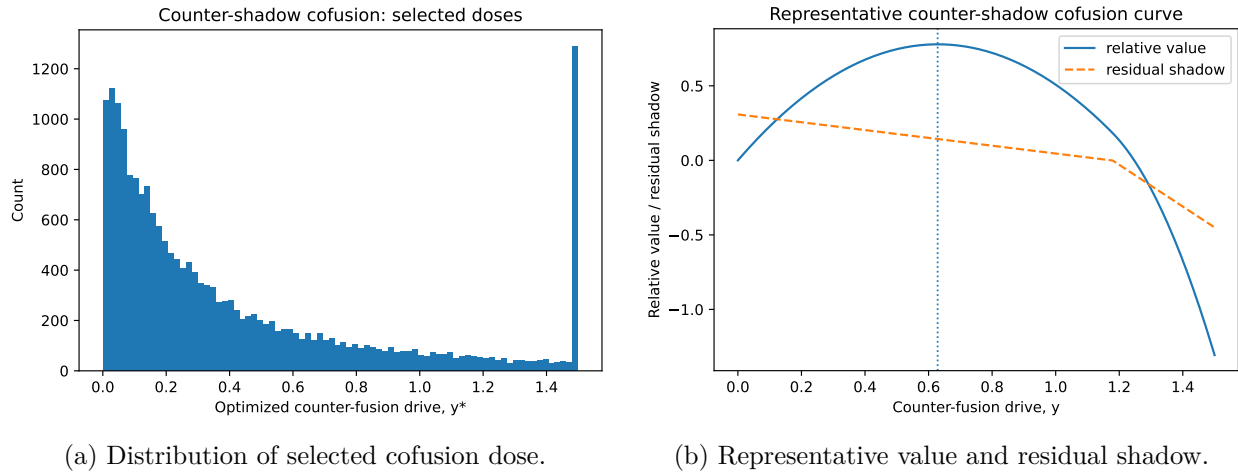


Figure 62: Counter-shadow cofusion. A secondary product can offset primary shadow, but excessive cofusion can overshoot and create a new residual.

## 14.7 Ruin-Limited Fusion Portfolio Law

Many fusion objectives maximize expected yield. Repeated operation, however, is often limited by ruin: rare wall damage, instability, catastrophic model error, irreversible cell damage, brittle crack growth, or locked-in bias. Let  $f_i \geq 0$  be the fraction of effort assigned to pathway  $i$  with  $\sum_i f_i = 1$ . Let  $\mu$  be expected useful return and  $Q$  the covariance or second-moment matrix of shadow losses, including tail-risk surrogates. A minimal repeated-fusion objective is

$$G(f) = \mu^\top f - \frac{\chi}{2} f^\top Q f, \quad f \in \Delta. \quad (162)$$

For an active set  $A$ , the interior optimum is

$$f_A^* = \frac{1}{\chi} Q_A^{-1} (\mu_A - \nu \mathbf{1}), \quad \nu = \frac{\mathbf{1}^\top Q_A^{-1} \mu_A - \chi}{\mathbf{1}^\top Q_A^{-1} \mathbf{1}}, \quad (163)$$

with inactive weights set to zero if the active-set solution is feasible. A portfolio  $f$  beats the highest-mean single pathway  $j$  when

$$\frac{\chi}{2} (Q_{jj} - f^\top Q f) > \mu_j - \mu^\top f. \quad (164)$$

**Proposed law 39** (Ruin-Limited Fusion Portfolio Law). *When product shadows, failures, or distribution shifts compound over repeated fusion cycles, the best fusion strategy can be a diversified portfolio of lower-yield pathways. The portfolio is selected by shadow covariance and tail risk, not by single-shot yield ranking.*

This law is related to but distinct from shadow-vector balancing. Shadow-vector balancing cancels mean residual products. Ruin-limited portfolios reduce variance and tail exposure even when the mean shadow vector is not canceled.

## 14.8 Virtual experiment 40: ruin-limited portfolios

Four-pathway virtual catalogs were sampled. Each pathway had an expected useful return, signed factor loadings that generated a positive-semidefinite shadow covariance matrix, pathway-specific tail surrogates, and a random risk-aversion parameter. The active-set solution of Eq. (163) was compared with the highest-mean pathway and the best single risk-adjusted pathway.

In 6,000 catalogs, a diversified optimum appeared in 97.2% of cases. The median active pathway count was four, and the median normalized entropy was 0.785. The portfolio beat the highest-mean single pathway in 97.2% of cases and beat the best single risk-adjusted pathway in 97.2%. The median positive log-value gain over the highest-mean pathway was 0.154, and the median variance reduction relative to the highest-mean pathway was 73.0%. These percentages are not proposed as universal frequencies; they reflect a deliberately risk-rich virtual environment. The robust qualitative result is that repeated fusion should not be optimized only by the highest expected yield.

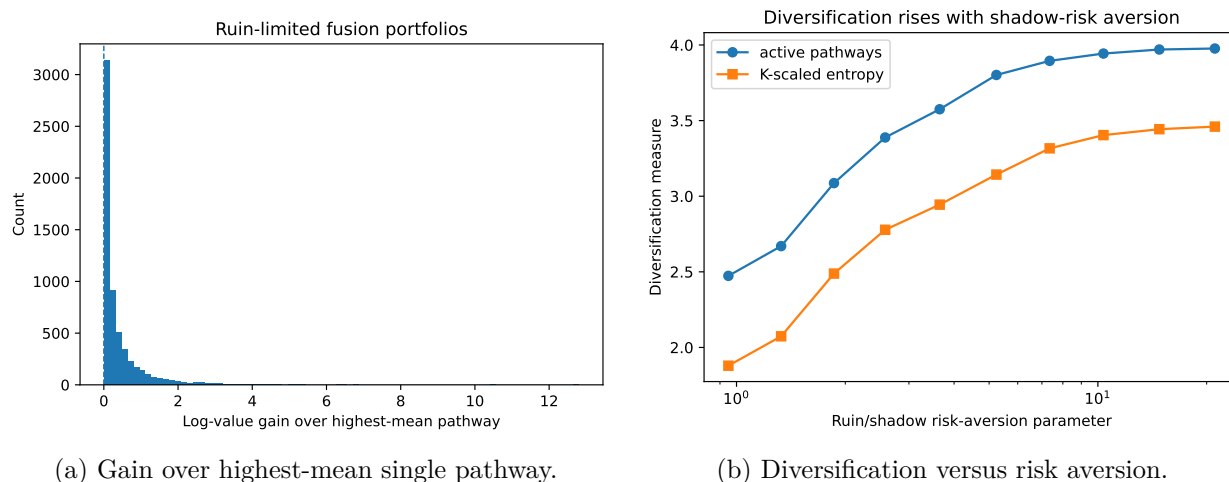


Figure 63: Ruin-limited fusion portfolios. When shadow covariance and tail risk matter, mixed pathway allocation can outperform the nominally best single path.

## 14.9 Tenth-sprint summary table

Table 11: Tenth-sprint virtual experiments. Percentages describe the sampled toy environments and should not be read as natural frequencies.

| Experiment              | Sample size | Main result                                                                                                                     |
|-------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| Witness scaffolds       | 30,000      | Positive witness allocation in 70.9%; interior optimum in 61.1%; median positive $\omega^* = 0.236$ ; threshold accuracy 99.8%. |
| Action geodesics        | 160         | Median action saving 46.1% versus straight path; median path length ratio 1.154.                                                |
| Counter-shadow cofusion | 30,000      | Positive cofusion in 65.4%; median residual-shadow reduction 45.1% among positive cases; threshold accuracy 100%.               |
| Ruin-limited portfolios | 6,000       | Diversified optimum in 97.2%; median active count 4; median shadow-variance reduction 73.0% versus highest-mean path.           |

### 14.10 Tenth-sprint synthesis

The tenth sprint suggests four additional design inversions:

1. A non-fusing witness can be a fusion scaffold if it makes shadow observable.
2. The best scaffold path can be longer in geometric distance while shorter in action.
3. A low-yield secondary fusion channel can be valuable when its product has the opposite sign in shadow space.
4. Repeated fusion should often be optimized as a survival portfolio, not as a maximum-yield event.

These findings extend the central thesis. Fusion is not only scaffolded, incomplete, product-managed, pulsed, path-dependent, sink-assisted, compliance-partitioned, and dopant-windowed. It is also observability-limited, action-geodesic, counter-shadow-coupled, and ruin-portfolio-limited.

### 14.11 Targets for further study from the tenth sprint

1. **Witness dose-response curves.** Add a non-fusing reporter, diagnostic, calibration head, coupon, or synthetic twin. Estimate  $H$ ,  $v_0$ ,  $\kappa$ , and  $c$ , then test the boundary  $\mathcal{W}_0 = 1$ .
2. **Action tomography.** Map local barrier density  $a(x)$  across scaffold, interface, alignment, or operating-state space. Compare direct paths with geodesics predicted by Eq. (157).
3. **Counter-shadow sign assays.** Identify products whose effect vector opposes a dominant product shadow. Sweep secondary drive  $y$  and test whether positive cofusion appears when  $\mathcal{C}_0 > 1$ .

4. **Fusion portfolio frontiers.** Estimate expected useful return  $\mu$ , shadow covariance  $Q$ , and tail-risk surrogates for multiple fusion modes. Compare maximum-yield operation with the frontier in Eq. (163).
5. **Coupling tests.** Test whether witnesses alter the action landscape, whether geodesic paths reduce ruin covariance, whether counter-shadow products become dopants or sinks, and whether portfolios should allocate witness capacity as well as fusion drive.

## 14.12 Tenth-sprint applications

**Nuclear fusion and plasma-facing systems.** Witness scaffolds map to diagnostics, synthetic diagnostics, digital twins, tracer pulses, witness samples, and uncertainty-aware control. Action geodesics map to ramp trajectories, target-compression histories, plasma-shape paths, detachment trajectories, and material-gradient routes through operating space. Counter-shadow cofusion maps to deliberately selected reaction, impurity, or exhaust products that neutralize ash, heat-flux, retention, or wall-damage channels. Ruin-limited portfolios map to risk-adjusted mixtures of operating scenarios, pulse schedules, fuel cycles, wall conditions, and fallback regimes.

**Materials joining.** Witness scaffolds map to embedded coupons, acoustic/thermal/electrical monitors, tracer layers, and sacrificial metrology strips. Action geodesics map to composition gradients, thermal paths, pressure schedules, and surface-treatment sequences. Counter-shadow cofusion maps to getters, reactive interlayers, transient products that consume brittle phases, and local chemistry that offsets residual stress. Ruin-limited portfolios map to hybrid joining strategies chosen for fatigue, crack, delamination, and residual-stress covariance rather than peak joint strength alone.

**Biological fusion and condensates.** Witness scaffolds map to fluorescent reporters, microenvironment probes, FRAP-like transport assays, reversible labels, and sentinel molecules. Action geodesics map to ordered membrane states, hemifusion intermediates, cytoskeletal remodeling paths, and condensate material trajectories. Counter-shadow cofusion maps to chaperones, lipid species, ions, or accessory proteins that offset crowding, aging, or gelation products. Ruin-limited portfolios map to combinations of fusogens, repair pathways, and reversible docking modes that avoid irreversible syncytia or aggregate lock-in.

**Information fusion and machine learning.** Witness scaffolds map to calibration heads, uncertainty probes, disagreement scorers, abstention heads, and small diagnostic submodels. Action geodesics map to curriculum alignment, latent-space bridges, staged adapters, and progressive representation merger. Counter-shadow cofusion maps to de-biasing modalities, retrieval channels, adversarial checks, or negative-control features that offset correlated errors. Ruin-limited portfolios map to ensembles and routing policies chosen for tail-risk covariance under distribution shift rather than leaderboard accuracy alone.

## 14.13 Tenth-sprint limitation statement

The witness model assumes that uncertainty decreases monotonically with witness fraction and that the witness does not perturb the underlying fusion state. Real witnesses can be invasive. The geodesic model assumes a measurable local action density, but many systems have nonlocal history dependence. The cofusion model assumes a scalar counter-shadow; real product spaces can be high-dimensional and overshoot can be dangerous. The portfolio model uses a quadratic risk

surrogate; true ruin may have discontinuous thresholds or fat tails. These limitations identify the next experiments: invasive-witness tests, nonlocal path-memory tests, high-dimensional counter-shadow maps, and stress tests against rare catastrophic shadow modes.

## 15 Eleventh virtual sprint: spectral aperture, selective moats, curvature windows, and extreme-tail shadow

The tenth sprint treated observability, path geometry, counter-products, and repeated-operation risk as hidden fusion variables. The eleventh sprint searches a neighboring blindspot: useful fusion may require deliberate constraints on which contacts are allowed, which modes can couple, which interface shapes are used, and which rare tails are tolerated. Four variables were isolated: spectral aperture  $\Gamma$ , anti-fusion moat strength  $m$ , interface curvature  $\kappa$ , and heavy-tail shadow index  $\alpha_T$ .

The unifying hypothesis is that some of the most valuable fusion controls look like refusals or distortions rather than stronger contact. A narrow spectral aperture refuses off-resonant coupling. A moat refuses parasitic fusion. A finite curvature concentrates the interface without allowing unbounded damage. A tail-aware fusion degree refuses high-yield operation when rare product shadow dominates repeated use.

### 15.1 Mode-Locked Spectral Aperture Law

Many fusion-like systems couple through modes: electromagnetic modes, acoustic modes, conformational modes, oscillatory membrane modes, synchronization modes, or latent computational modes. Let  $\Delta_i$  be the detuning of useful target mode  $i$  and  $\delta_j$  be the detuning of parasitic mode  $j$ . A simple aperture kernel is

$$K_\Gamma(x) = \exp\left(-\frac{x^2}{2\Gamma^2}\right), \quad (165)$$

where  $\Gamma$  is a spectral, temporal, or representational aperture. Define target and parasitic capture fractions

$$C_T(\Gamma) = \frac{1}{n_T} \sum_{i=1}^{n_T} K_\Gamma(\Delta_i), \quad C_P(\Gamma) = \frac{1}{n_P} \sum_{j=1}^{n_P} K_\Gamma(\delta_j). \quad (166)$$

A minimal value function is

$$V(\Gamma) = BC_T(\Gamma) - LC_P(\Gamma)^q - c\Gamma^r - \frac{\eta}{\Gamma}, \quad q > 1, \quad (167)$$

where  $\eta/\Gamma$  is the control cost of an excessively narrow high- $Q$  aperture and  $c\Gamma^r$  is broad-aperture cost, leakage, or heating. An interior optimum satisfies

$$\frac{B}{n_T} \sum_i \frac{\Delta_i^2}{\Gamma^{*3}} K_{\Gamma^*}(\Delta_i) - \frac{LqC_P(\Gamma^*)^{q-1}}{n_P} \sum_j \frac{\delta_j^2}{\Gamma^{*3}} K_{\Gamma^*}(\delta_j) - cr\Gamma^{*(r-1)} + \frac{\eta}{\Gamma^{*2}} = 0. \quad (168)$$

**Proposed law 40** (Mode-Locked Spectral Aperture Law). *When useful and parasitic fusion channels occupy overlapping mode spectra, the optimal interface has a finite aperture. Excessively narrow apertures miss detuned useful modes and require high control cost; excessively broad apertures admit parasitic coupling and product shadow. Useful fusion should peak at a mode-locking aperture satisfying Eq. (168) when such an interior point is feasible.*

This law separates “stronger drive” from “better spectral permission.” A system can underperform not because the drive is weak, but because the aperture is either too pure to capture real targets or too broad to reject shadow-producing side channels.

## 15.2 Virtual experiment 41: mode-locked spectral aperture

A virtual mode catalog was sampled with two to five useful modes and eight to twenty parasitic modes. Useful modes were usually near resonance but not exactly aligned. Parasitic modes contained both a broad background and a near-resonant minority. Equation (167) was optimized over  $0.02 \leq \Gamma \leq 2.0$ .

In 25,000 sampled systems, an interior spectral aperture appeared in 98.2% of cases and beat both grid-edge apertures in the same fraction. The median selected aperture among interior cases was  $\Gamma^* = 0.492$ , the median gain over the best edge aperture was 0.514 dimensionless units, and the 90th percentile gain was 1.477. The high interior frequency is a property of the deliberately detuned toy catalog, not a claimed natural frequency. The exceptional lesson is qualitative: spectral selectivity is not monotone. Both idealized purity and permissive broadband coupling can be inferior to finite mode-lock.

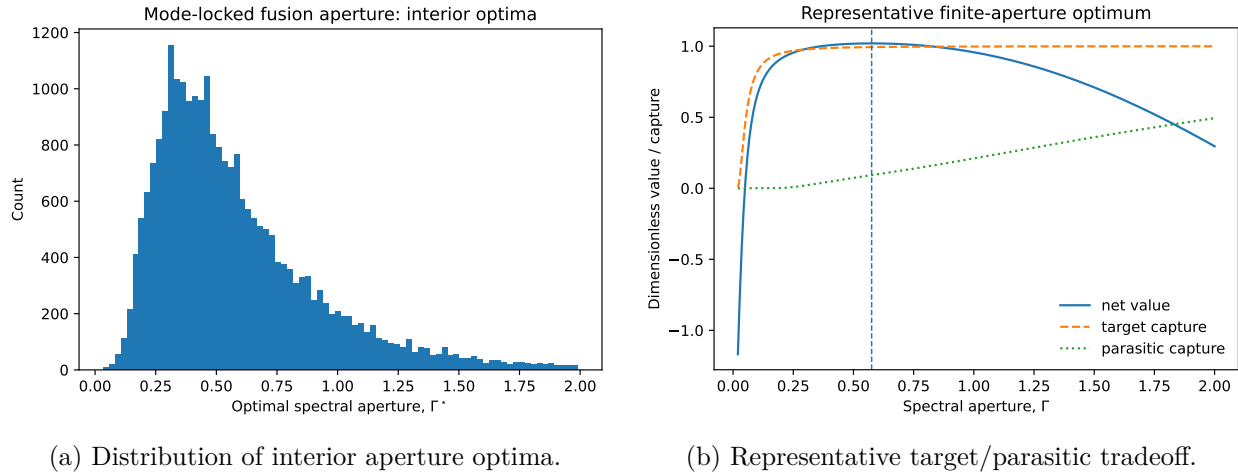


Figure 64: Mode-locked spectral aperture. A finite aperture captures useful detuned modes while limiting parasitic mode admission and high- $Q$  control cost.

## 15.3 Selective Moat Fusion Law

Previous laws emphasized scaffolds that promote desired fusion. A complementary variable is an anti-fusion moat that suppresses undesired contacts more strongly than desired contacts. Let  $m \in [0, 1]$  be moat strength. A minimal target/parasitic model is

$$V(m) = U_0 e^{-a_T m} - L_0 e^{-a_P m} - c m^r, \quad a_P > a_T, \quad (169)$$

where  $U_0$  is target value,  $L_0$  is parasitic shadow,  $a_T$  is target sensitivity to the moat, and  $a_P$  is parasitic sensitivity. For  $r > 1$ , the initial condition for positive moat use is

$$\boxed{a_P L_0 > a_T U_0.} \quad (170)$$

An interior moat satisfies

$$a_P L_0 e^{-a_P m^*} - a_T U_0 e^{-a_T m^*} = c r m^{*(r-1)}. \quad (171)$$

**Proposed law 41** (Selective Moat Fusion Law). *An anti-fusion boundary can increase useful fusion when it attenuates parasitic contacts faster than target contacts. The moat should be increased until the marginal shadow avoided equals the marginal target loss plus moat cost.*

This is a sign inversion in scaffold design. Some of the highest-value fusion structures may be repulsive, insulating, steric, masking, gating, or quarantine-like. Their purpose is not to fuse, but to make the remaining fusion selective.

## 15.4 Virtual experiment 42: selective moats

Equation (169) was tested in 50,000 random systems. Target value, parasitic loss, target sensitivity, parasitic sensitivity, moat cost, and cost exponent were sampled over broad ranges, with  $a_P > a_T$ . The moat was optimized on a grid over  $0 \leq m \leq 1$ .

Positive moat strength appeared in 72.3% of sampled systems, with an interior moat in 45.4%. Conditional on positive moat use, the median moat was  $m^* = 0.644$ , the median gain over no moat was 0.511, and the 90th percentile gain was 4.216. The local threshold in Eq. (170) agreed with the grid optimum in 99.3% of cases. The exceptional result is that deliberate anti-fusion can unlock net fusion value when parasitic fusion is the hidden limiter.

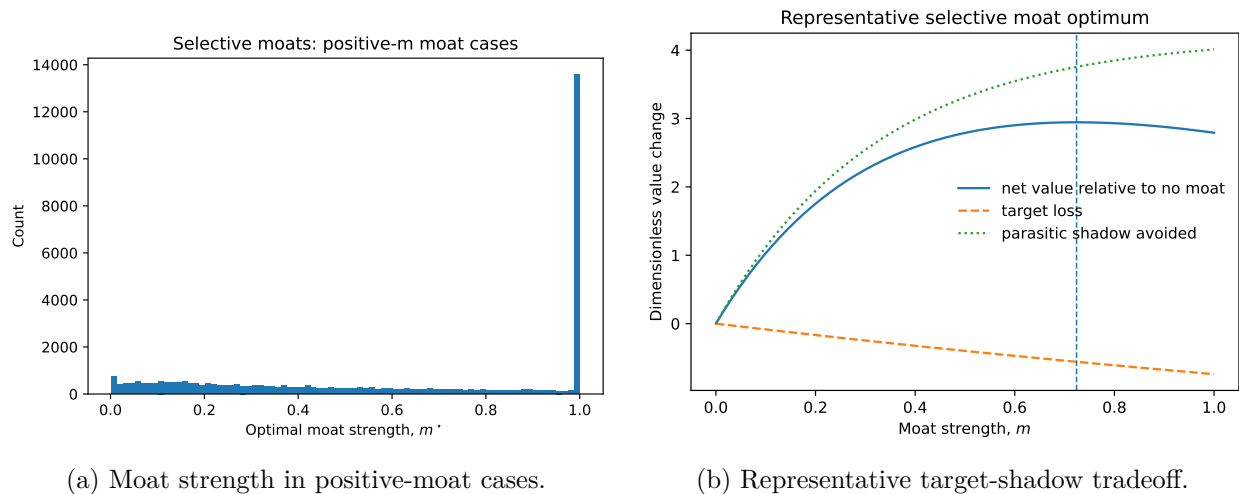


Figure 65: Selective moats. The moat decreases target value slowly while suppressing parasitic shadow quickly, creating a finite optimum.

## 15.5 Curvature-Enhanced Interface Window Law

Interfaces are rarely geometry-neutral. Curvature can concentrate electric field, stress, membrane bending, local temperature, acoustic intensity, contact pressure, or attention density. Curvature can also create damage, fracture, leakage, or uncontrolled product shadow. Let  $\kappa \geq 0$  be a dimensionless

curvature or field-concentration index. A minimal window model is

$$V(\kappa) = B(1 - e^{-\eta\kappa}) - L\kappa^q - c\kappa, \quad q > 1. \quad (172)$$

The initial curvature number is

$$\mathcal{K}_0 = \frac{B\eta}{c}. \quad (173)$$

For  $\mathcal{K}_0 > 1$ , small curvature is initially useful. An interior curvature satisfies

$$B\eta e^{-\eta\kappa^*} = Lq\kappa^{*(q-1)} + c. \quad (174)$$

**Proposed law 42** (Curvature-Enhanced Interface Window Law). *When curvature lowers a local fusion barrier but curvature-induced shadow grows superlinearly, useful fusion should peak at finite curvature. Flat interfaces under-concentrate the fusion event; excessively sharp interfaces convert barrier relief into damage or parasitic product generation.*

This law suggests searching not only over materials, fuels, modalities, or catalysts, but also over local interface shape. The relevant variable may be tip radius, pore rim curvature, patterned roughness, acoustic focality, magnetic geometry, or latent-manifold concentration.

## 15.6 Virtual experiment 43: curvature windows

Equation (172) was tested in 50,000 random systems over  $0 \leq \kappa \leq 5$ . Barrier-relief strength, curvature sensitivity, damage scale, superlinear damage exponent, and linear cost were sampled independently.

Positive curvature appeared in 92.4% of cases, and an interior curvature optimum appeared in 88.0%. Conditional on positive curvature, the median selected curvature was  $\kappa^* = 1.086$ , the median gain over a flat interface was 0.608, and the 90th percentile gain was 3.584. The threshold  $\mathcal{K}_0 > 1$  matched the grid optimum in 99.3% of cases. The exceptional lesson is that interface shape can be a first-order fusion variable, not merely a fabrication detail.

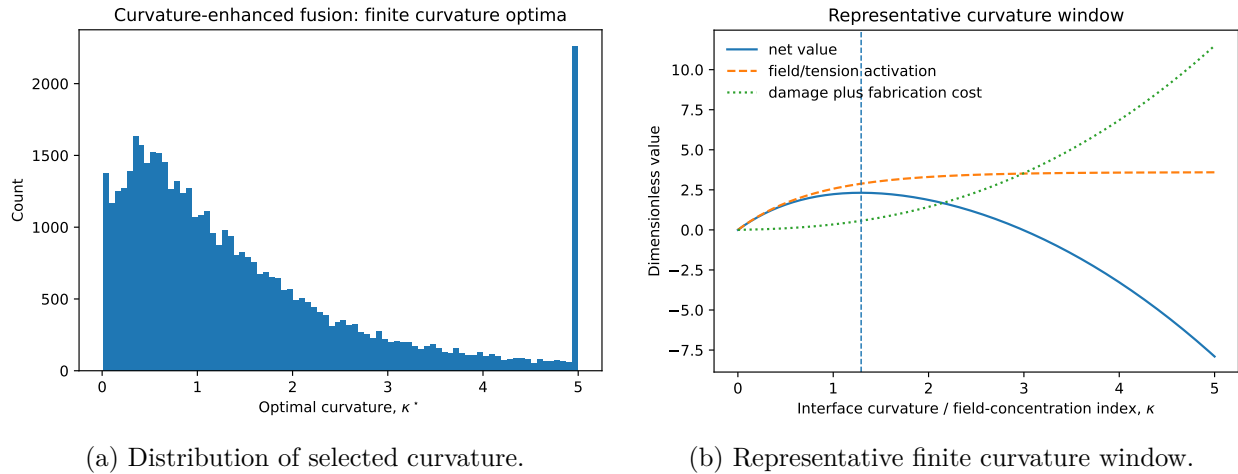


Figure 66: Curvature-enhanced interface windows. Curvature improves activation at first but eventually creates damage and shadow.

## 15.7 Extreme-Tail Product Shadow Law

Product shadow has been treated mostly through mean accumulation and residence time. A rarer failure mode appears when product shadow is heavy-tailed. Let a rare product-shadow loss  $Z$  satisfy a Pareto-like tail

$$\Pr(Z > z) \sim \left(\frac{z_0}{z}\right)^{\alpha_T}. \quad (175)$$

For  $N$  repeated fusion cycles, the largest shadow scales approximately as  $N^{1/\alpha_T}$ . For  $\alpha_T > 1$ , a standard extreme-value approximation gives the per-cycle maximum-shadow scale

$$C_{\alpha_T}(N) \approx \Gamma\left(1 - \frac{1}{\alpha_T}\right) N^{1/\alpha_T - 1}. \quad (176)$$

For  $\alpha_T \leq 1$ , the untruncated expectation is not a stable design quantity; finite reserves, hard cutoffs, or explicit ruin constraints are required. A tail-adjusted fusion-degree objective is

$$V_N(\lambda) = b\lambda - a\lambda^p - \chi C_{\alpha_T}(N)\lambda^q. \quad (177)$$

An interior tail-aware optimum satisfies

$$\boxed{b - ap\lambda^{*(p-1)} - \chi q C_{\alpha_T}(N)\lambda^{*(q-1)} = 0.} \quad (178)$$

**Proposed law 43** (Extreme-Tail Product Shadow Law). *When product shadow is heavy-tailed, useful fusion degree is limited by the extreme tail rather than by the mean shadow. As  $\alpha_T$  approaches one, repeated operation should shift from maximum-yield fusion toward lower degree, stronger refusal, truncation, quarantine, or portfolio control.*

The law is not a replacement for ruin-limited portfolios. It is the scalar local version: before choosing a portfolio, a single pathway may already require tail-aware throttling.

## 15.8 Virtual experiment 44: extreme-tail shadow

Equation (177) was tested in 50,000 random systems. The ordinary optimum of  $b\lambda - a\lambda^p$  was compared with the tail-aware optimum of Eq. (177). Tail index  $\alpha_T$  ranged from 0.65 to 3.5 and the number of repeated cycles ranged from 10 to  $10^6$ . For  $\alpha_T \leq 1.05$ , the simulation used a capped severity proxy rather than an infinite expectation, so the result should be read as a stress test rather than a literal extreme-value estimate.

Tail-aware optimization reduced the selected fusion degree in 37.5% of sampled systems. Severe reductions, defined by  $\lambda_T^* < 0.5\lambda_0^*$ , appeared in 20.7% of systems. When tail-aware reduction occurred, the median relative reduction was 61.5% and the median tail-aware value gain over the ordinary optimum, evaluated under tail stress, was 0.808 dimensionless units. Severe-reduction cases had median tail index  $\alpha_T = 0.964$ . The exceptional lesson is that rare product-shadow tails can make the ordinary optimum actively misleading: the system is not over-fused on average, but over-fused in the tail.

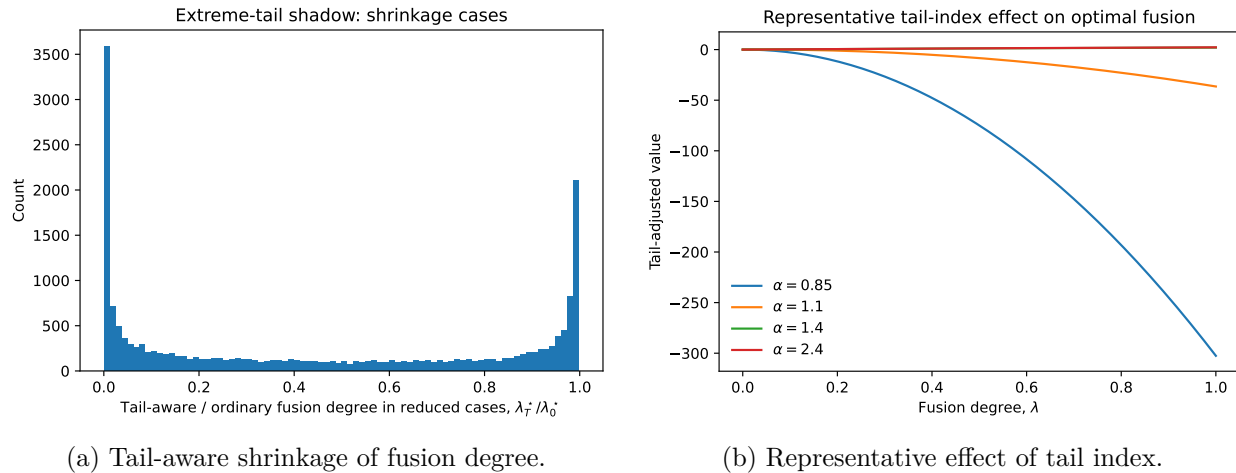


Figure 67: Extreme-tail product shadow. Heavy-tailed rare shadow can move the optimum from full or high fusion toward throttled fusion.

## 15.9 Eleventh-sprint summary table

| Experiment           | Main result                                                                                | Candidate design implication                                                                  |
|----------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Mode-locked aperture | Interior aperture in 98.2% of sampled mode catalogs; median gain over edge aperture 0.514. | Tune spectral/temporal/latent aperture; do not assume narrowest or broadest coupling is best. |
| Selective moat       | Positive moat in 72.3%; threshold Eq. (170) matched 99.3%.                                 | Add anti-fusion boundaries when they suppress parasitic contacts faster than target contacts. |
| Curvature window     | Interior curvature in 88.0%; median gain over flat interface 0.608.                        | Treat interface shape and field concentration as primary optimization variables.              |
| Extreme-tail shadow  | Tail-aware fusion degree shrank in 37.5%; severe shrinkage in 20.7%.                       | Optimize repeated fusion against rare product-shadow tails, not only mean yield.              |

Table 12: Eleventh-sprint virtual experiments. Percentages describe sampled toy environments and are not estimates of natural prevalence.

## 15.10 Eleventh-sprint synthesis

The eleventh sprint suggests four additional design inversions:

1. A fusion interface can fail by having too much permission. Spectral aperture, not only drive strength, controls useful coupling.
2. Anti-fusion structures can be pro-fusion at the system level when they remove parasitic contacts.
3. Local shape can substitute for global force, but only up to the curvature-shadow window.
4. Rare product-shadow tails can dominate repeated operation even when mean product-shadow metrics look acceptable.

Together these results extend the central theory from scaffolded incomplete fusion to *permitted scaffolded incomplete fusion*: useful fusion requires explicit rules for what is allowed to couple, where, at what shape, and under what tail-risk budget.

### 15.11 Targets for further study from the eleventh sprint

**Spectral aperture atlases.** For oscillatory, pulsed, acoustic, electromagnetic, conformational, or representation-space fusion systems, measure target and parasitic detuning distributions. The target output is an empirical curve  $V(\Gamma)$  and an aperture map separating missed targets from parasitic admission.

**Moat dose-response curves.** Add repulsive, insulating, masking, steric, filtering, quarantine, or abstention boundaries and measure whether parasitic shadow decays faster than target value. The target test is Eq. (170).

**Curvature sweeps.** Fabricate or simulate interfaces with controlled tip radius, pore curvature, roughness, focality, or latent concentration. The target test is the finite optimum in Eq. (174).

**Tail-shadow metrology.** Measure the tail index, cutoff, and clustering of rare product-shadow events. The target output is not only mean shadow but a tail budget  $C_{\alpha_T}(N)$  for repeated operation.

### 15.12 Eleventh-sprint applications

**Nuclear and plasma systems.** Spectral aperture maps suggest tuning drive, heating, wave coupling, pulse spectra, and control bandwidth to reject parasitic channels while preserving target coupling. Selective moat logic maps to separatrix, divertor, impurity-screening, and wall-conditioning concepts when they suppress damaging contacts faster than they suppress useful confinement. Curvature windows map to field geometry, target roughness, pellet or liner shape, and plasma-facing microstructure. Tail-shadow logic maps to rare disruption, wall-load, activation, and component-fatigue events.

**Biological fusion and condensates.** Aperture maps correspond to matching fusogen state, membrane mechanics, and timing. Moats correspond to steric glycocalyx barriers, local inhibitors, receptor masks, and spatial quarantine. Curvature windows correspond to pore rims, membrane tubules, BAR-domain-like curvature control, and condensate surface tension. Tail-shadow logic corresponds to rare irreversible syncytia, aggregate nucleation, or pathological aging events.

**Materials joining.** Aperture maps correspond to ultrasonic, thermal, laser, or pressure frequency windows. Moats correspond to masks, diffusion barriers, wetting inhibitors, and intermetallic-suppression layers. Curvature windows correspond to surface texturing, asperity design, powder morphology, crack-tip avoidance, and stress focusing. Tail-shadow logic corresponds to rare brittle phases, void chains, fatigue cracks, or extreme residual-stress events.

**Information fusion and machine learning.** Aperture maps correspond to attention bandwidth, retrieval radius, cross-modal alignment temperature, and adapter selectivity. Moats correspond to abstention gates, negative controls, firewalls between modalities, and quarantine memories. Curvature windows correspond to representation bottleneck sharpness, embedding concentration, and local decision-boundary geometry. Tail-shadow logic corresponds to rare but severe hallucination, correlated failure, distribution-shift, and safety-tail events.

### 15.13 Eleventh-sprint limitation statement

The aperture model assumes Gaussian spectral kernels and separable parasitic load. Real mode coupling can be nonlinear, phase-dependent, and history-dependent. The moat model assumes parasitic channels are more moat-sensitive than target channels; if this sign is wrong, moats reduce useful fusion. The curvature model assumes monotone saturating benefit and convex damage; real geometries can have multiple local windows. The tail model uses Pareto-like extreme scaling and a capped proxy near infinite-mean regimes; empirical tail estimation is difficult and can be invalid under nonstationarity. These limitations suggest the next tests: non-Gaussian apertures, adaptive moats, multi-window curvature landscapes, and tail-index estimation under controlled distribution shift.

## 16 Blindspots and proposed research directions

### 16.1 Blindspot 1: optimizing contact instead of the path to contact

Many fusion systems emphasize peak conditions at contact: temperature, pressure, power density, or raw data volume. The Bridge-Fusion Law suggests that the fusion path can be more important than the peak. If barriers are convex, intermediate states can reduce required action even when they impose cost. Examples include magnetic-field shaping, hohlraum symmetry control, graded materials, fusogen conformational intermediates, hemifusion states, and latent alignment models.

### 16.2 Blindspot 2: assuming full merger is the endpoint

The Optimal Incomplete Fusion Theorem suggests that many useful systems should retain boundaries. In nuclear systems, this means strong coupling between plasma confinement and exhaust rather than maximal confinement of everything. In cells, it means controlled local membrane merger rather than uncontrolled syncytia. In information systems, it means gates, uncertainty estimates, and modality-specific memory.

### 16.3 Blindspot 3: treating products as outputs rather than future boundary conditions

The Product-Shadow Saturation Law predicts that high raw fusion rate can saturate if products accumulate. This shifts attention from momentary gain to product residence. In reactor language this includes ash exhaust, alpha control, neutron damage, heat rejection, tritium processing, and blanket integration. In condensates it includes aging and gelation. In data fusion it includes model bias, overconfidence, and loss of independent error modes.

### 16.4 Blindspot 4: missing the scaffold lifetime variable

A scaffold that remains too long becomes part of the product shadow. Let scaffold alignment benefit saturate as

$$B_S(\tau_S) = \gamma(1 - e^{-\tau_S/\tau_a}), \quad (179)$$

## 15A Twelfth virtual sprint: coded interfaces, moving fronts, product-age harvesting, and nucleation seeds

The eleventh sprint treated permission, mode selectivity, moat boundaries, interface curvature, and rare tail risk. The twelfth sprint searches a different blindspot: many fusion systems are not point events. They are coded, propagated, aged, and nucleated. A useful fusion process may fail because its interfaces lack identity codes, because its joining front advances at the wrong speed, because its products are harvested at the wrong age, or because its nucleation seeds are too sparse or too dense.

Four variables were isolated: scaffold-code dimension  $d$ , fusion-front velocity  $v_f$ , product harvest age  $t_h$ , and seed density  $s$ . The unifying hypothesis is that exceptional fusion often depends on structured initiation: before or during merger, the system must decide which partners recognize each other, how quickly the interface moves, how long products remain before extraction, and how many nucleation sites are permitted.

### 15A.1 Scaffold-Code Orthogonality Law

A scaffold can be more than a physical bridge; it can be an identity code. In biological systems this resembles receptor-ligand specificity, fusogen compatibility, or spatial tagging. In materials it resembles interlayer chemistry and lattice-orientation matching. In multimodal information fusion it resembles keys, adapters, latent tokens, gating vectors, or uncertainty tags.

Let  $d$  be the dimension of a scaffold code and let  $M$  be the number of possible parasitic partners in the local fusion field. For approximately random normalized codes, the maximum accidental correlation among  $M$  parasitic candidates scales as:

$$\rho_{\max}(d, M) \approx \min[1, \sqrt{2 \log(M) / d}]. \quad (185)$$

A minimal coded-interface value is:

$$V(d) = B(1 - \exp(-\eta d)) - L(\exp(\beta \rho_{\max}(d, M)) - 1 + \epsilon) - c d^r. \quad (186)$$

In the uncapped interior regime, a finite optimum satisfies:

$$B \eta \exp(-\eta d^*) + [L \beta / (2 d^*)] \sqrt{2 \log(M) / d^*} \exp(\beta \sqrt{2 \log(M) / d^*}) = c r (d^*)^{(r-1)}.$$

Proposed law. When many non-target partners can accidentally fuse, the interface should carry enough code dimension to suppress extreme cross-talk, but not so much code that recognition and maintenance costs dominate. The optimal code dimension grows with the logarithm of the parasitic catalogue and is set by marginal cross-talk reduction, target capture, and code burden.

Virtual experiment 45 sampled a catalogue of desired and parasitic contacts, with  $M$  ranging from 4 to 5000 and integer dimensions  $1 \leq d \leq 128$ . Positive code beyond  $d=1$  was selected in 99.978% of systems. An interior code dimension appeared in 53.402%. Among interior cases, the median selected dimension was  $d^*=67$ , the median gain over the best edge dimension was 1.231, and the 90th percentile gain was 8.286. The median selected maximum accidental correlation was 0.365. The exceptional result is that a moderate code can make a dense fusion environment behave as if it were sparse, because accidental collisions are suppressed by orthogonality rather than by physical separation.

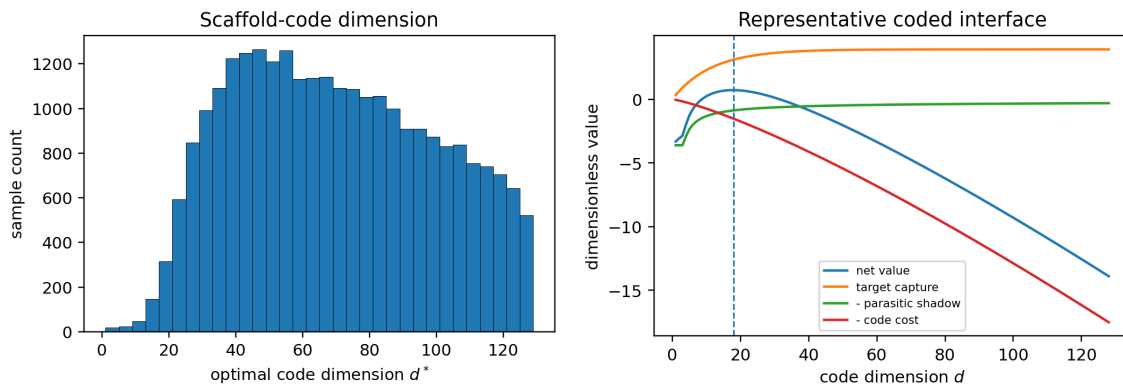


Figure 15A.1. Scaffold-code orthogonality. Increasing code dimension improves target recognition and suppresses extreme accidental correlation, but eventually code burden dominates.

## 15A.2 Fusion-Front Velocity Window Law

Fusion often propagates through a front rather than occurring everywhere at once: a burn front, weld front, pore-expansion front, reaction-diffusion front, condensate coalescence front, curriculum front, or model-integration front. Let  $v_f$  be the dimensionless velocity of such a front. A minimal value function is:

$$V(v_f) = B(1 - \exp(-\eta v_f)) - L v_f^q - A/v_f - c v_f, \quad q > 1. \quad (188)$$

Here  $A/v_f$  represents losses from excessive dwell time: aging, leakage, premature quenching, thermal diffusion, contamination, or unobserved shadow accumulation. The term  $L v_f^q$  represents damage from advancing too quickly: shock, trapped voids, turbulent mismatch, tearing, correlated model error, or unstable burn. A finite velocity satisfies:

$$B \eta \exp(-\eta v_f^*) + A/(v_f^*)^2 = L q (v_f^*)^{(q-1)} + c. \quad (189)$$

Proposed law. When fusion advances through a spatial, temporal, or computational front, useful fusion peaks at a finite front velocity if slow-front dwell losses and fast-front damage both exist. The front should move until marginal activation plus avoided dwell loss equals marginal damage plus drive cost.

Virtual experiment 46 tested Eq. (188) in 50,000 random systems over  $0.05 \leq v_f \leq 10$ . An interior velocity appeared in 99.698% of sampled systems. Among interior cases, the median selected velocity was  $v_f^* = 1.924$ , the median gain over the better edge velocity was 6.331, and the 90th percentile gain was 23.809. The exceptional lesson is that fronts may need speed governors rather than maximum drive.

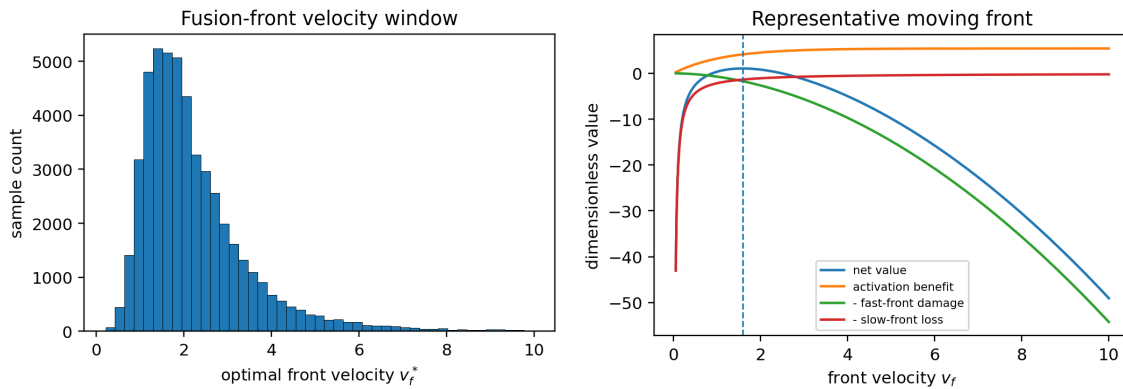


Figure 15A.2. Fusion-front velocity window. Slow fronts suffer dwell, aging, and leakage; fast fronts create damage and trapped shadow.

### 15A.3 Product-Age Harvest Law

The product-shadow framework often suggests rapid removal. That is incomplete if the product must mature before it is useful. Alpha energy must be transferred before ash removal; condensate products may require residence before catalytic output; a joined material may require dwell before strength develops; multimodal information may require delayed aggregation before confidence is calibrated. Let  $t_h$  be product harvest age and define a maturation function:

$$H(t_h) = H_0 + (1 - H_0)(1 - \exp(-\mu t_h)). \quad (190)$$

With useful decay rate  $\delta$ , shadow-aging rate  $\sigma$ , and holding cost  $c$ , a minimal harvest objective is:

$$V(t_h) = Y H(t_h) \exp(-\delta t_h) - L(\exp(\sigma t_h) - 1) - c t_h. \quad (191)$$

The local condition for nonzero holding is:

$$H_{\text{number}} = Y(1 - H_0)\mu / [L \sigma + c + Y H_0 \delta] > 1. \quad (192)$$

An interior harvest age satisfies:

$$Y \exp(-\delta t_h^*) [(1 - H_0)\mu \exp(-\mu t_h^*) - \delta H(t_h^*)] = L \sigma \exp(\sigma t_h^*) + c. \quad (193)$$

Proposed law. Product residence should not always be minimized. If product usefulness matures before product shadow dominates, the optimal removal policy has a finite harvest age. The correct control variable is not residence time alone but the age distribution of useful and shadow products.

Virtual experiment 47 optimized Eq. (191) in 50,000 random systems over  $0 \leq t_h \leq 12$ . Positive holding time was selected in 77.550% of systems, and an interior harvest age appeared in 77.516%. Conditional on positive holding, the median selected age was  $t_h^* = 1.248$ . Among interior cases, the median gain over the better edge policy was 1.249 and the 90th percentile gain was 4.938. The threshold condition in Eq. (192) agreed with the grid optimum in 99.798% of cases. The exceptional result is that rapid exhaust and long confinement are both incomplete heuristics. The stronger target is an age-resolved product distribution with a controllable peak.

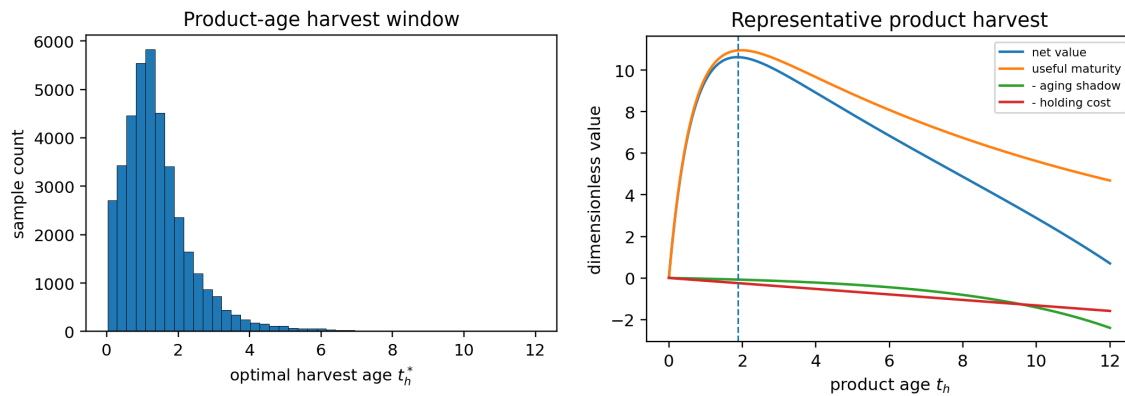


Figure 15A.3. Product-age harvesting. Some products require residence to mature, but aging and holding costs eventually turn product value into shadow.

## 15A.4 Nucleation Seed-Density Window Law

Many fusion systems require a seed: a hot spot, nucleation site, pore, defect, pilot arc, grain, catalyst patch, prototype, or alignment anchor. Seeds lower initial barriers, but too many seeds can fragment the fused product, create defects, consume scaffold budget, or launch parasitic fronts. Let  $s$  be dimensionless seed density. A minimal value function is:

$$V(s) = B(1 - \exp(-a s)) - L s^q - h s^2/(1+s) - c s, \quad q > 1. \quad (194)$$

The local condition for positive seeding is  $S_0 = B a / c > 1$ . A finite seed density satisfies:

$$B a \exp(-a s^*) = L q (s^*)^{(q-1)} + h s^*(2+s^*)/(1+s^*)^2 + c. \quad (195)$$

Proposed law. When seeds lower the fusion barrier but also generate fragmentation or parasitic fronts, useful fusion peaks at finite seed density. Seed density should increase until marginal nucleation relief equals marginal fragmentation, interference, and seed cost.

Virtual experiment 48 tested Eq. (194) in 50,000 random systems over  $0 \leq s \leq 8$ . Positive seed density was selected in 99.936% of cases, and an interior seed density appeared in 98.640%. Conditional on positive seeding, the median selected density was  $s^*=1.888$ . Among interior cases, the median gain over the better edge policy was 2.728 and the 90th percentile gain was 8.496. The seed threshold agreed with the grid optimum in 99.978% of cases. The exceptional result is a seed-density inversion: a fusion system may be seed-starved and over-seeded at different scales simultaneously. The target is not a maximum number of initiation sites, but a controlled seed field.

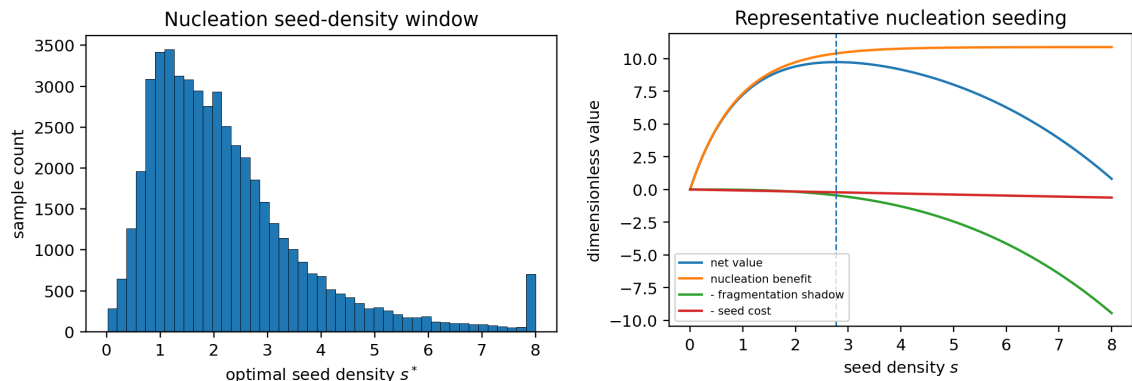


Figure 15A.4. Nucleation seed-density window. Seeds relieve barriers at first, but excessive seed density fragments the product and creates shadow.

## 15A.5 Summary and synthesis

| Experiment          | Main virtual result                                                                             | Design implication                                                                              |
|---------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Coded interfaces    | Positive code in 99.978%; interior code dimension in 53.402%; median interior $d^*=67$ .        | Add identity codes to scaffolds when accidental cross-talk dominates dense fusion environments. |
| Moving fronts       | Interior front velocity in 99.698%; median $v_f^*=1.924$ ; median edge gain 6.331.              | Control the speed of the fusion permission boundary rather than maximizing drive speed.         |
| Product-age harvest | Positive holding in 77.550%; harvest-number threshold matched 99.798%.                          | Replace single residence-time targets with age-distribution control and harvest windows.        |
| Seed density        | Interior seed density in 98.640%; median positive $s^*=1.888$ ; seed threshold matched 99.978%. | Tune seed fields; avoid both seed starvation and fragmentation from over-seeding.               |

Table 15A.1. Twelfth-sprint virtual experiments. Percentages describe sampled toy environments and are not estimates of natural prevalence.

The twelfth sprint adds four design inversions. First, fusion can fail because identities are not coded; orthogonal scaffold codes suppress extreme accidental contacts without requiring full separation. Second, fusion can fail because the front moves at the wrong speed; maximum drive can be inferior to finite front velocity. Third, fusion products can have a useful age; exhaust should be rapid only after useful maturity has been extracted. Fourth, fusion can fail at the nucleation layer; too few seeds leave barriers high while too many seeds fragment the result.

Together these results extend permissioned scaffolded incomplete fusion into coded kinetic fusion: useful fusion depends on identity coding, propagation velocity, product-age sorting, and seed-field design.

## 15A.6 Targets for further study

Scaffold-code atlases. Measure target and non-target affinity distributions for candidate scaffold codes. The target output is an extreme cross-talk curve  $\rho_{\max}(d, M)$  and an empirical value curve  $V(d)$ .

Front-velocity sweeps. In systems with spatial or temporal fusion fronts, vary front speed while measuring activation, dwell loss, trapped shadow, damage, and recovery. The target test is Eq. (189).

Product-age histograms. Replace scalar product residence with an age distribution. Measure useful maturity, useful decay, and shadow aging as functions of product age. The target test is Eq. (192).

Seed-field maps. Vary seed density, seed spacing, and seed heterogeneity while measuring barrier relief, fragmentation, and parasitic initiation. The target test is Eq. (195).

## 15A.7 Possible applications

Nuclear and plasma systems. Scaffold-code logic maps to wave phasing, polarization, isotope/fuel selectivity, impurity-state selectivity, pellet composition, diagnostic tags, and control-channel separation. Front velocity maps to burn-front, pellet-ablation, liner,

current-ramp, and detachment-front control. Product-age harvesting maps to alpha-energy extraction before ash removal and to divertor/exhaust residence control. Seed-density logic maps to hot-spot design, pellet seeding, impurity seeding, magnetic perturbation seeding, and controlled initiation of benign fronts.

Biological fusion and condensates. Scaffold codes map to fusogen identity, receptor-ligand specificity, glycocalyx patterning, and intracellular localization tags. Front velocity maps to pore expansion, syncytium growth, wound closure, and condensate coalescence. Product-age harvesting maps to condensate maturation, enzymatic output, and aging-induced loss of function. Seed density maps to nucleation centers, BAR-domain curvature patches, viral entry sites, and aggregate initiation sites.

Materials joining. Scaffold-code dimension maps to interlayer chemistry, crystallographic orientation tags, surface patterning, and joining atmospheres that distinguish target interfaces from contaminating phases. Front velocity maps to weld speed, thermal-gradient speed, sintering-front speed, and diffusion-bonding dwell. Harvest age maps to post-join dwell, quench timing, heat-treatment windows, and volatile byproduct evacuation. Seed density maps to grain refiners, nucleants, additive particles, pores, and controlled roughness.

Information fusion and machine learning. Scaffold-code dimension maps to adapter keys, modality tags, retrieval keys, uncertainty codes, and latent routing signatures. Front velocity maps to curriculum fusion, staged model merging, progressive retrieval expansion, and online fusion update speed. Product-age harvesting maps to delayed decision fusion, evidence accumulation, calibration windows, and memory consolidation. Seed density maps to exemplars, prototypes, anchors, retrieved neighbors, and initialization points.

## 15A.8 Reproducibility files

The twelfth-sprint code is stored at `code/simulate_fusion_principles_v12.py`. Summary files are `data/simulation_summary_v12.json` and `data/simulation_summary_v12.csv`. Additional outputs include `data/scaffold_code_orthogonality_sample.csv`, `data/fusion_front_velocity_sample.csv`, `data/product_age_harvest_sample.csv`, `data/nucleation_seed_density_sample.csv`, and representative-curve CSV files for each experiment. Figures are stored in `figures/` with both PDF and PNG versions. The random seed is 12012026.

## 15A.9 Limitation statement

The coded-interface model uses an extreme-correlation approximation for random codes; real codes can be structured, learned, adversarially correlated, or physically constrained. The front-velocity model assumes separable slow-loss and fast-damage terms; real fronts can branch, extinguish, accelerate, or couple to control latency. The product-age model assumes a single maturation curve and single shadow-aging curve; real products can have multiple useful and harmful age classes. The seed-density model assumes homogeneous seed fields; real seeding patterns can be anisotropic, hierarchical, adaptive, or history-dependent. These limitations define immediate next tests: nonrandom scaffold codes, branching fronts, multi-species product-age distributions, and spatially patterned seed fields.

and let scaffold residue cost be

$$C_S(\tau_S) = \delta \tau_S^q, \quad q > 1. \quad (180)$$

Then the optimal scaffold lifetime solves

$$\boxed{\frac{\gamma}{\tau_a} e^{-\tau_S/\tau_a} = \delta q \tau_S^{q-1}}. \quad (181)$$

This predicts neither permanent scaffolds nor instantaneous scaffolds, but turnover-matched scaffolds: present long enough to align, absent soon enough not to poison.

### 16.5 Blindspot 5: assuming continuous contact is automatically superior

The third sprint suggests that some systems should be optimized as clocks rather than furnaces. Continuous contact can maintain alignment, but it can also prevent product shadow from clearing. Pulsed operation is valuable when the activation function is convex, the product-shadow variable decays on an accessible time scale, and the alignment state has memory. The measurable design variables are duty fraction  $\delta$ , period  $T$ , product residence  $\tau_P$ , alignment build time  $\tau_A$ , and alignment memory  $\tau_M$ .

### 16.6 Blindspot 6: treating diversity as a count rather than an error geometry

The Diversity-Limited Fusion Law indicates that source count is a weak proxy for fusion value. A system with many correlated modalities may have low effective diversity. In information systems, the missing measurement is the covariance of errors under realistic distribution shifts. In biological and material systems, the analogous variable is diversity of failure modes: multiple components are not truly independent if they fail under the same product shadow.

### 16.7 Blindspot 7: classifying products as either useful output or waste

The Shadow-to-Scaffold Recycling Law suggests a third category: products that should be transiently converted into scaffolds, heat, reaction layers, or control signals. The missing measurement is not only product residence  $\tau_P$ , but product fate: discard, retain, recycle, convert, or gate. The predicted failure mode is over-recycling, where a once-useful product becomes an occluding scaffold.

### 16.8 Blindspot 8: assuming quiet deterministic interfaces are always best

The Interface Annealing Theorem predicts that some fusion interfaces need controlled fluctuation. In such systems, zero noise freezes the interface into high-action alignment paths, while excessive noise destroys the alignment. The overlooked variable is annealing amplitude  $\sigma$ , including thermal, mechanical, chemical, temporal, or computational perturbation.

### 16.9 Blindspot 9: treating selectivity as a scalar rather than a channel geometry

The Orthogonal Gate Fusion Theorem predicts that product shadow should be represented as a channel direction relative to useful output. A scalar conductance can only increase or decrease

everything. A vector gate can transmit useful modes while rejecting product-shadow modes when their angle  $\vartheta$  is large enough. The missing experimental map is the useful-shadow angle.

### **16.10 Blindspot 11: treating fusion products as only waste**

The fifth sprint suggests that some products become future scaffolds while others become shadow. A single residence-time variable can hide this sign split. Future work should estimate signed product memory: retained beneficial scaffold versus retained poison.

### **16.11 Blindspot 12: letting the same network carry fusion and shadow**

A network that carries useful fusion efficiently may also carry product shadow efficiently. Safe extensive fusion requires separate control of  $c_F$  and  $c_P$ . This blindspot appears in physical exhaust paths, biological syncytia, organizational integration, and data-sharing systems.

### **16.12 Blindspot 13: maximizing diversity without an alignment anchor**

Diversity is valuable only when it remains alignable. Systems with no shared anchor may be impossible to fuse without imposing a costly or biased map. The anchor-complementarity window predicts an interior optimum between independence and redundancy.

### **16.13 Blindspot 14: resetting interfaces without preserving memory**

Resets, purges, anneals, cleanings, and dropouts can improve performance by removing product shadow, but they can also erase the memory that lowered the barrier. Selective reset should be treated as an independent design variable.

### **16.14 Blindspot 15: treating product shadow as scalar**

The sixth sprint suggests that product shadow may be a vector in a channel space. A pathway that is poor alone may be valuable because it cancels, rotates, or distributes the residual shadow of another pathway.

### **16.15 Blindspot 16: optimizing feedback gain without latency number**

Feedback can have the correct sign and still destabilize fusion when measurement, computation, actuation, or transport delays cross the latency boundary. Fusion-control papers should report latency budgets as design variables, not only prediction accuracy or gain.

### **16.16 Blindspot 17: maximizing contact area**

More interface can increase product shadow faster than useful binding. Sparse, punctate, porous, islanded, or gated contact topologies may outperform maximal overlap.

### 16.17 Blindspot 18: ignoring module size

In many-entity fusion, the optimum may be neither atomized nor flat. A finite module size can minimize the sum of within-module convex shadow and cross-module coordination cost.

### 16.18 Blindspot 19: assuming fusion is associative

When product shadow and interface adaptation are path-dependent, the final fused set is not enough to specify the system. The missing variable is the fusion path  $\pi$ . Experiments should compare different sequences that reach the same nominal final composition, geometry, or model architecture.

### 16.19 Blindspot 20: committing without scouting

Fusion programs often treat diagnostic probes as overhead rather than option-generating actions. The Scout-Fusion Value Law predicts that cheap probes are most valuable near proceed/refuse boundaries. The missing curve is  $V_{\text{scout}}(e) - V_0$ .

### 16.20 Blindspot 21: ranking candidates separately instead of reciprocally

The best isolated fuel, material, modality, fusogen, or catalyst is not necessarily part of the best fusion pair. A candidate should be characterized by what it provides, what it needs, and what shadow it adds. The missing map is reciprocal scaffold complementarity.

### 16.21 Blindspot 22: smearing scarce scaffolds uniformly

Scaffold, attention, catalyst, active interface, and diagnostic bandwidth often have shadow prices. Equal allocation can waste scaffold on low-return interfaces. The missing control variable is the Lagrange price  $\nu$  that determines which interfaces are active.

### 16.22 Blindspot 23: undervaluing non-fusing sinks

A sink, buffer, decoy, getter, quarantine memory, or exhaust pathway can look parasitic if only direct fusion yield is counted. The Sacrificial Sink Law predicts that such components can increase useful fusion by reducing self-shadow, even when they produce no primary output.

### 16.23 Blindspot 24: refusing to defuse before fusing

Some interfaces begin in a harmful pre-fused state: contaminated, passivated, stressed, correlated, crowded, or locked. The Defusion Preconditioning Law predicts that temporary separation, cleaning, loosening, or de-correlation should be tested before increasing final drive.

### 16.24 Blindspot 25: treating the interface as zero-thickness

A boundary layer can distribute mismatch, create escape volume, and buffer shadow. The Finite Boundary-Layer Thickness Law predicts an interior interphase thickness when mismatch relief is convex but layer volume creates its own cost.

### 16.25 Blindspot 26: optimizing average candidate quality instead of acceptance thresholds

Candidate fusions are often heterogeneous. The Post-Selective Fusion Rectification Law predicts that diagnostic thresholds can convert a mixed candidate population into a useful accepted subset. The missing quantity is the marginal accepted-event value  $q(t)$ .

### 16.26 Blindspot 27: assuming symmetric deformation is fair or efficient

The Compliance-Partition Fusion Theorem predicts that equal adaptation can be wasteful. Interfaces should report who pays the mismatch action: participant A, participant B, scaffold, boundary layer, adapter, or product channel. The missing measurement is the compliance/action vector  $\mathbf{a}$ .

### 16.27 Blindspot 28: synchronizing fusion events by default

Synchrony is often engineered because it is easy to specify and measure. The Anti-Coincidence Fusion Scheduling Law predicts that synchrony can be harmful when product shadow is convex in instantaneous load. The missing variable is the phase vector  $\varphi$ , not only average drive.

### 16.28 Blindspot 29: treating reversibility as indecision

A reversible reserve can be a productive component of fusion design when hidden product shadow is only revealed after partial merger. The missing curve is expected value as a function of rollback, repair, detachment, or routing reserve  $r$ .

### 16.29 Blindspot 30: treating impurity as monotone

Some additives, impurities, dopants, noise sources, and interlayers lower barriers at low dose and become product shadow at high dose. The missing experiment is the finite dose-response map  $V(y)$ , especially the small-dose slope and high-dose poisoning exponent.

## 17 Falsifiable predictions

1. **Bridge scaling.** In systems with measured convex mismatch cost  $D^p$ , the optimal number of intermediate scaffold states should scale approximately as  $((p-1)D^p/c)^{1/p}$  after discretization.
2. **Partial-fusion optimum.** When fusion benefit grows sub-quadratically but mismatch or product cost grows superlinearly, a measurable interior optimum  $0 < \lambda^* < 1$  should appear. Systems engineered only for  $\lambda = 1$  should underperform.

3. **Product-residence dominance.** When product suppression is approximately exponential in product concentration, useful rate should follow Lambert-W saturation,  $R = W(\alpha\tau_P R_0)/(\alpha\tau_P)$ , rather than scaling linearly with raw rate.
4. **No-harm fusion.** In multimodal or multi-sensor models under distribution shift, constrained uncertainty-aware fusion should weakly dominate fixed fusion against the better single modality, provided the error second-moment estimate is calibrated.
5. **Scaffold turnover.** Biological, material, and computational systems should show maximum useful fusion at finite scaffold residence time, not at maximal scaffold persistence.
6. **Selective confinement.** Systems with separate useful-activation and product-shadow variables should be better predicted by split residence times  $(\tau_E, \tau_P)$  than by either residence time alone.
7. **Critical network fusion.** If global merger destroys diversity or control, useful network fusion should peak near the first percolation transition rather than at maximum connectivity.
8. **Complementarity ridge.** When complementarity rises with difference but alignment survival falls with difference, successful fusion pairings should cluster near an interior distance  $d^*$ , not at the nearest-neighbor or most-different endpoints.
9. **Pulse-shadow optimum.** Under equal average drive, systems with convex activation and product shadow should show an interior or constrained duty optimum satisfying  $KU^q\delta^{1-q} \approx 1$ .
10. **Recovery window.** Pulsed systems with measurable product residence and alignment memory should peak when recovery time clears products without destroying alignment memory, up to external fatigue and output-smoothness constraints.
11. **Marginal-action bridge placement.** In heterogeneous interfaces, nonuniform bridge placement should outperform equal spacing, with step sizes scaling as  $a_i^{-1/(p-1)}$ .
12. **Diversity ceiling.** In multi-source information fusion, gains should scale with  $N_{\text{eff}}$ , not nominal source count. Systems with high pairwise error correlation should saturate near the ceiling  $1/\rho$ .
13. **Product-recycling window.** When products can be converted into scaffolds or control signals, useful rate should peak at the finite recycling fraction in Eq. (58); full recycling should underperform when over-scaffold occlusion dominates.
14. **Interface annealing.** Systems satisfying  $Aa > \mu$  should show a finite optimal agitation amplitude satisfying Eq. (63); systems with  $Aa \leq \mu$  should prefer the cold or least-noisy interface.
15. **Impedance-matched scaffolds.** In multiplicative transmission interfaces, optimal graded-layer count should scale approximately as  $|\ln z|/(2\sqrt{c}) - 1$ , after integer rounding and external constraints.
16. **Orthogonal gating.** When useful output and product shadow occupy separable channels, low-rank directional gates should outperform isotropic gates. When  $\lambda_{\max} \leq c_g$ , the optimal gate should refuse fusion.

**Prediction 17: Shadow-vector mixtures.** In systems where product effects can be represented in multiple channels, a mixture of pathways satisfying  $\|Hw\| < \min_i \|h_i\|$  should outperform the best single pathway even when no individual pathway is optimal on its own.

**Prediction 18: Latency boundary.** Closed-loop fusion systems with delayed product-shadow suppression should transition from steady operation to oscillatory shadow debt near  $\mathcal{L} = \Delta/\Delta_c = 1$ , with  $\Delta_c$  given by Eq. (102) under the single-loop approximation.

**Prediction 19: Sparse-contact optimum.** When cross-contact product shadow grows with exponent  $\gamma$  greater than the benefit exponent  $\beta$ , patterned sparse contact should beat full contact with optimum near Eq. (107).

**Prediction 20: Finite granularity.** Many-entity fusion should show finite module-size optima near Eq. (110) when within-module shadow is convex and cross-module integration cost decreases with module size.

**Prediction 21: Sequence dependence.** Systems with convex mismatch action or cumulative product shadow should show measurable performance differences between different fusion paths that reach the same final nominal state. The order penalty should increase with barrier convexity  $p$  and shadow strength  $\alpha$ .

**Prediction 22: Scout-value peak.** Pre-fusion probes should have maximum expected value near sign uncertainty  $4p(1-p) \approx 1$ , after adjusting for probe cost and noise. Clearly good and clearly bad fusions should show lower probe value.

**Prediction 23: Hidden reciprocal pairs.** In candidate libraries with reciprocal scaffold exchange, pairwise search should find high-value fusions that are missed by individual ranking. The miss rate should increase with the variance of  $u_i^\top n_j + u_j^\top n_i$  relative to isolated score variance.

**Prediction 24: Water-filled scaffold allocation.** When scaffold responses are saturating and heterogeneous, optimal scaffold allocation should be sparse at low budget and should activate additional interfaces only when their marginal return exceeds the shadow price  $\nu$ .

**Prediction 25: Sink-unlocked fusion.** In self-shadowed systems satisfying Eq. (124), adding a product sink should increase useful rate according to the Lambert-W response in Eq. (125); the optimal sink dose should satisfy Eq. (127).

**Prediction 26: Defusion threshold.** Temporary preconditioning should improve final fusion only when  $\mathcal{F}_D > 1$  under the local barrier-relief approximation. Below that threshold, more preconditioning should waste effort unless a nonlocal transition is present.

**Prediction 27: Finite interphase optimum.** When mismatch cost decreases as  $h^{1-p}$  and layer cost grows as  $h^q$ , the best boundary layer should occur near Eq. (134), not at zero thickness or maximum thickness.

**Prediction 28: Marginal-value acceptance.** In thresholded candidate fusion, the optimal diagnostic threshold should occur where the conditional value of the marginal accepted event crosses zero:  $q(t^*) = 0$ . Accepting below this threshold should reduce useful value even if it increases nominal fusion count.

**Prediction 29: Compliance partition.** In convex mismatch interfaces, measured deformation or adaptation shares should scale as  $a_i^{-1/(p-1)}$  after normalization. Equal-share designs should show excess action close to Eq. (141) relative to optimized compliant interfaces.

**Prediction 30: Anti-coincidence scheduling.** If product shadow is convex in instantaneous product load and useful yield is phase-insensitive, phase-staggered fusion should reduce shadow by a factor approaching Eq. (143) in the narrow-pulse limit. Synchrony should be retained only when the measured coincidence bonus exceeds the shadow saving.

**Prediction 31: Reversibility reserve.** Fusion protocols with hidden after-contact shadow should select a positive rollback or repair reserve when  $\mathcal{R}_{\text{rev}} > 1$  and should show interior optima satisfying Eq. (146).

**Prediction 32: Controlled dopant window.** Systems where low-dose impurity lowers barrier but high-dose impurity creates convex poisoning should exhibit a finite optimum  $y^*$  satisfying Eq. (149). Positive dosing should appear when the measured initial dopant number  $\mathcal{D}_0 > 1$ .

**Prediction 33: Witness-scaffold threshold.** Non-fusing diagnostic, reporter, sentinel, calibration, or synthetic-twin scaffolds should improve useful fusion when  $\mathcal{W}_0 > 1$  in Eq. (155). Below that threshold, the witness should reduce value unless it also lowers the physical barrier.

**Prediction 34: Action-geodesic detours.** Heterogeneous fusion interfaces should exhibit lower-action paths that are geometrically longer than direct contact when Eq. (158) is satisfied. Observed optimal paths should follow local action-density maps more closely than Euclidean distance.

**Prediction 35: Counter-shadow cofusion.** A secondary low-yield fusion channel should become beneficial when its product opposes primary shadow strongly enough that  $\mathcal{C}_0 > 1$  in Eq. (161). Excess secondary drive should eventually create overshoot shadow or independent shadow.

**Prediction 36: Ruin-limited portfolios.** In repeated fusion systems with correlated shadow risk, mixed pathway portfolios should outperform the highest-mean single path when Eq. (164) holds. The portfolio advantage should grow with shadow covariance, tail-risk penalty, and operating repetition.

**Prediction 37: Spectral aperture optimum.** Fusion systems with overlapping useful and parasitic mode spectra should show a finite aperture optimum satisfying Eq. (168). Narrowing the aperture below this point should lose useful detuned targets; broadening it above this point should admit parasitic product shadow.

**Prediction 38: Selective moat threshold.** Anti-fusion moats should improve useful fusion when parasitic attenuation exceeds target attenuation according to Eq. (170). Reversing the sign, so that target contacts are more moat-sensitive than parasitic contacts, should destroy the moat benefit.

**Prediction 39: Curvature window.** Interface shape sweeps should reveal a finite curvature optimum when  $\mathcal{K}_0 > 1$  and curvature-induced damage grows superlinearly. The measured optimum should satisfy Eq. (174) under the local saturation approximation.

**Prediction 40: Extreme-tail throttling.** Repeated fusion systems with heavy-tailed product shadow should choose lower fusion degree than mean-field optimization predicts. The shrinkage should grow as the tail index approaches one and as the repeated-operation tail budget  $\mathcal{C}_{\alpha_T}(N)$  increases.

## 18 Potential applications

### 18.1 Nuclear fusion design

The framework suggests that gain metrics should be paired with product-residence metrics. A candidate design index is

$$\text{FVI} = \frac{\Lambda_0}{1 + \alpha R(T) \tau_P} \exp \left[ -\frac{\mathcal{A}_0/k^{p-1} + (k-1)c}{\Theta_{\text{eff}}} \right], \quad (182)$$

where FVI denotes a Fusion Viability Index. It is not intended to replace plasma-physics simulation; it is a screening heuristic for asking whether a design is under-investing in scaffolds or exhaust.

## 18.2 Materials joining

For dissimilar materials,  $D$  could be a vector norm combining thermal-expansion mismatch, melting-point mismatch, lattice mismatch, chemical incompatibility, and intermetallic brittleness. The Bridge-Fusion Law predicts that optimal interlayer count and composition gradient should scale with  $D^p/c$ , where  $c$  includes added process complexity and new defect risks.

## 18.3 Biological fusion control

For membrane fusion,  $D$  includes membrane spacing, hydration, curvature, surface crowding, charge, lipid composition, and membrane tension. The theory predicts that fusogens, actin protrusions, and local mechanics act as time-limited scaffolds. Eq. (181) suggests that persistent fusogenic states can become pathological.

## 18.4 Machine learning and sensor fusion

For multimodal AI,  $\lambda$  is the coupling strength between modalities and  $\phi$  is the learned alignment map. The no-harm theorem implies that production fusion systems should include calibrated fallback to a single modality or subsystem. The product-shadow analogue is negative transfer or excessive confidence in correlated errors.

## 18.5 Condensates and reversible mesoscale fusion

For biomolecular condensates, the second sprint suggests measuring not only coalescence rate but also material aging, internal diffusion, and exchange lifetime. Recent material-property frameworks emphasize that condensate states span viscoelastic, gel-like, glass-like, and quasi-solid regimes that are coupled to biological function [31]. In the present notation, this means condensate usefulness should be mapped against  $(\lambda, \tau_S, \tau_P, c)$ : degree of mixing, residence time of organizing interactions, residence time of aged or trapped products, and network connectivity.

## 18.6 Networked and organizational fusion

The Critical Fusion Percolation Principle suggests a design target for systems composed of many partially fused units: operate near enough to the percolation threshold to share function, but below the point where a giant component destroys independence. Candidate applications include model ensembles, sensor swarms, modular robotics, synthetic tissues, cooperative organizations, and sintering networks.

## 18.7 Complementarity search and invention

The Exceptional Fusion Ridge suggests a method for invention: construct a difference metric  $d$ , estimate the complementarity exponent  $m$ , estimate the alignment penalty  $\mu d^p$ , and search for

pairings near  $d^* = (m/\mu p)^{1/p}$ . This applies naturally to dissimilar materials, enzyme-scaffold design, hybrid physical-computational devices, and multimodal model architectures. In information fusion, recent methods that use information bottlenecks, redundancy suppression, and gated attention already point in this direction by treating redundancy and misalignment as first-class design constraints [32, 33].

## 18.8 Rhythmic fusion protocols

The third sprint suggests a new application class: rhythmic fusion protocols. In physical systems this means pulsed drive plus explicit chamber or interface recovery. In biology it means time-limited fusogen presentation and recovery phases. In machine learning it means periodic modality dropout, recalibration, and no-harm fallback checks rather than always-on feature merging. The target is not pulsing for its own sake, but pulse timing that reduces product-shadow residence while preserving alignment memory.

## 18.9 Diversity engineering

For information, sensor, and model fusion, the design target becomes decorrelated competence. The most direct application is to select or train modalities so that their errors are not exchangeable. In safety-critical systems, reporting  $N_{\text{eff}}$  may be more informative than reporting nominal sensor count. The same idea extends to materials and biological systems as diversity of failure modes: components that share the same product-shadow failure channel are less diverse than they appear.

## 18.10 Product-to-control loops

The fourth sprint suggests a new application pattern: close the loop from product measurement to scaffold control. Fusion products, residuals, reaction layers, and error signals should be classified by whether they are best exhausted, retained, recycled, or converted into a gate command. This may apply to alpha-particle heating versus ash removal, controlled intermetallic formation, post-fusion biological signaling, and residual-driven uncertainty calibration in multimodal systems.

## 18.11 Annealed and impedance-matched interfaces

Interfaces that fail under direct contact may benefit from two independent interventions: finite annealing amplitude  $\sigma$  and graded impedance scaffolds  $k$ . In materials this suggests combining vibration or thermal cycling with nonuniform interlayers. In biological systems it suggests timed fluctuation of fusogen, membrane tension, or crowding. In computational systems it suggests curriculum alignment, representation noise, and geometric latent bridges.

## 18.12 Product-selective gates

The Orthogonal Gate Fusion Theorem reframes boundary design as an eigenvector problem. A gate should be evaluated by useful transmission, product-shadow leakage, and selectivity, not only by total conductance. Applications include divertor-like exhaust concepts, semipermeable synthetic

membranes, condensate permeability control, dissimilar-material diffusion barriers, and multimodal attention gates with no-harm refusal.

### 18.13 Fusion memory engineering

Repeated fusion systems should be engineered around signed memory. Beneficial memory should be preserved, amplified, or recycled; harmful shadow should be exhausted, neutralized, or quarantined. The key measurement is the net memory coefficient  $g$ , together with its decomposition into scaffold and shadow terms.

### 18.14 Dual-channel quarantine and rectification

The dual-percolation law suggests applications in modular reactor subsystems, immune-safe synthetic biology, networked organizations, and multimodal data pipelines. The design principle is to let useful influence percolate while keeping shadow influence below its giant-component threshold.

### 18.15 Anchor design for multimodal fusion

Information fusion systems should include explicit anchor design. Examples include shared timestamps, calibrated spatial frames, paired labels, contrastive alignment sets, common physical constraints, or domain-specific invariants. The target is not maximal shared representation but an overlap near  $\rho^*$ .

### 18.16 Memory-selective reset protocols

Selective reset protocols can be translated into plasma exhaust and wall conditioning, materials heat treatment, condensate rejuvenation, cell-fusion recovery periods, and machine-learning recalibration. The reset is useful when Eq. (90) is positive.

### 18.17 Shadow-vector balanced operating regimes

When product loads occupy multiple channels, operational planning should search for mixtures that reduce residual shadow norm, not only for the highest-yield single mode. This may apply to plasma scenarios, material-process schedules, ensemble-model mixtures, and condensate reaction environments.

### 18.18 Latency-budgeted fusion control

Plasma controllers, robotic sensor-fusion stacks, biological inhibition loops, and adaptive multimodal models should estimate a latency number. A controller outside the latency budget can transform correction into oscillatory product-shadow debt.

### 18.19 Patterned sparse interfaces

Dissimilar-material joining, membrane fusion, condensate coalescence, and information-fusion attention maps can use sparse active sites to preserve escape channels and reduce cross-contact shadow coupling.

### 18.20 Finite-granularity fusion architecture

Large fusion systems should test module size as an explicit variable. Small modules may stabilize internal alignment before higher-level joining, reducing the cost of both atomized fragmentation and flat merger.

### 18.21 Fusion sequence engineering

Fusion recipes should report path variables, not only final states. Candidate applications include plasma scenario ramp design, staged target preparation, graded material deposition, sequential fusogen activation, condensate client loading, and staged multimodal feature alignment.

### 18.22 Scout-fusion diagnostics

A new diagnostic class is suggested: small probes designed specifically to change the proceed/refuse decision. This may include diagnostic shots, micro-joins, pilot co-cultures, reversible pre-fusion contacts, low-dose perturbations, model-disagreement probes, and short-horizon online tests.

### 18.23 Reciprocal complementarity discovery

Candidate libraries should store provide-vectors and need-vectors. Discovery algorithms should search for reciprocal scaffold exchange and shadow cancellation, not just high individual merit. The target output is a ranked list of pairs or sequences with high  $u_i^\top n_j + u_j^\top n_i$  and low  $\|p_i + p_j\|$ .

### 18.24 Scaffold water-filling controllers

Resource-constrained fusion systems can implement water-filling controllers. In physical systems this means prioritizing heating, catalysts, interlayers, diagnostics, or active sites by marginal return. In computational systems it means allocating attention and compute to modalities or cross-modal links only when their expected return exceeds a learned price.

### 18.25 Sacrificial sink design

Fusion systems should include components whose value is measured by product-shadow relief rather than direct yield. Candidate implementations include exhaust paths, getters, buffers, decoys, sacrificial phases, chaperones, uncertainty reservoirs, and quarantine memories.

### 18.26 Defusion preconditioning protocols

Preparation steps should be measured as part of the fusion path. Cleaning, stress relief, de-correlation, partial dissolution, reversible docking, and low-amplitude shake-down can be scored by the defusion number  $\mathcal{F}_D$  before final drive is optimized.

### 18.27 Finite boundary-layer engineering

Interlayers, shells, adapters, and translation modules should be scanned by thickness. The target is not the thinnest possible interface, but the thickness that balances mismatch distribution against volume-shadow cost.

### 18.28 Fusion rectifiers

Candidate fusions should be routed through thresholded diagnostics when candidate quality is heterogeneous. A rectifier is useful when screening cost is lower than the value gained by rejecting negative marginal events.

### 18.29 Compliance-partitioned interface design

A practical application is to add a deliberately compliant element and then tune its compliance rather than merely minimizing mismatch. Examples include graded interlayers, flexible adapters, soft biological fusion zones, elastic plasma-facing structures, and low-risk latent adapters in multimodal systems. The design target is Eq. (140).

### 18.30 Anti-coincident product scheduling

Systems with repeated or distributed fusion events can be scheduled to flatten product load. Applications include pulsed plant cadence, distributed heat-load management, staggered material deposition, temporally dispersed biological fusion events, and asynchronous multimodal model updates. The target is not lower average activity but lower convex product coincidence.

### 18.31 Reversibility-reserve engineering

A reversible reserve can be implemented as rollback capacity, removable fixtures, reversible hemifusion arrest, model-routing fallback, repair bandwidth, detachment margin, or abortable pulse sequences. It is most valuable near hidden-shadow uncertainty rather than in clearly safe or clearly unsafe regimes.

### 18.32 Controlled impurity and dopant windows

Dopants, impurities, alloying elements, radiators, lipids, noise, dropout, calibration heads, and retrieval modules should be swept as finite-dose variables. The central application is to replace purity dogma with a dose-response measurement: low-dose barrier relief, high-dose shadow, and an optimum between them.

## 19 Limitations

The virtual tests use stylized distributions and dimensionless models. They cannot establish physical truth in any domain. The proposed variables may be difficult to measure directly, and the same symbol may correspond to different mechanisms in different fields. The equations are best interpreted as generative hypotheses: they suggest which experiments to perform and which design variables may be overlooked.

Another limitation is that not all barriers are convex, not all product effects are suppressive, not all scaffolds are independent, and not all pulsed systems can tolerate high peak drive or long recovery gaps. Product recycling, interface annealing, impedance matching, orthogonal gating, signed fusion memory, dual percolation, anchor overlap, selective reset, sequencing, scouting, reciprocal complementarity, scarce allocation, sacrificial sinks, defusion preconditioning, boundary-layer thickness, and post-selection may also interact in nonlinear ways not captured by the separable toy models. Some systems may have cooperative, threshold, chaotic, or history-dependent behavior that violates the assumptions. Those violations are useful: they identify where the theory must be extended.

The ninth sprint adds another limitation: allocation variables can interact. A dopant can change compliance coefficients; phase staggering can alter recovery memory; reversibility reserves can increase interface residence time; and a compliant scaffold can become a product sink or a product source. Therefore the four new laws should first be tested in controlled one-variable sweeps and only then combined in higher-dimensional design spaces.

The tenth sprint adds a further limitation: infrastructure variables can be mistaken for physics variables. A witness scaffold can change the system it measures; an apparent geodesic can be an artifact of coarse action tomography; counter-shadow can produce delayed independent shadow; and portfolio diversification can hide low-frequency common-cause failures. Therefore observability, geodesic, cofusion, and portfolio claims should be tested with perturbation controls, held-out operating regimes, and explicit rare-event stress tests.

The eleventh sprint adds another limitation: permission variables can be mistaken for universal improvements. Narrow apertures can miss the desired state, moats can block targets, curvature can localize damage, and tail-aware throttling can under-use safe regimes if the tail is overestimated. Therefore aperture, moat, curvature, and tail claims require sign tests, dose-response sweeps, and held-out rare-event validation.

## 20 Conclusion

A cross-domain pattern emerges from the surveyed evidence and the virtual tests:

Useful fusion is optimized by scaffolded alignment, bounded merger, selective product management, channel-selective gates, and path-aware control.

(183)

The most concise proposed design law is

|                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{Optimize } (k, \tau_S, \lambda, \tau_P, c, d, \delta, T, N_{\text{eff}}, \xi, \sigma, z, \vartheta, g, c_F, c_P, \rho, q_P/q_M, \pi, e, u, n, \nu, S, \\ s, \mathcal{F}_D, h_B, t^*, \mathbf{a}, \boldsymbol{\varphi}, r, y, \omega, \gamma, y_c, \mathbf{f}, \Gamma, m_{\text{moat}}, \kappa, \alpha_T)$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

(184)

Here  $k$  is scaffold count,  $\tau_S$  scaffold lifetime,  $\lambda$  fusion degree,  $\tau_P$  product residence,  $c$  network or bridge connectivity,  $d$  complementarity distance,  $\delta$  duty fraction,  $T$  pulse period,  $N_{\text{eff}}$  effective diversity,  $\xi$  product-recycling fraction,  $\sigma$  annealing amplitude,  $z$  impedance ratio,  $\vartheta$  useful-shadow channel angle,  $g$  signed memory coefficient,  $(c_F, c_P)$  useful and shadow connectivity,  $\rho$  anchor overlap,  $q_P/q_M$  reset selectivity,  $\pi$  fusion path,  $e$  scout effort,  $(u, n)$  reciprocal provide/need maps,  $\nu$  scaffold shadow price,  $S$  scaffold budget,  $s$  sink capacity,  $\mathcal{F}_D$  defusion number,  $h_B$  boundary-layer thickness,  $t^*$  post-selection threshold,  $\mathbf{a}$  compliance/action vector,  $\boldsymbol{\varphi}$  phase-offset vector,  $r$  reversibility reserve,  $y$  dopant or impurity dose,  $\omega$  witness-scaffold fraction,  $\gamma$  action-geodesic path,  $y_c$  counter-shadow cofusion dose, and  $\mathbf{f}$  ruin-limited portfolio weights.

The highest-value next step is to convert this theory from analogy into measurement. For each domain, define an empirical mismatch distance  $D$ , estimate barrier convexity  $p$ , measure scaffold cost and lifetime, control fusion degree  $\lambda$ , independently vary product residence  $\tau_P$ , scan duty and recovery period, map local interface stiffness, estimate effective diversity, identify product-recycling windows, scan finite annealing amplitude, measure impedance-like mismatch, estimate useful-shadow channel angle, measure signed memory  $g$ , separate useful and shadow connectivity  $(c_F, c_P)$ , scan anchor overlap  $\rho$ , test reset selectivity  $q_P/q_M$ , compare fusion paths  $\pi$ , measure scout-value curves over effort  $e$ , map reciprocal scaffold vectors  $(u, n)$ , allocate scaffold by shadow price  $\nu$ , tune sink capacity  $s$ , estimate defusion number  $\mathcal{F}_D$ , vary boundary-layer thickness  $h_B$ , tune post-selection threshold  $t^*$ , estimate compliance/action vector  $\mathbf{a}$ , vary phase offsets  $\boldsymbol{\varphi}$ , preserve reversibility reserve  $r$ , sweep dopant dose  $y$ , scan witness fraction  $\omega$ , map action-geodesic paths  $\gamma$ , test counter-shadow cofusion dose  $y_c$ , estimate ruin-limited portfolio weights  $\mathbf{f}$ , tune spectral aperture  $\Gamma$ , sweep selective moat strength  $m$ , map curvature  $\kappa$ , and estimate tail-shadow index  $\alpha_T$ . If the predicted scalings survive those tests, the framework may become a useful general theory of fusion-like processes.

## A Supplementary virtual-environment details

The companion simulation files store the random seed, code, summary tables, and parameter subsets. The seed used for the reported virtual tests is 20260507. The random tests are reproducibility checks, not claims about natural parameter distributions.

### A.1 Information fusion model

The estimator error matrix was sampled by choosing random noise scales  $s_1, s_2$ , correlation  $\rho$ , and biases  $b_1, b_2$ . The second moments were

$$e_{11} = s_1^2 + b_1^2, \quad e_{22} = s_2^2 + b_2^2, \quad e_{12} = \rho s_1 s_2 + b_1 b_2. \quad (185)$$

The equal-weight estimate used  $w = 1/2$ . The adaptive estimate used Eq. (21).

## A.2 Bridge model

The random bridge test sampled  $D$ ,  $p$ , and  $c$ , then minimized Eq. (6) over integer  $k \in \{1, \dots, 60\}$ . The direct barrier was  $D^p$ . Barrier reduction was  $1 - \mathcal{A}_{k^*}/D^p$ .

## A.3 Incomplete fusion model

The partial-fusion test sampled  $a, b, p$  and computed Eq. (13). The reported partial fraction is the fraction of cases with  $\lambda^* < 1$ .

## A.4 Product-shadow model

The product-shadow test used

$$r(T) = T^2 \exp(-1/T), \quad A = \tau_P r(T), \quad (186)$$

with heating  $Gr(T) \exp[-\alpha A]$  and loss  $T + \beta\sqrt{T}$ . The threshold raw gain was computed by minimizing over  $T$ .

## A.5 Turnover-matched scaffold model

The second sprint sampled  $\eta$ ,  $\tau_A$ ,  $\tau_P$ , and  $\gamma$  log-uniformly and used Eq. (29) to compute  $\tau_S^*$ . The data file `data/scaffold_lifetime_sample.csv` stores a 10,000-row sample. The figures are `fig_scaffold_lifetime_distribution` and `fig_scaffold_lifetime_curve`.

## A.6 Selective confinement model

The second sprint evaluated Eq. (35) on a  $120 \times 120$  grid of  $(\tau_E, \tau_P)$ . The output files are `data/selective_confinement_grid.csv` and `data/selective_confinement_boundary.csv`. The threshold heatmap and boundary figures are stored as `fig_selective_confinement_heatmap` and `fig_selective_confinement_boundary`.

## A.7 Critical network-fusion model

The second sprint used Eq. (38) with random  $a, b, d, q$  and minimized over  $c \in [0, 5]$ . The giant-component curve used Eq. (37). The data file is `data/percolation_fusion_sample.csv`; the figures are `fig_percolation_opt_c_distribution`, `fig_percolation_giant_component`, and `fig_percolation_value_curve`.

## A.8 Exceptional-fusion ridge model

The second sprint sampled  $m, p, \mu$  and computed Eq. (40). The data file is `data/exceptional_fusion_ridge_sample.csv`; the figures are `fig_exceptional_ridge_curve` and `fig_exceptional_ridge_dstar_distribution`.

## A.9 Second sprint reproducibility files

The additional code is stored at `code/simulate_fusion_principles_v02.py`. The summary tables are `data/simulation_summary_v02.json` and `data/simulation_summary_v02.csv`. The random seed for all second-sprint experiments is 20260507, with deterministic offsets per experiment.

## A.10 Third sprint reproducibility files

The third-sprint code is stored at `code/simulate_fusion_principles_v03.py`. Summary files are `data/simulation_summary_v03.json` and `data/simulation_summary_v03.csv`. The random seed is 20260510.

## A.11 Pulse-shadow duty model

The pulse-shadow Monte Carlo sampled average drive  $U$ , activation exponent  $q$ , shadow coefficient  $K$ , and peak-ratio cap. It computed Eq. (43) and stored `data/pulse_shadow_sample.csv`. The figures are `fig_pulse_shadow_curve` and `fig_pulse_shadow_delta_hist`.

## A.12 Recovery-window model

The recovery-window scan evaluated Eq. (49) on a grid of  $(\delta, T)$ . A 15,000-row sample is stored in `data/recovery_window_grid.csv`; the representative trajectory is `data/recovery_window_optimal_trace.csv`. Figures are `fig_recovery_window_heatmap` and `fig_recovery_window_trace`.

## A.13 Nonuniform bridge model

The nonuniform bridge test sampled random heterogeneous coefficients  $a_i$ , exponent  $p$ , and total mismatch  $D$ . It compared equal-distance allocation with Eq. (52). Outputs are `data/nonuniform_bridge_sample.csv` and `data/nonuniform_bridge_representative.csv`. Figures are `fig_nonuniform_bridge_reduction_hist`, `fig_nonuniform_bridge_allocation`, and `fig_nonuniform_bridge_cv_scatter`.

## A.14 Diversity-limited fusion model

The diversity-limited fusion test sampled nominal source count  $N$  and common error correlation  $\rho$ , then computed Eq. (56). Outputs are `data/diversity_limited_fusion_sample.csv` and `data/random_covariance_fusion_rank.csv`. Figures are `fig_diversity_correlation_ceiling`, `fig_diversity_neff_hist`, and `fig_diversity_rank_scatter`. The random-covariance diagnostic was retained as a warning: simple covariance effective rank is not a universal substitute for calibrated error covariance and no-harm fusion.

### A.15 Fourth sprint reproducibility files

The fourth-sprint code is stored at `code/simulate_fusion_principles_v04.py`. Summary files are `data/simulation_summary_v04.json` and `data/simulation_summary_v04.csv`. The random seed is 20260517. All fourth-sprint experiments are dimensionless toy models designed to generate falsifiable hypotheses.

### A.16 Shadow-to-scaffold recycling model

The recycling test sampled product load  $L$ , scaffold load  $M$ , shadow strength  $\alpha$ , scaffold benefit  $\beta$ , occlusion coefficient  $\gamma$ , and conversion cost  $c$ , then computed Eq. (58). Outputs are `data/shadow_to_scaffold_recycling_sample.csv`, `fig_shadow_to_scaffold_curve`, and `fig_shadow_to_scaffold_xi_hist`.

### A.17 Interface-annealing model

The annealing test sampled cold barrier action  $A$ , annealing efficacy  $a$ , and noise damage  $\mu$ , then computed Eq. (63) under  $\sigma_{\max} = 3$ . Outputs are `data/interface_annealing_sample.csv`, `fig_interface_annealing_curve`, and `fig_interface_annealing_sigma_hist`.

### A.18 Impedance-matched scaffold model

The impedance-matching test sampled impedance ratio  $z$  and per-layer loss  $c$ , evaluated Eq. (66) for  $k \in \{0, \dots, 80\}$ , and compared the exact discrete optimum with Eq. (68). Outputs are `data/impedance_matched_scaffold_sample.csv`, `fig_impedance_matching_curve`, `fig_impedance_matching_k_hist`, and `fig_impedance_matching_heatmap`.

### A.19 Orthogonal-gate fusion model

The orthogonal-gate test sampled useful-shadow channel angle, product-shadow penalty  $\kappa$ , and gate cost  $c_g$  in an eight-dimensional channel space. It computed the leading eigenvalue in Eq. (71), compared directional and forced-isotropic gates, and measured useful/product selectivity. Outputs are `data/orthogonal_gate_fusion_sample.csv`, `fig_orthogonal_gate_value_curve`, `fig_orthogonal_gate_selectivity_scatter`, and `fig_orthogonal_gate_selectivity_hist`.

### A.20 Fifth sprint reproducibility files

The fifth-sprint code is stored at `code/simulate_fusion_principles_v05.py`. Summary files are `data/simulation_summary_v05.json` and `data/simulation_summary_v05.csv`. The random seed is 20260517.

### A.21 Autocatalytic scaffold-shadow model

The autocatalytic experiment sampled the terms in Eq. (75), classified each case by the sign of  $g$  and the fold condition  $gR_0 = e^{-1}$ , and stored `data/autocatalytic_bifurcation_sample.csv`.

The branch curve is stored in `data/autocatalytic_bifurcation_branches.csv`. Figures are `fig_autocatalytic_bifurcation_branches` and `fig_autocatalytic_feedback_ratio_hist`.

## A.22 Dual-percolation model

The dual-percolation experiment evaluated Eq. (83) over a grid of  $(c_F, c_P)$ , using the giant-component equation  $S = 1 - e^{-cS}$ . Outputs are `data/dual_percolation_grid.csv`, `data/dual_percolation_sample.csv`, and `data/dual_percolation_rectification_curves.csv`. Figures are `fig_dual_percolation_value_heatmap` and `fig_dual_percolation_rectification_curves`.

## A.23 Anchor-complementarity model

The anchor-complementarity experiment sampled  $(U, A, \epsilon)$  in Eq. (84) and computed Eq. (86). Outputs are `data/anchor_complementarity_sample.csv` and `data/anchor_complementarity_representative_curve.csv`. Figures are `fig_anchor_complementarity_curve` and `fig_anchor_complementarity_rhostar_hist`.

## A.24 Memory-selective reset model

The reset experiment used a vectorized cyclic model with beneficial memory  $M$ , product shadow  $P$ , and four policies: no reset, blind adaptive reset, selective adaptive reset, and over-reset. Outputs are `data/memory_selective_reset_sample.csv` and `data/memory_selective_reset_representative_trace.csv`. Figures are `fig_memory_selective_reset_trace` and `fig_memory_selective_reset_gain_hist`.

## A.25 Sixth sprint reproducibility files

The sixth-sprint code is stored at `code/simulate_fusion_principles_v06.py`. Summary files are `data/simulation_summary_v06.json` and `data/simulation_summary_v06.csv`. The random seed is 20260525.

## A.26 Shadow-vector balancing model

The shadow-vector experiment sampled five pathways with four-dimensional product-shadow vectors and optimized Eq. (94) over the simplex by projected-gradient ascent. Outputs are `data/shadow_vector_balancing_sample.csv`, `fig_shadow_vector_gain_hist`, and `fig_shadow_vector_cancellation_scatter`.

## A.27 Latency-instability model

The latency experiment simulated Eq. (100) with random  $\eta, \lambda_0, \kappa, \alpha, \Delta$ , computed the linearized critical delay in Eq. (102), and measured late-time shadow and output oscillations. Outputs are the latency-instability CSV files in `data/` and figures `fig_latency_instability_boundary` and `fig_latency_instability_trajectories`.

### A.28 Sparse-contact topology model

The sparse-contact experiment sampled  $M, b, a, \beta, \gamma$  in Eq. (106), rounded the continuous optimum in Eq. (107), and compared against single-contact and full-contact baselines. Outputs are `data/sparse_contact_topology_sample.csv`, `data/sparse_contact_representative_curve.csv`, `fig_sparse_contact_fraction_hist`, and `fig_sparse_contact_value_curve`.

### A.29 Hierarchical-granularity model

The granularity experiment sampled  $N, A, B, p, q$  in Eq. (109), rounded Eq. (110), and compared against atomized ( $m = 1$ ) and flat ( $m = N$ ) fusion. Outputs are the hierarchical-granularity CSV files in `data/` and figures `fig_hierarchical_granularity_fraction_hist` and `fig_hierarchical_granularity_cost_curve`.

### A.30 Seventh sprint reproducibility files

The seventh-sprint code is stored at `code/simulate_fusion_principles_v07.py`. Summary files are `data/simulation_summary_v07.json` and `data/simulation_summary_v07.csv`. The random seed is 20260601.

### A.31 Non-associative sequence model

The sequencing experiment used Eq. (114) with seven components. The exact optimum was computed by dynamic programming over subsets because the centroid and cumulative shadow depend only on the already-fused set. Outputs are `data/nonassociative_sequence_sample.csv`, `data/nonassociative_sequence_representative.csv`, `fig_nonassociative_order_penalty_hist`, and `fig_nonassociative_representative_trace`.

### A.32 Scout-fusion model

The scouting experiment sampled Gaussian prior values, probe noise, and effort costs, then optimized Eq. (119) over a finite effort grid. Outputs are `data/scout_fusion_value_sample.csv`, `data/scout_fusion_value_binned.csv`, `fig_scout_value_vs_uncertainty`, and `fig_scout_effort_hist`.

### A.33 Reciprocal scaffold complementarity model

The reciprocal experiment sampled candidate libraries with output vectors, need vectors, product-shadow vectors, benefits, and barriers, then evaluated Eq. (120). Outputs are `data/reciprocal_scaffold_complementarity_sample.csv`, `data/reciprocal_scaffold_representative_pairs.csv`, `fig_reciprocal_pair_gain_hist`, and `fig_reciprocal_hidden_pair_scatter`.

### A.34 Scarce scaffold allocation model

The allocation experiment solved Eq. (122) using the water-filling condition in Eq. (123). Outputs are `data/scarce_scaffold_allocation_sample.csv`, `data/scarce_scaffold_allocation_binned.csv`, `data/scarce_scaffold_representative_allocation.csv`, `fig_scarce_scaffold_gain_hist`, `fig_scarce_scaffold_active_fraction`, and `fig_scarce_scaffold_representative_allocation`.

### A.35 Ninth-sprint virtual experiments

The compliance-partition experiment sampled mismatch distance, convexity, and three action coefficients, then evaluated the exact optimum in Eq. (140). Outputs are `data/compliance_partition_sample.csv`, `data/compliance_partition_representative_curve.csv`, `fig_compliance_partition_penalty_hist`, and `fig_compliance_partition_contrast_curve`.

The anti-coincidence experiment sampled three to eight product-producing pathways with periodic Gaussian product pulses, random amplitudes, random pulse widths, and random convex shadow exponent  $q$ . It compared synchronous phases with evenly staggered phases and stored the narrow-pulse prediction in Eq. (143). Outputs are `data/anticoincidence_scheduling_sample.csv`, `data/anticoincidence_representative_traces.csv`, `fig_anticoincidence_load_traces`, `fig_anticoincidence_gain_hist`, and `fig_anticoincidence_theory_curve`.

The reversibility-reserve experiment sampled hidden-shadow probability, loss scale, protection efficiency, reserve costs, and convex cost exponent, then optimized Eq. (145). Outputs are `data/reversibility_reserve_sample.csv`, `data/reversibility_reserve_binned.csv`, `data/reversibility_reserve_representative_curve.csv`, `fig_reversibility_reserve_rstar_hist`, `fig_reversibility_reserve_threshold_curve`, and `fig_reversibility_reserve_representative_curve`.

The controlled-dopant experiment sampled saturating barrier-relief strength, saturation dose, poisoning strength, high-dose exponent, direct dose cost, and baseline rate, then optimized Eq. (148). Outputs are `data/controlled_dopant_window_sample.csv`, `data/controlled_dopant_window_binned.csv`, `data/controlled_dopant_representative_curve.csv`, `fig_controlled_dopant_hstar_hist`, `fig_controlled_dopant_representative_curve`, and `fig_controlled_dopant_threshold_curve`. Summary outputs are `data/simulation_summary_v09.csv` and `data/simulation_summary_v09.json`.

### A.36 Tenth-sprint virtual experiments

The witness-scaffold experiment sampled hidden-shadow uncertainty, witness informativeness, and witness cost, then optimized Eq. (153). Outputs are `data/witness_scaffold_identifiability_sample.csv`, `data/witness_scaffold_representative_curve.csv`, `fig_witness_scaffold_gain_hist`, and `fig_witness_scaffold_value_curve`.

The action-geodesic experiment generated two-dimensional action-density fields, computed a Dijkstra minimum-action path, and compared it with a direct midline path using Eq. (156). Outputs are `data/geodesic_scaffold_path_sample.csv`, `data/geodesic_scaffold_representative_field.csv`, `data/geodesic_scaffold_representative_path.csv`, `fig_geodesic_scaffold_representative_path`, and `fig_geodesic_scaffold_savings_hist`.

The counter-shadow cofusion experiment optimized the quadratic value in Eq. (159) over primary and secondary drive. Outputs are `data/counter_shadow_cofusion_sample.csv`, `data/counter_shadow_cofusion_representative_curve.csv`, `fig_counter_shadow_cofusion_y_hist`, and `fig_counter_shadow_cofusion_curve`.

The ruin-limited portfolio experiment sampled four-pathway catalogs with useful returns, covariance-like shadow matrices, and tail-risk surrogates, then solved Eq. (163). Outputs are `data/ruin_limited_portfolio_sample.csv`, `data/ruin_limited_portfolio_binned.csv`, `data/ruin_limited_portfolio_representative.csv`, `fig_ruin_limited_portfolio_entropy_hist`, and `fig_ruin_limited_portfolio_representative`. Summary outputs are `data/simulation_summary_v10.csv` and `data/simulation_summary_v10.json`.

### A.37 Eleventh-sprint virtual experiments

The mode-locked aperture experiment sampled useful and parasitic detuned modes, then optimized Eq. (167). Outputs are `data/mode_locked_aperture_sample.csv`, `data/mode_locked_aperture_parameter_sample.csv`, `data/mode_locked_aperture_representative_curve.csv`, `fig_mode_locked_gamma_hist`, and `fig_mode_locked_representative_curve`.

The selective moat experiment sampled target value, parasitic shadow, attenuation rates, and moat cost, then optimized Eq. (169). Outputs are `data/selective_moat_sample.csv`, `data/selective_moat_representative_curve.csv`, `fig_selective_moat_mstar_hist`, and `fig_selective_moat_representative_curve`.

The curvature-window experiment sampled curvature activation, damage, and cost parameters, then optimized Eq. (172). Outputs are `data/curvature_window_sample.csv`, `data/curvature_window_representative_curve.csv`, `fig_curvature_kstar_hist`, and `fig_curvature_representative_curve`.

The extreme-tail shadow experiment sampled ordinary fusion benefits and costs, heavy-tail indices, cycle counts, and tail severity, then compared the ordinary optimum with Eq. (177). Outputs are `data/extreme_tail_shadow_sample.csv`, `data/extreme_tail_shadow_binned.csv`, `data/extreme_tail_shadow_representative_curves.csv`, `fig_extreme_tail_lambda_ratio_hist`, and `fig_extreme_tail_representative_curves`. Summary outputs are `data/simulation_summary_v11.csv` and `data/simulation_summary_v11.json`. The eleventh-sprint code is `code/simulate_fusion_principles_v11.py`; the random seed is 110107.

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## Addendum 1.3: Control Allocation, Tensor Directionality, Backpressure, and Abort-Gated Fusion

Scaffolded Incomplete Fusion Theory: A Cross-Domain Interface-Barrier-Product Framework

*Research sprint 13. Candidate theory and virtual experiments.*

### Abstract of this addendum

This addendum extends the interface-barrier-product framework by testing four new blindspots in fusion-like systems: shared control budgets, anisotropic fusion directions, downstream transport backpressure, and dynamic abort gates. The central finding is that useful fusion is often limited less by the absence of a fusion event than by misallocation of control effort, choosing the wrong direction of contact, exceeding downstream product-handling capacity, or failing to stop after early shadow signals. Four virtual experiments were run and preserved as CSV/JSON data and executable source. The results support four additional candidate laws: the Coupled Scaffold-Exhaust Budget Law, the Anisotropic Fusion Tensor Theorem, the Backpressure-Capped Fusion Law, and the Abort-Gated Fusion Stopping Law.

The exceptional pattern across these tests is that optimal fusion may require spending resources on non-fusion functions: exhaust, diagnostics, direction selection, refusal, and rollback. In this sense, fusion design is not merely reaction design. It is a control allocation problem over attraction, resistance, transport, and stopping.

### 1 Motivating blindspots

Earlier sections developed the scaffold-shadow picture: barriers can be reduced by intermediate scaffolds, complete merger is often inferior to partial merger, and product shadow can dominate primary yield. The new question is whether this theory remains under-specified in four ways.

First, many systems draw scaffold formation and product removal from the same finite control budget. In magnetic-confinement fusion, for example, power exhaust and boundary control are active research constraints, as illustrated by recent Super-X divertor exhaust-control demonstrations and roadmap emphasis on plasma-facing components and exhaust systems [1, 2]. In biological fusion, surface crowding can create a tunable energetic barrier to cell-cell fusion, showing that the non-fusing interface environment can govern whether fusion occurs at all [3]. In multimodal information fusion, missing or noisy modalities show that alignment and quality estimation are as important as the nominal merge operation [4].

Second, mismatch is not a scalar. A fusion interface can have directions: membrane curvature directions, crystal axes, magnetic-field-line directions, time-frequency modes, latent-space axes, or organizational modules. Treating all mismatch as a scalar hides the possibility that the raw attraction direction is also the direction of maximum damage.

Third, downstream product transport can create backpressure. Product shadow is not only a local penalty; it may be a capacity bottleneck across scales. A microscopic fusion rate can be too high for macroscopic exhaust, heat removal, memory bandwidth, immune clearance, or quality-control capacity.

Fourth, fusion systems need abort gates. A system that starts fusing but observes early signs of product shadow may gain more by stopping than by pursuing nominal completion. This is distinct from pre-fusion scouting: it is post-initiation optimal stopping.

## 2 Candidate law 35: Coupled Scaffold-Exhaust Budget Law

**Candidate law 1** (Coupled Scaffold-Exhaust Budget Law). *When a common control budget must be divided between barrier reduction and product-shadow reduction, useful fusion is optimized by equalizing marginal logarithmic returns, not by allocating all control to the apparent fusion event.*

Let a total control budget  $B$  be divided into scaffold/control effort  $u$  and exhaust/shadow-control effort  $v = B - u$ . A minimal useful-rate model is

$$\log R(u) = \log R_0 - \frac{A}{\Theta + s(u)} - \frac{H}{1 + e(B - u)}, \quad (1)$$

where  $A$  is primary barrier action,  $\Theta$  is effective fluctuation or temperature scale,  $H$  is the product-shadow burden,  $s(u)$  is scaffold efficacy, and  $e(v)$  is exhaust efficacy. For the common saturating-root case,

$$s(u) = \sigma u^a, \quad e(v) = \eta v^b, \quad 0 < a, b < 1, \quad (2)$$

any interior optimum satisfies

$$\boxed{\frac{A\sigma a u^{a-1}}{(\Theta + \sigma u^a)^2} = \frac{H\eta b (B - u)^{b-1}}{(1 + \eta(B - u)^b)^2}} \quad (3)$$

The left side is the marginal log-rate gain from additional scaffold control. The right side is the marginal log-rate gain from additional product-shadow control. The law predicts that all-to-scaffold allocation is optimal only if the left marginal return stays above the right marginal return across the whole budget interval; all-to-exhaust allocation is optimal only under the opposite inequality.

This law reframes a common design error. Increasing the probability of fusion may reduce useful fusion if the additional product stream cannot be exported. The best reactor, synthetic cell, weld process, or multimodal model may therefore reserve a large fraction of resources for what looks like non-fusion overhead.

### Virtual experiment 13.1

Fifty thousand random systems were sampled with

$$\log R = -\frac{A}{\Theta + \sigma\sqrt{u}} - \frac{H}{1 + \eta\sqrt{B - u}}, \quad 0 \leq u \leq B. \quad (4)$$

Grid optimization over the budget fraction found an interior split in 92.740% of systems. The median optimum allocated 66.750% of the budget to scaffold/control and 33.250% to exhaust/shadow control. The optimized split produced a median useful-rate gain of  $1.280\times$  relative to the better of the two boundary policies, with a 90th percentile gain of  $4.554\times$ . All-to-scaffold was optimal in only 2.814% of sampled systems; all-to-exhaust was optimal in 0.254%.

## 3 Candidate law 36: Anisotropic Fusion Tensor Theorem

**Candidate theorem 1** (Anisotropic Fusion Tensor Theorem). *If gain, barrier, and product shadow are directional tensors, the optimal infinitesimal fusion mode is the leading generalized eigenmode of the net fusion tensor. It need not align with the direction of largest raw complementarity.*

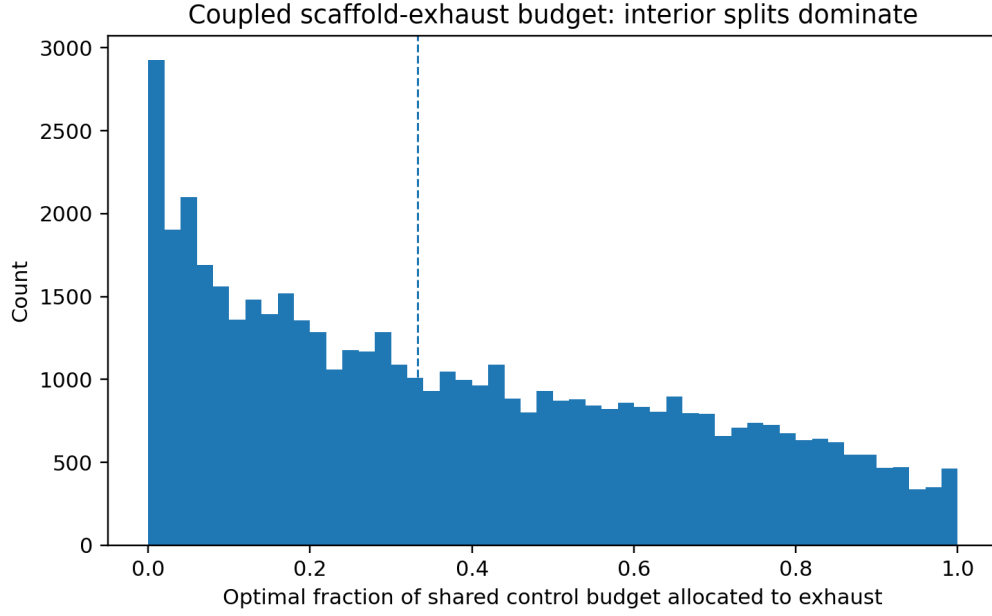


Figure 1: Optimal exhaust allocation in the shared-budget virtual experiment. The broad interior mass indicates that product-shadow control is not an afterthought; it is often part of the optimal fusion mechanism.

Let  $z$  be a normalized local fusion displacement, contact mode, spectral aperture, crystal-orientation move, membrane-deformation mode, or latent-space coupling direction. Define symmetric tensors  $G$ ,  $B$ , and  $P$  for fusion gain, barrier action, and product-shadow cost. Under a metric  $W$ , the local useful fusion increment is

$$\Delta V(z) = z^T (G - B - P)z, \quad z^T Wz = 1. \quad (5)$$

Therefore

$$z^* = \arg \max_{z \neq 0} \frac{z^T (G - B - P)z}{z^T Wz}, \quad (6)$$

and  $z^*$  solves

$$(G - B - P)z^* = \lambda_{\max} Wz^*. \quad (7)$$

A local fusion mode is viable if and only if  $\lambda_{\max} > 0$ . The raw highest-affinity mode, given by the leading eigenvector of  $G$  alone, can be a *shadow trap* when the shadow tensor is correlated with gain.

This theorem is almost certainly a better local description than scalar mismatch whenever fusion has orientation: crystal grains, anisotropic membranes, magnetic field geometry, laser-target asymmetries, cross-attention heads, multiple sensors, or tissue interfaces. It also suggests a practical diagnostic: measure not only the strongest coupling direction but the covariance of coupling with shadow.

## Virtual experiment 13.2

Sixteen thousand anisotropic systems were generated in five dimensions. The virtual catalog deliberately included a superlinear correlation between raw complementarity and product shadow, testing the hypothesis that the most attractive direction may also be the most dangerous. The tensor optimum was positive in 86.181% of systems. In 16.244% of all sampled systems, the naive highest-complementarity direction had

non-positive utility while the tensor-optimal direction had positive utility. Conditional on a positive tensor optimum, the optimal direction selected a non-top raw-complementarity eigenaxis in 36.645% of systems.

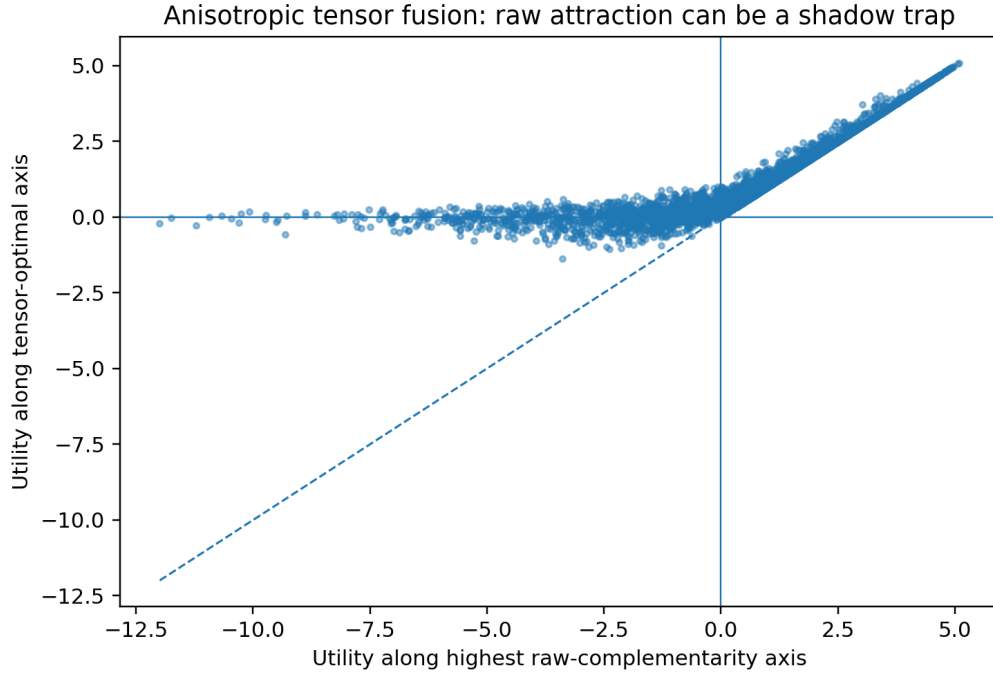


Figure 2: Naive raw-affinity direction versus tensor-optimal direction. Points above the diagonal are systems where accounting for directional barrier and shadow improves utility. The quadrant with naive utility below zero and tensor utility above zero represents rescued fusion modes.

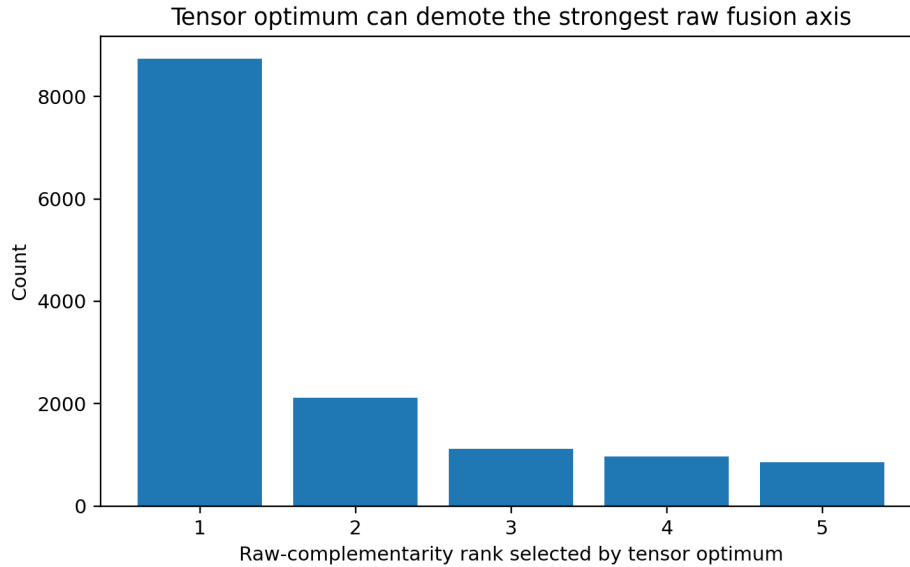


Figure 3: Raw-complementarity rank selected by the tensor optimum. Rank 1 is the strongest raw-affinity axis. Non-top selections are cases where the best useful fusion direction is hidden below the strongest nominal attraction direction.

## 4 Candidate law 37: Backpressure-Capped Fusion Law

**Candidate law 2** (Backpressure-Capped Fusion Law). *When fusion products must pass through a downstream transport or processing capacity  $C$ , useful output is often maximized near the capacity boundary rather than at maximum upstream fusion drive.*

Let an upstream drive variable  $x$  produce raw fusion rate  $R(x)$  with  $R'(x) > 0$  and saturation at  $R_{\max}$ . Let downstream capacity be  $C$ . Define useful output

$$U(x) = R(x) \exp[-\gamma \max(0, R(x) - C)^q - cx], \quad q > 1. \quad (8)$$

For  $R(x) > C$ , an interior optimum satisfies

$$\frac{R'(x)}{R(x)} - \gamma q [R(x) - C]^{q-1} R'(x) - c = 0. \quad (9)$$

Equivalently, the excess over capacity is approximately

$$R^* - C \approx \left[ \frac{1/R^* - c/R'(x^*)}{\gamma q} \right]^{1/(q-1)} \quad (10)$$

when the numerator is positive. For strong convex backpressure,  $R^*$  lies close to  $C$ .

The discovery here is that product-shadow management is not only a residence-time problem. It is also a throughput-matching problem. Excess production can reduce useful output even if the primary fusion event is efficient.

### Virtual experiment 13.3

Thirty thousand systems were sampled with

$$R(x) = R_{\max} (1 - e^{-kx}), \quad U(x) = R(x) e^{-\gamma \max(0, R(x) - C)^q - cx}. \quad (11)$$

Downstream capacity was active in 88.707% of systems. Conditional on active capacity, the optimum production-to-capacity ratio had median  $R^*/C = 1.113$ , and 61.269% of active-capacity systems lay in the interval  $0.8 < R^*/C < 1.2$ . Maximum drive was optimal in only 1.957% of all systems. The median log-gain of optimized drive versus maximum drive was 1.556, equivalent to a clipped median gain of  $4.740\times$ .

## 5 Candidate law 38: Abort-Gated Fusion Stopping Law

**Candidate law 3** (Abort-Gated Fusion Stopping Law). *A fusion process with observable product-shadow signals should continue only while expected marginal gain exceeds expected marginal shadow cost. The abort gate is part of the fusion mechanism, not merely a safety feature.*

Let  $S_t$  be accumulated product shadow and  $b_t$  be expected incremental benefit from one more fusion interval. For product penalty  $P(S)$ , the continuation condition is

$$b_t > \mathbb{E}[P(S_t + \Delta S_t) - P(S_t) \mid \mathcal{F}_t]. \quad (12)$$

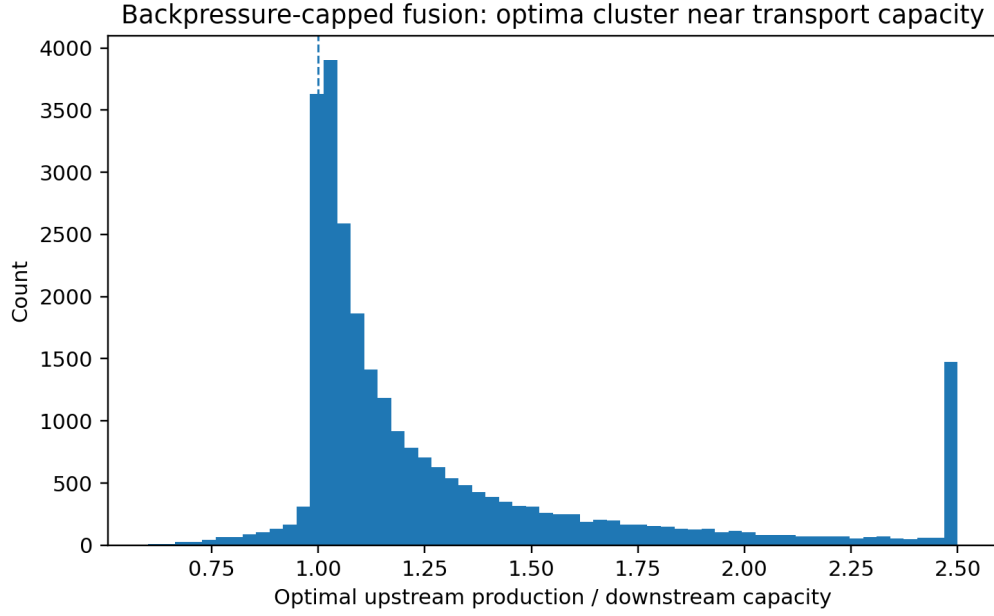


Figure 4: Distribution of optimal upstream production divided by downstream capacity. The peak near unity supports a capacity-matching principle: many fusion systems should run at the edge of product-processing capacity rather than at maximum primary drive.

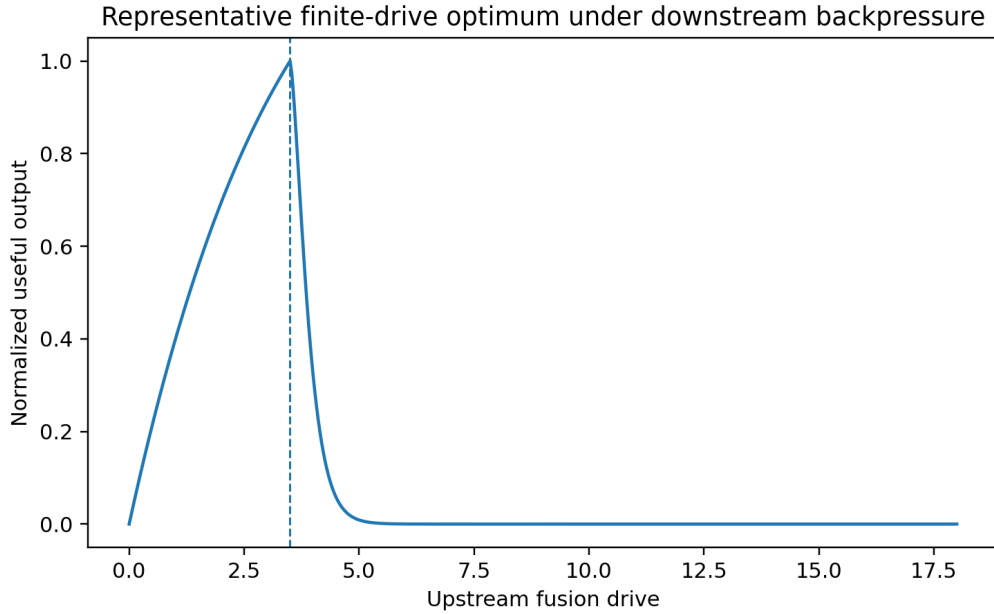


Figure 5: Representative finite-drive optimum. The useful output rises with drive until product transport capacity is reached, then collapses as backpressure shadow grows.

For small increments,

$$b_t \gtrsim P'(S_t) \mathbb{E}[\Delta S_t \mid \mathcal{F}_t]. \quad (13)$$

With  $P(S) = \kappa S^p$  and mean shadow increment  $\bar{s}$ , the deterministic threshold is

$$S^* \approx \left( \frac{b}{\kappa p \bar{s}} \right)^{1/(p-1)}. \quad (14)$$

Fusion should stop, pause, quench, defuse, or route to repair when the observed shadow estimate exceeds this threshold.

This law unifies several design actions that are often treated separately: early termination of a failed weld, pulse termination in plasma control, immune checkpointing around fused cells, stopping multimodal inference when a modality becomes adversarial, and quenching a chemical/material fusion front before scarring dominates.

### Virtual experiment 13.4

Two thousand five hundred stochastic fusion processes were simulated. Each case contained 240 sample paths and up to 40 fusion intervals. Shadow increments were lognormal and observed through additive noise. Threshold policies stopped the process when the observed product-shadow signal exceeded a chosen threshold. Optimized abort gates improved value over no-abort fusion in 68.480% of systems. The median gain versus no-abort was 5.873 value units, with a 90th percentile gain of 340.549. The median optimized process stopped after 18.427 of 40 intervals. The analytic marginal threshold retained a median 99.824% of the optimized threshold's value advantage relative to no abort.

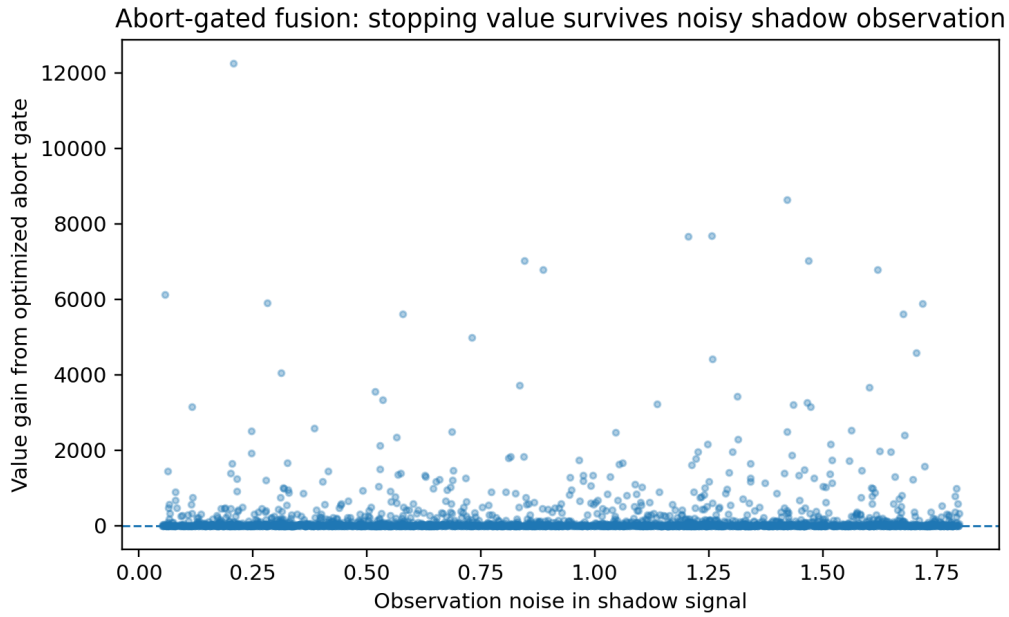


Figure 6: Value gain from optimized abort gating as a function of shadow-observation noise. Abort value persists across a broad range of signal quality, though high-noise systems show greater variance.

## 6 Cross-law synthesis: fusion as resource-priced contact

The new laws suggest a stronger formulation of the theory. A fusion design should be evaluated not only by whether two systems can be joined, but by whether the contact is correctly priced across four coupled

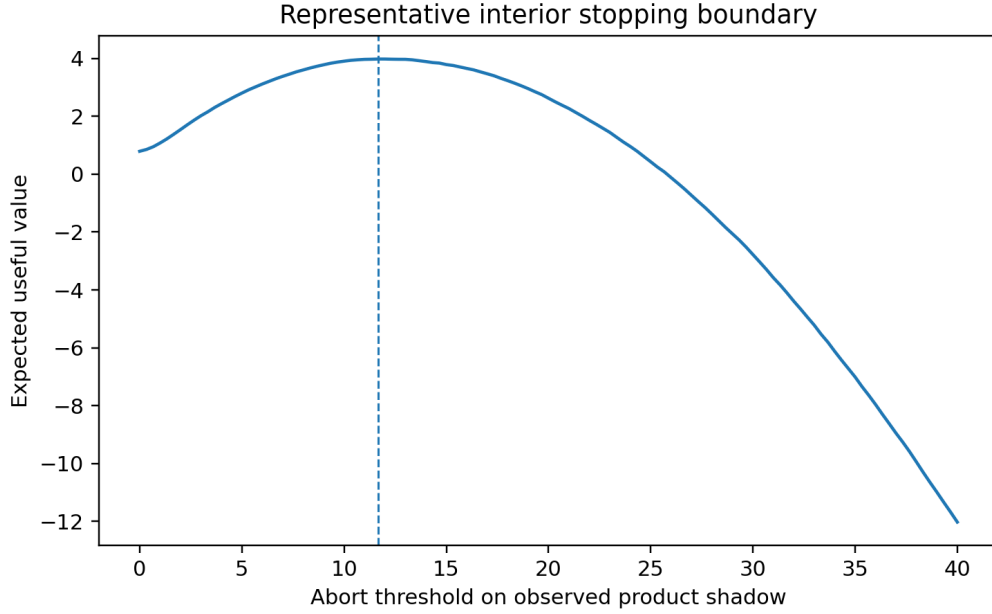


Figure 7: Representative interior stopping boundary. Too-low thresholds abandon useful fusion prematurely; too-high thresholds allow product shadow to dominate.

axes:

$$\mathcal{V} = \max_{u,z,x,\pi} \mathbb{E} \left[ \sum_t B_t(z, x, u) - A_t(z, u) - P_t(S_t, x) - C_t(u, x, \pi) \right], \quad (15)$$

where  $u$  allocates shared control budget,  $z$  selects fusion direction,  $x$  controls upstream production drive, and  $\pi$  is an abort/continue policy. This expression extends the earlier  $(k, \lambda, \tau_P)$  triad. The expanded design vector is

$$(k, \lambda, \tau_P, u, z, x, \pi), \quad (16)$$

with  $k$  controlling scaffold granularity,  $\lambda$  fusion degree,  $\tau_P$  product residence,  $u$  control allocation,  $z$  contact direction,  $x$  rate relative to capacity, and  $\pi$  stopping policy.

The strongest new conceptual result is the **shadow-trap effect**: the direction or policy that maximizes raw fusion probability can also maximize damage. This effect appeared in two different forms. In the tensor experiment, the highest-complementarity axis was non-viable while a hidden tensor axis remained viable in 16.244% of sampled systems. In the backpressure experiment, maximum drive was optimal in only 1.957% of sampled systems. These are not proofs about physical fusion reactors or biological cells; they are robust warnings from abstract fusion models.

## 7 Targets for further study

### Target 1: Scaffold-exhaust elasticity maps

For a given fusion system, estimate the two marginal elasticities

$$E_s(u) = \left. \frac{\partial \log R}{\partial u} \right|_{\text{scaffold}}, \quad E_e(v) = \left. \frac{\partial \log R}{\partial v} \right|_{\text{exhaust}}. \quad (17)$$

The immediate experimental target is to map where  $E_s = E_e$ . Applications include heating versus divertor-control allocation in plasmas, fusogen density versus membrane-clearance machinery in cells, heat input versus cooling/extraction in materials joining, and alignment compute versus quality-estimation compute in information fusion.

### Target 2: Fusion tensor atlases

Measure or infer  $G$ ,  $B$ , and  $P$  for candidate fusion directions. In materials, these may be grain-orientation and thermal-stress tensors. In biological membranes, they may be curvature, crowding, lipid composition, and tension modes. In multimodal AI, they may be modality-pair subspaces, attention heads, or latent dimensions. The target is a catalog of shadow traps: axes with large raw gain but negative net eigenvalue.

### Target 3: Backpressure capacity curves

For any product-generating fusion process, measure the downstream capacity  $C$  and the exponent  $q$  governing overload. The proposed falsifiable prediction is that useful output peaks near  $R/C \approx 1$  when  $q > 1$  and  $\gamma$  is large. Deviations from this prediction should identify either hidden buffering, weak overload nonlinearity, or incorrect product-shadow measurement.

### Target 4: Abort-signal calibration

Develop shadow reporters that can trigger stopping before irreversible damage. A minimal experiment is to vary abort thresholds and compare final useful output. The analytic threshold

$$S^* \approx \left( \frac{b}{\kappa p \bar{s}} \right)^{1/(p-1)} \quad (18)$$

should be tested against empirical optimal thresholds.

## 8 Possible applications

**Nuclear and plasma fusion.** The coupled-budget law suggests experiments that deliberately trade heating/current-drive/control effort against exhaust diagnostics, impurity control, divertor configuration, and ash removal. The backpressure law predicts that increasing core performance without matching exhaust capacity can reduce useful operating margin. The abort law motivates predictive pulse termination or mode switching based on early shadow signals such as impurity accumulation, wall loading, or confinement degradation.

**Biological and synthetic cell fusion.** The tensor theorem implies that membrane fusion should be controlled by directional state variables: curvature, surface crowding, cortical tension, local composition, and fusogen patterning. The budget law predicts an optimum between fusogen activation and clearance of crowding molecules or post-fusion stress. Abort-gated fusion suggests reversible or self-limiting fusogen systems that shut down when early damage signals accumulate.

**Materials joining and manufacturing.** Welding, sintering, bonding, and additive-manufacturing fusion fronts can be interpreted as rate-limited by heat flow, stress transport, impurity segregation, and defect removal. The backpressure law predicts that excessive local fusion drive can produce worse joints when

cooling, diffusion, or stress-relaxation capacity is exceeded. The tensor theorem suggests orientation-aware joining routes rather than scalar temperature schedules.

**Information fusion and multimodal AI.** The shared-budget law predicts that compute spent on alignment alone can be inferior to compute split among alignment, uncertainty estimation, gating, and modality repair. The tensor theorem maps naturally to selecting latent subspaces for fusion: the best subspace may not be the one with strongest raw cross-modal correlation if that subspace carries bias, hallucination risk, or noisy-modality shadow. Abort-gated fusion becomes a refusal, early-exit, or re-query policy.

**Organizational and social fusion.** Mergers, interdisciplinary collaborations, and institutional integrations often fail because all effort is spent on contact formation rather than product-shadow handling: integration debt, cultural mismatch, decision bottlenecks, and loss of diversity. The proposed laws predict benefits from explicit control-budget splits, direction-selective integration, capacity-limited rollout, and abortable pilot phases.

## 9 Falsifiable predictions added by version 1.3

1. In systems with shared scaffold and exhaust resources, the optimum will be interior when the two marginal elasticity curves cross. Measured all-to-fusion policies should underperform when exhaust elasticity is initially high.
2. In anisotropic systems, axes with maximum raw attraction will often be demoted when gain and shadow are positively correlated. The frequency of demotion should increase with shadow-gain covariance.
3. In systems with convex downstream overload, useful output should peak near product transport capacity and decline under maximum drive.
4. In stochastic fusion processes with observable product-shadow signals, threshold abort policies should outperform no-abort policies whenever marginal shadow cost crosses marginal benefit before nominal completion.

## 10 Limitations

All four experiments are deliberately abstract and dimensionless. They do not claim predictive quantitative validity for any specific reactor, cell, material, or AI architecture. Their purpose is to test whether candidate principles remain coherent under randomized virtual environments. The reported percentages are properties of the sampled toy distributions, not universal constants. The value of the addendum lies in the candidate equations, falsifiable predictions, and cross-domain targets.

## 11 Reproducibility note

The accompanying source bundle contains the Python script `run_v13_experiments.py`, generated CSV files for all four virtual experiments, the JSON/CSV summary tables, all figures, this L<sup>A</sup>T<sub>E</sub>X source, and the merged manuscript PDF. The random seed for this sprint is fixed at 13013.

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